

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Received

Well File No.
28654

FEB 18 2016

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

ND Oil & Gas Division

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed October 15, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Kline Federal 5300 41-18 13T2X					
Footages	Qtr-Qtr	Section	Township	Range	
401 F S L 237 F W L	LOT4	18	153 N	100 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 10/15/15 the above referenced well is on pump.

End of Tubing: 2-3/8" L-80 tubing @ 8200.39'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 8037.79'

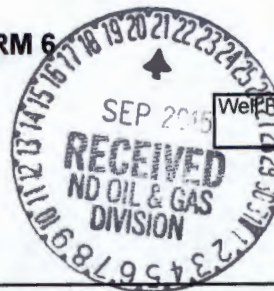
Company Oasis Petroleum North America LLC		Telephone Number 281 404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Specialist	Date February 17, 2016		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 3-3-2016	
By 	
Title TAYLOR ROTH	
Engineering Technician	

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28654**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 13T2X		Spacing Unit Description Sec. 17/18/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Pool Bakken	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension

LOCATION OF WELL

At Surface 401 F S L	237 F W L	Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date September 16, 2014	Date TD Reached February 7, 2015	Drilling Contractor and Rig Number Nabors 486		KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2159	13 1/2	36			1210	0
Vertical Hole	Intermediate	7	0	11185	8 3/4	32			815	3500
Lateral1	Liner	4 1/2	10271	20754	6	13.5			515	10271

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20759	Perforations	11185 20754	13100		06/09/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11185' to 20754'						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) August 27, 2015			Producing Method Flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) Producing		
Date of Test 08/27/2015	Hours Tested 24	Choke Size 38 /64	Production for Test		Oil (Bbbls) 1359	Gas (MCF) 915	Water (Bbbls) 3110	Oil Gravity-API (Corr.) °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 1250		Calculated 24-Hour Rate	Oil (Bbbls) 1359	Gas (MCF) 915	Water (Bbbls) 3110	Gas-Oil Ratio 673		

PLUG BACK INFORMATION

[illegible][illegible]

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
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Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 06/09/2015	Stimulated Formation Three Forks Second Bench	Top (Ft) 11185	Bottom (Ft) 20754	Stimulation Stages 50	Volume 126978	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 4230380	Maximum Treatment Pressure (PSI) 9122		Maximum Treatment Rate (BBLS/Min) 39.0	
Details 100 Mesh White: 219480 20/40 White: 1882960 40/70 White: 1515060 20/40 Resin Coated: 612880						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Email Address jswenson@oasispetroleum.com
Signature 		Date 09/18/2015
Printed Name Jennifer Swenson		Title Regulatory Specialist



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE -- Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No. 28654
NDIC CTS No. To be assigned

228654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number KLINE FEDERAL 5300 41-18 13T2X	Qtr-Qtr LOT4	Section 18	Township 153	Range 100	County Mckenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective August 22, 2015
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Field Address	City	State	Zip Code
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Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective August 22, 2015
---	---	-----------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
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The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
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Power Crude Transport	25%	August 22, 2015
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Other Transporters Transporting From This Lease	% Transported	Date Effective
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		August 22, 2015
--	--	------------------------

Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date August 22, 2015
--	--------------------------------

Signature <i>Dina Barron</i>	Printed Name Dina Barron	Title Mktg. Contracts Administrator
---------------------------------	------------------------------------	---

Above Signature Witnessed By:		
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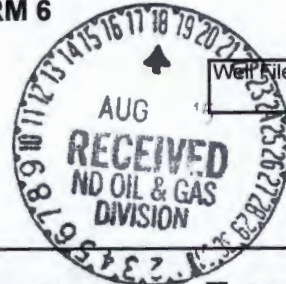
Signature <i>Jeremy Harris</i>	Printed Name Jeremy Harris	Title Marketing Scheduler
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FOR STATE USE ONLY

Date Approved SEP 18 2015
By <i>Ernie Robinson</i>
Title Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28654**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 13T2X		Spacing Unit Description Sec. 17/18/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		Pool Bakken	
State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface 401 F S L	237 F WL	Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date September 16, 2014	Date TD Reached February 7, 2015	Drilling Contractor and Rig Number Nabors 486		KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD; CBL from int. TD to surface						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2159	13 1/2	36			1210	0
Vertical Hole	Intermediate	7	0	11185	8 3/4	32			815	2850
Lateral1	Liner	4 1/2	10271	20754	6	13.5			515	10271

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20759	Perforations		13100					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)		
Date Well Completed (SEE INSTRUCTIONS)		Producing Method		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In)		
Date of Test	Hours Tested	Choke Size /64	Production for Test	Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Oil Gravity-API (Corr.)	Disposition of Gas
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Gas-Oil Ratio

PLUG BACK INFORMATION

[illegible]

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
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Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address jswenson@oasispetroleum.com	Date 08/17/2015
	Signature 	Printed Name Jennifer Swenson

Industrial Commission of North Dakota
Oil and Gas Division

Well or Facility No

28654

Verbal Approval To Purchase and Transport Oil

Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Todd Hanson	(701) 577-1632

WELL INFORMATION

Well Name					Inspector
KLINE FEDERAL 5300 41-18 13T2X					Richard Dunn
Well Location	QQ	Sec	Twp	Rng	County
	LOT4	18	153 N	100 W	MCKENZIE
Footages	401	Feet From the	S	Line	Field
	237	Feet From the	W	Line	BAKER
Date of First Production Through Permanent Wellhead					Pool
					BAKKEN
					This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser	Transporter
Kinder Morgan	Kinder Morgan

TANK BATTERY

Central Tank Battery Number : 228654-01
--

SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
15000	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

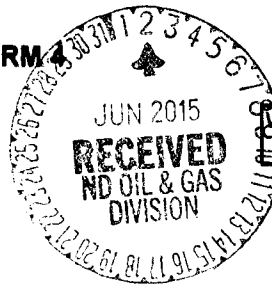
DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date **8/25/2015**
Date Approved **8/25/2015**
Approved By **Richard Dunn**

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 1**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
800 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 6749 (09-2008)



Well File No. **28654**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date June 2, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Kline Federal 5300 41-18 13T2X					
Footages 401 F S L 237 F W L		Qtr-Qtr LOT4	Section 18	Township 163 N	Range 100 W
Field Baker		Pool BAKKEN		County McKenzie	

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective Immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed

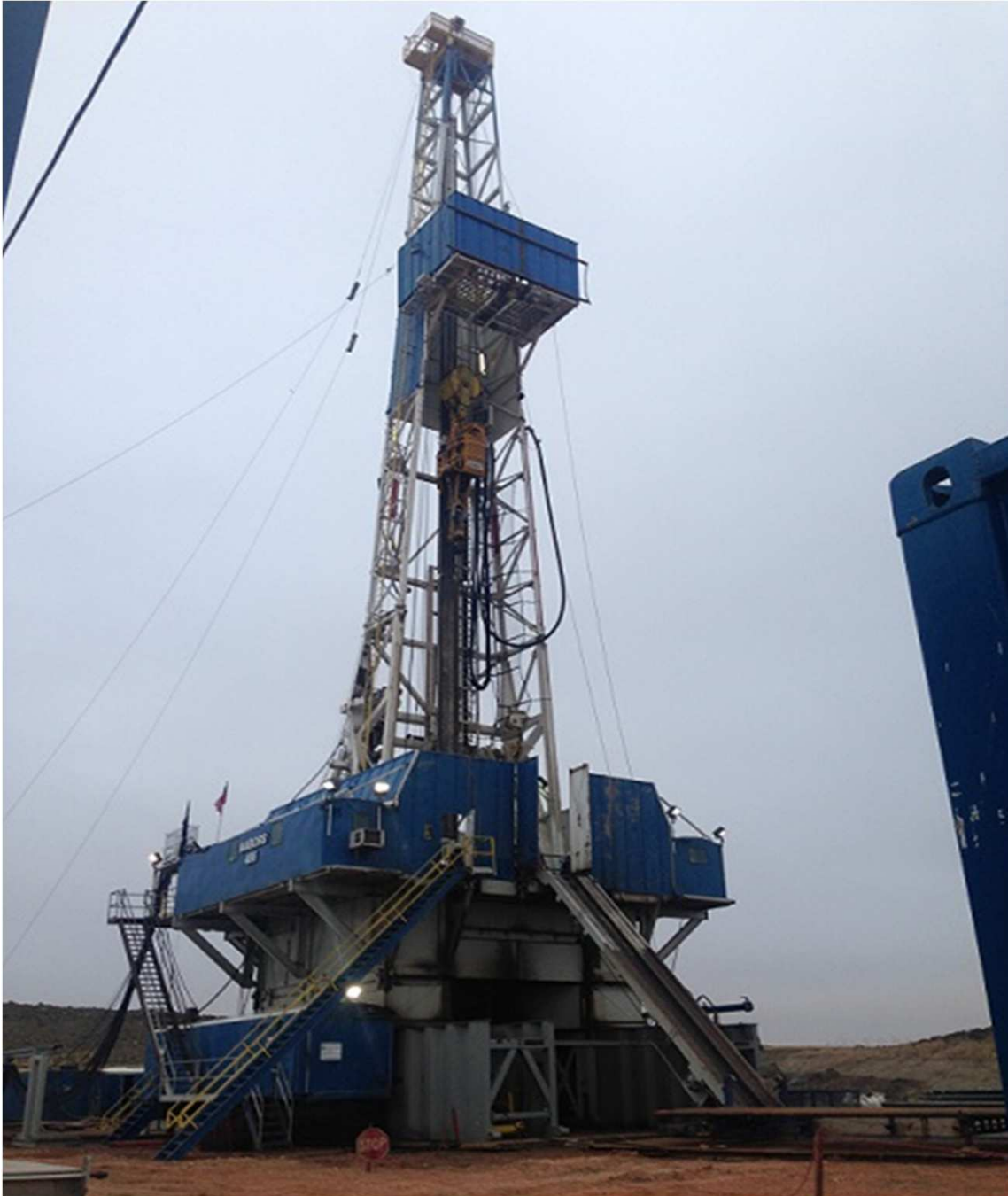
OFF CONFIDENTIAL 12/02/15.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date June 2, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6/03/15	
By 	
Title Engineering Technician	

Oasis Petroleum North America

Kline Federal 5300 41-18 13T2X



Services Performed For:
Andrew Nelson, Brendan Hargrove
Oasis Petroleum North America
1001 Fannin Suite 202
Houston, TX 77002

Onsite Geology Performed by:
Devin Wood, Bob Witmer
RPM Geologic
geology@rpmconsultinginc.com
(303) 595-7625

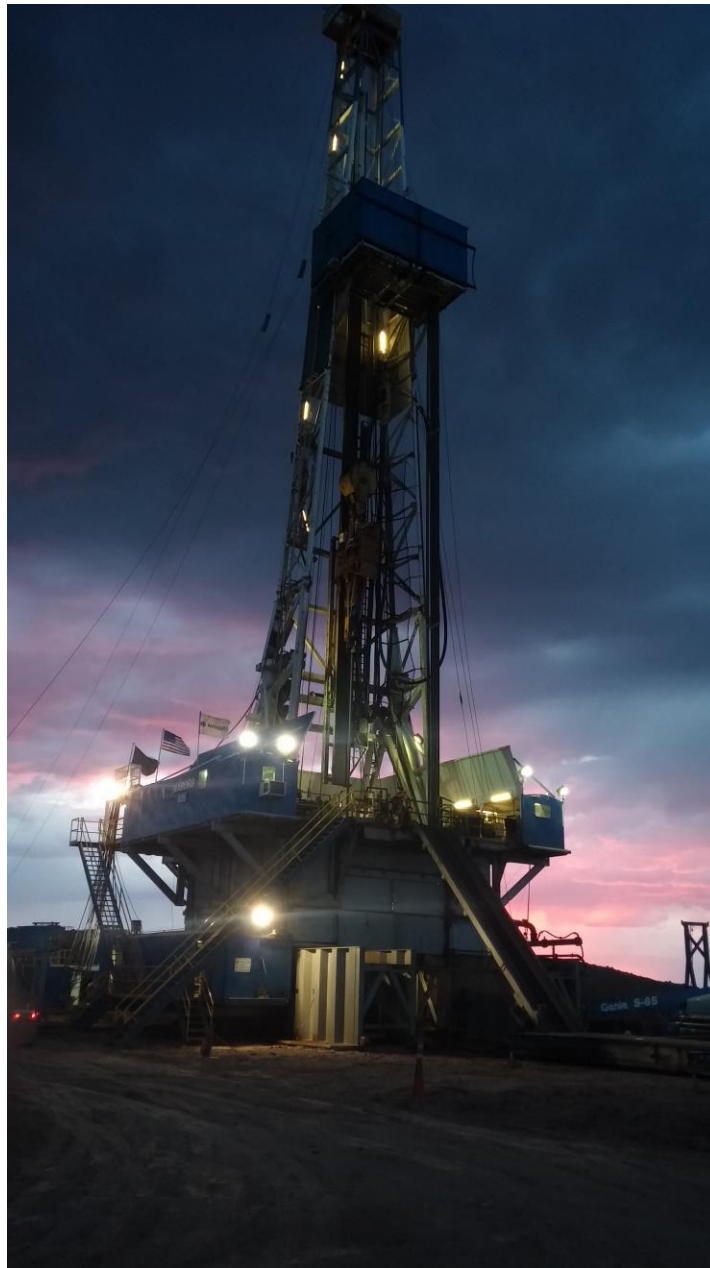
WELL EVALUATION

Oasis Petroleum

Kline Federal 5300 41-18 13T2X

SW SW Sec. 18, T153N, R100W

Williams County, North Dakota



Synopsis

Oasis Petroleum *Kline Federal 5300 41-18 13T2X* [SW SW Sec. 18 T153N R100W] is located ~6 miles Northwest of the town of Williston, North Dakota. *Kline Federal 5300 41-18 13T2X* is situated within the Baker Field. A single lateral leg trending 90° was proposed to be drilled from the SW SW corner of Section 18 to the SE SE corner of Section 17 targeting the porous dolostone in the second bench of the **Three Forks**.

Control Wells

Two completed wells were used as the control offsets to the *Kline Federal 5300 41-18 13T2X*.

The primary offset considered was the *Kline Federal 5300 41-18 14BX* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 41-18 14BX* is located approximately 33 feet south of *Kline Federal 5300 41-18 13T2X*.

The other offset considered was the *Kline Federal 5300 11-18H* [SW NW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 11-18H* is located approximately half a mile north of *Kline Federal 5300 41-18 13T2X*.

During the curve, the gamma ray that was produced was constantly compared to the *Kline Federal 5300 41-18 14BX* and the *Kline Federal 5300 11-18H* wells. Gamma ray was used to help determine a proper landing depth.

To assist in the choosing of the landing target, an isopach table (Table 1) was constructed to measure the distance of formation tops to the target depth determined from the *Kline Federal 5300 41-18 14BX* and *Kline Federal 5300 11-18H* offset wells. Comparison of gamma ray markers showed the most consistent isopach for up-hole formations to be the **Middle Bakken**. The same information was coalesced into a single chart (Table 2) displaying offset tops and an average of the estimated landing point based on the offset wells. This chart was used to corroborate the landing point, but isopach thickness was the main consideration in choosing a landing point.

<u>Control Well</u>								
Operator:	Oasis Petroleum				Oasis Petroleum			
Well Name:	<i>Kline Federal 5300 41-18 14BX</i>				<i>Kline Federal 5300 11-18H</i>			
Location:	SW SW Sec. 18 T153N R100W McKenzie Co., ND Approximately 33 ft South				SW NW Sec. 18 T153N R100W McKenzie Co., ND Approximately .5 miles North			
Elevation:	KB: 2,082'				KB: 2,078'			
Formation/ Zone	Est. TVD	MSL Datum	Thickness	Dist to target	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382'	-6,300'	182'	2,471'	8,369	-6,291'	147'	2,467'
Charles Salt	8,564'	-6,482'	641'	2,289'	8,516	-6,438'	700'	2,320'
Base Last Salt	9,205'	-7,123'	223'	1,648'	9,216	-7,138'	224'	1,620'
Mission Canyon	9,428'	-7,346'	548'	1,425'	9,440	-7,362'	556'	1,396'
Lodgepole	9,976'	-7,894'	728'	877'	9,996	-7,918'	684'	840'
False Bakken	10,704'	-8,622'	8'	149'	10,680	-8,602'	13'	156'
Upper Bakken Shale	10,712'	-8,630'	16'	141'	10,693	-8,615'	18'	143'
Middle Bakken	10,728'	-8,646'	-	125'	10,711	-8,633'	-	125'

Table 1: Distance from Formation tops in the control wells were utilized to determine an initial drilling target.

Kline Federal 12TX Target Landing Chart										
Operator:	Oasis Petroleum			Oasis Petroleum			Oasis Petroleum			
Well Name:	Kline Federal 5300 41-18 12TX			Kline Federal 5300 41-18 13TX			Kline Federal 5300 41-18 14BX			
Location:	SW SW Sec. 18 T153N, R100W McKenzie County, ND			SW SW Sec. 18 T153N, R100W McKenzie County, ND			SW SW Sec. 18 T153N, R100W McKenzie County, ND			
Elevation:	KB: 2,082'			KB: 2,082'			KB: 2,082'			
Formation / Zone	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Primary Tops TVD (gamma)	Primary Tops MSL	Thickness	Proj. Target Landing
Kibbey	8,384'	-6,302'	172'	8,382'	-6,300'	178'	8,382'	-6,300'	182'	10,819'
Charles	8,556'	-6,474'	650'	8,560'	-6,478'	646'	8,564'	-6,482'	641'	10,826'
Base Last Salt	9,206'	-7,124'	222'	9,206'	-7,124'	222'	9,205'	-7,123'	223'	10,819'
Mission Canyon	9,428'	-7,346'	547'	9,428'	-7,346'	547'	9,428'	-7,346'	548'	10,820'
Lodgepole	9,975'	-7,893'	115'	9,975'	-7,893'	112'	9,976'	-7,894'	114'	10,821'
LPA	10,090'	-8,008'	75'	10,087'	-8,005'	70'	10,090'	-8,008'	74'	10,822'
LPB	10,165'	-8,083'	46'	10,157'	-8,075'	54'	10,164'	-8,082'	45'	10,823'
LPC	10,211'	-8,129'	188'	10,211'	-8,129'	186'	10,209'	-8,127'	194'	10,818'
LPD	10,399'	-8,317'	161'	10,397'	-8,315'	162'	10,403'	-8,321'	154'	10,825'
LPE	10,560'	-8,478'	65'	10,559'	-8,477'	67'	10,557'	-8,475'	68'	10,818'
LPF	10,625'	-8,543'	79'	10,626'	-8,544'	79'	10,625'	-8,543'	77'	10,820'
False Bakken	10,704'	-8,622'	9'	10,705'	-8,623'	7'	10,702'	-8,620'	10'	10,818'
Upper Bakken Shale	10,713'	-8,631'	17'	10,712'	-8,630'	18'	10,712'	-8,630'	16'	10,820'
Middle Bakken	10,730'	-8,648'	41'	10,730'	-8,648'	-	10,728'	-8,646'	-	10,818'
Lower Bakken	10,771'	-8,689'	12'	10,770'	-	-	-	-	-	-
Pronghorn	10,783'	-8,701'	16'	10,771'	-	-	-	-	-	-
Three Forks	10,799'	-8,717'	21'	10,799'	-	-	-	-	-	-
Target	10,820'	-8,738'	-10,820'	10,820'	-8,738'	-	10,817'	-8,735'	-	10,817'
									Average -->	10,820'

Table 2: Estimated landing point based on offset well formation tops.

Geologic Assessment

Methods

Geologic supervision of *Kline Federal 5300 41-18 13T2X* was provided by experienced RPM well site geologists. Gas and chromatograph levels were measured using NOV WellSight Gas Detection real time gas detector and chromatograph system. The WellSight gas detection system uses non-dispersive infrared and chemical sensor gas detection to evaluate gases liberated from the formation by the drill bit and carried to the surface by the drilling fluid.

The gas detector was interfaced with a RigWatch electronic data recorder system. RigWatch provided drill rate, on-off bottom and pump strokes to the gas detector and received total gas information from the gas detector for viewing on location and remotely.

Under the direction of RPM well site geologists, rig crews were instructed to catch lagged drill cutting samples at 30' intervals in the vertical hole and curve build from 8,350' to 11,200' MD and 50' samples from 11,200' to TD (20,583').

Sampled drill cuttings were examined wet and dry under a binocular microscope using both plain (broad spectrum) and transmitted light. Cuttings were evaluated for hydrocarbon "cut" by immersion in Acetone and inspection under a UV fluoroscope. 10% hydrochloric acid and alizarin red were used to determine the calcareous and dolomitic content of rocks and cementing.

RPM Geologic, Inc. (RPM) well site geologists also closely examined MWD gamma-ray information and penetration rates to aid in steering decisions and dip degree estimations.

Vertical Operations

Overview

The *Kline Federal 5300 41-18 13T2X* was spud for surface drilling on 13 September 2014 utilizing Nabors 486 as the drilling contractor. Prior to commencement of RPM mud logging services, a 17 1/2" hole was drilled with fresh water to depth of 2,159' and isolated with 13 3/8" 36# J-55 casing cemented from 2,157' to surface. Due to uphole stability issues a second set of casing was set. Two 12 1/2" bits were used to drill to a depth of 6,135' and 9 5/8 40# HCL 80 casing was set @ 6,123'

RPM well site geologists began logging the vertical interval at 8,350'.

One PDC bit were used to drill out of secondary casing to vertical TD. PDC bit #1 (8 3/4" Security) was used to drill from 6,135' to 10,300' and averaged an ROP of 61.7 ft/hr in 67.5 hours of use.

Diesel invert drilling fluid with a mud weight ranging from 9.80-10.30 ppg was used for the remainder of the vertical hole and in the curve builds sections. The vertical interval was drilled to a kick off point (KOP) of 10,300', at which point vertical operations were ceased.

Lithology

The top of the **Kibbey Lime** [Mississippian Madison Group] was picked at 8,382' (-6,300'), level to the *Kline Federal 5300 41-18 14BX*. Samples from this interval (Figure 1) were described as:

LIMESTONE: wackestone, medium gray, friable to hard, finely crystalline, blocky, earthy to sucrosic, trace visible porosity



*Figure 1: Photograph of Limestone seen in sample from the **Kibbey Limestone** formation.*

The top of the **Charles Salt** [Mississippian Madison Group] was logged at 8,560' (-6,478'), 5' low to the *Kline Federal 5300 41-18 14BX*. The **Base Last Salt** was drilled at 9,206' (-7,124'), 1' low to the *Kline Federal 5300 41-18 14BX*. Samples from this interval (Figure 2) were described as:

SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

LIMESTONE: mudstone, light gray to medium gray, off white to cream, microcrystalline, soft to fair, chalky to crystalline texture, trace intercrystalline porosity, slightly light brown oil stain

ANHYDRITE: off white, cream, soft to firm, amorphous texture, no visible porosity



Figure 2: Photograph of Salt seen in sample from the Charles formation.

The top of the **Mission Canyon** formation of the Madison Group [Mississippian] was reached at 9,428' (-7,346'), level to the *Kline Federal 5300 41-18 13T2X*. Samples from this interval (Figure 3) were described as:

LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline



Figure 3: Photograph of limestone from the Mission Canyon formation.

The top of the **Lodgepole** formation of the Madison Group (Mississippian) was logged at 9,975' (-7,893'), 1' high to the *Kline Federal 5300 41-18 14BX*. Samples from this interval (Figure 4) were described as:

ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain



*Figure 4: Photograph of argillaceous limestone from **Lodgepole** formation.*

Directional Operations

Gyrodata provided equipment and personnel for MWD teams, while RPM Consulting provided directional personnel. RPM geologists worked closely with RPM directional as well as Gyrodata MWD teams to formulate steering decisions to maximize the footage of borehole in the pay zone.

Curve Build

Overview

One PDC bits was used to build the curve section. Bit #4 was an 8 ¾" SEC MMD55C bit used to drill the curve from a measured depth of 10,300' to 11,150' in 32.0 hours, averaging 27.66 ft/hr. 7" 32# P-110 29 lbs casing was set to 10,700' MD upon completion of the curve.

A Chart was constructed for the curve (Chart 1) that compared the isopach of certain easily recognizable gamma markers from the **Kibbey Lime** to the **Middle Bakken** on the control well *Kline Federal 5300 41-18 14BX* to the *Kline Federal 5300 11-18H* as marker picks were made. This chart shows the **Middle Bakken** top being the closest in thickness to target based on the offset well, and was the main point used to make changes to the target landing depth during the curve. A table (Table 2) was constructed comparing *Kline Federal 5300 41-18 13T2X* to the offset formation tops as well as an average of the estimated landing point based on the offset wells. This chart was used to corroborate the landing point, but isopach thickness was the main consideration in choosing a landing point. The curve was completed at a measured depth of 11,185' and TVD of 10,852', ~25 feet into the second bench of the **Three Forks**.

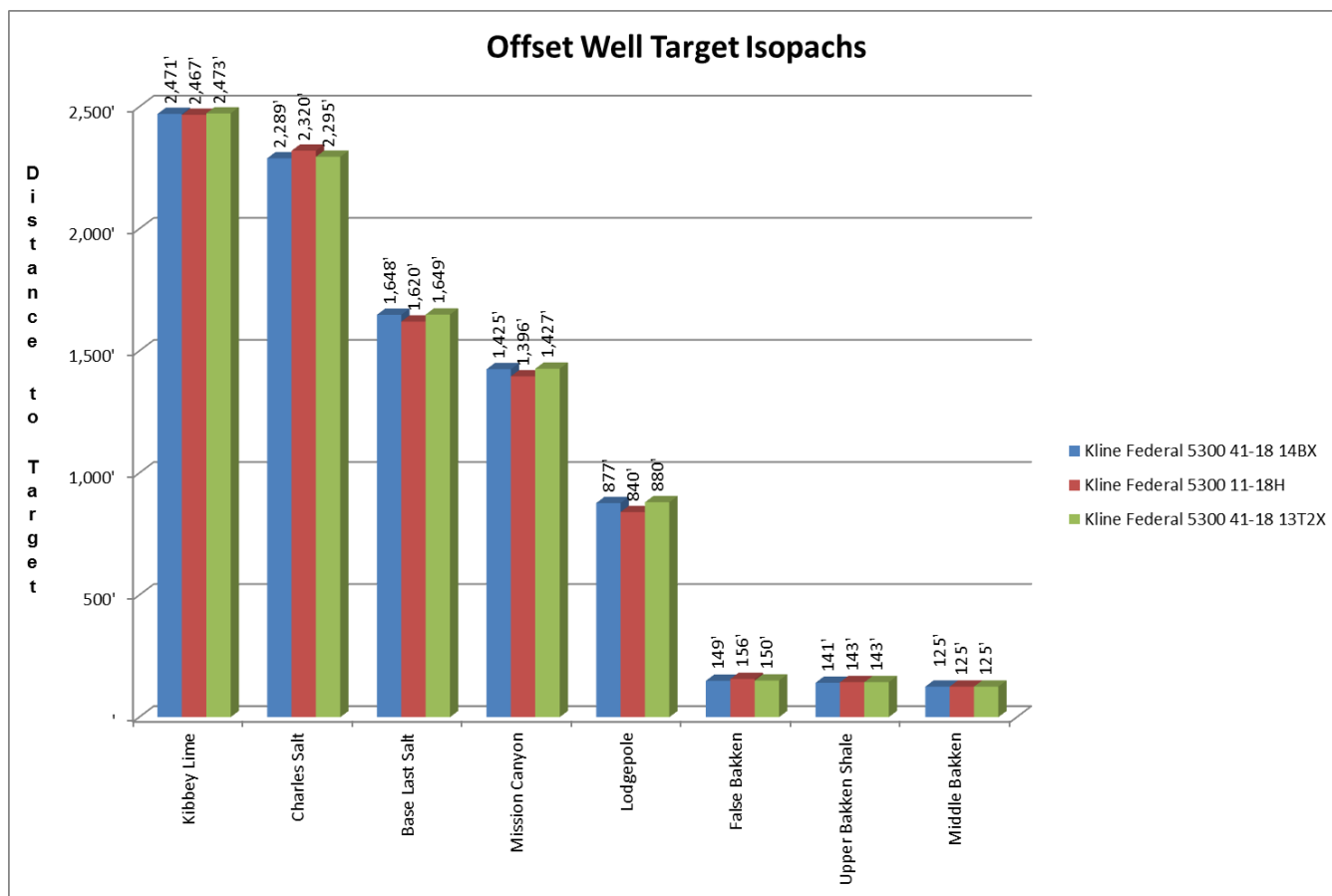


Chart 1: Comparing distances from gamma markers on the offset wells to markers in gamma in the curve.

Operator: Well Name: Location:	Oasis Petroleum Kline Federal 5300 41-18 13T2X SW SW Sec. 18 T153N, R100W McKenzie County, ND			Oasis Petroleum Kline Federal 5300 41-18 14BX SW SW Sec. 18 T153N, R100W McKenzie County, ND			Oasis Petroleum Kline 5300 11-18H SW NW Sec18 T153N, R100W McKenzie County, ND			
Elevation:	KB: 2,082'			KB: 2,082'			KB: 2,078'			
Formation / Zone	Primary Tops TVD (gamma)	Primary Tops MSL	Thickness	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Proj. Target Landing
Kibbey	8,382'	-6,300'	178'	8,382'	-6,300'	182'	8,369'	-6,291'	147'	10,841'
Charles	8,560'	-6,478'	646'	8,564'	-6,482'	641'	8,516'	-6,438'	700'	10,808'
Base Last Salt	9,206'	-7,124'	222'	9,205'	-7,123'	223'	9,216'	-7,138'	224'	10,865'
Mission Canyon	9,428'	-7,346'	547'	9,428'	-7,346'	548'	9,440'	-7,362'	556'	10,866'
Lodgepole	9,975'	-7,893'	112'	9,976'	-7,894'	114'	9,996'	-7,918'	112'	10,875'
LPA	10,087'	-8,005'	70'	10,090'	-8,008'	74'	10,108'	-8,030'	112'	10,874'
LPB	10,157'	-8,075'	54'	10,164'	-8,082'	45'	10,220'	-8,142'	38'	10,914'
LPC	10,211'	-8,129'	186'	10,209'	-8,127'	194'	10,258'	-8,180'	126'	10,902'
LPD	10,397'	-8,315'	162'	10,403'	-8,321'	154'	10,384'	-8,306'	158'	10,838'
LPE	10,559'	-8,477'	67'	10,557'	-8,475'	68'	10,542'	-8,464'	76'	10,838'
LPF	10,626'	-8,544'	79'	10,625'	-8,543'	77'	10,618'	-8,540'	62'	10,847'
False Bakken	10,705'	-8,623'	7'	10,702'	-8,620'	10'	10,680'	-8,602'	13'	10,831'
Upper Bakken Shale	10,712'	-8,630'	18'	10,712'	-8,630'	16'	10,693'	-8,615'	18'	10,835'
Middle Bakken	10,730'	-8,648'	40'	10,728'	-8,646'	125'	10,711'	-8,633'	125'	10,836'
Lower Bakken	10,770'	-8,688'	12'	-	-	-	-	-	-	-
Pronghorn	10,782'	-8,700'	17'	-	-	-	-	-	-	-
Three Forks	10,799'	-8,717'	25'	-	-	-	-	-	-	-
Claystone	10,824'	-8,742'	9'	-	-	-	-	-	-	-
Three Forks 2nd Bench	10,833'	-	-	-	-	-	-	-	-	-
Target	10,855'	-8,773'	-	10,853'	-8,771'	-	10,836'	-8,758'	-	10,836'
									Average -->	10,854'

+ Table 2: Estimated landing point based on offset well formation tops.

Lithology

The top of the “**False Bakken**” was drilled at a measured depth of 10,756’ (TVD 10,705’) (-8,623’) 1’ low to the *Kline Federal 5300 41-18 14BX*.

The *Upper Shale* of the **Bakken** formation [Mississippian – Devonian] was drilled at 10,764’ MD (10,712’ TVD) (-8,630’) with sample returns of black, carbonaceous, and petroliferous shale (Figure 5) characterized by gamma ray values in excess of 255 API counts. The *Upper Shale* was flat to the *Kline Federal 5300 41-18 14BX*. Samples were described as:

SHALE: black, firm, earthy to waxy texture, pyrite, carbonaceous, platy, visible fracture porosity



Figure 5: Photograph sample from the Upper Bakken Shale.

The *Middle Member* of the **Bakken** formation was penetrated at 10,789' MD, (10,730' TVD) (-8,648'), 2' low to the *Kline Federal 5300 41-18 14BX*. The *Middle Member* contained layers of sandstone and silty sandstones with varying amounts of porosity and oil stain (Figure 6). Samples from the *Middle Member* were described as:

SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain



*Figure 6: Photograph of sandstone and silty sandstone from the middle member of the **Bakken** formation.*

The *Lower Member* of the **Bakken** formation was penetrated at 10,852' MD, (10,770' TVD) (-8,688'). The *Lower Member* consists of a dark, petroliferous shale (Figure 7). Samples from the *Lower Member* were described as:

SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated



*Figure 7: Photograph of sandstone and silty sandstone from the lower member of the **Bakken** formation.
8 3/4" Bit & Diesel Invert Drilling Fluid*

The **Pronghorn** formation was drilled at 10,874' MD, (10,782' TVD) (-8,700').

The **Three Forks** formation was drilled at 10,909' MD, (10,799' TVD) (-8,717'). The **Three Forks** consists of a light colored, pyritic, slightly oil stained dolostone (Figure 8). Samples from the **Three Forks** were described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain



*Figure 8: Photograph of dolostone from the **Three Forks** formation.*

The **Three Forks** first bench claystone was drilled at 10,974' MD, (10,824' TVD) (-8,742'). Samples from the claystone were described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

The **Three Forks** second bench formation was drilled at 11,004' MD, (10,833' TVD) (-8,751'). The **Three Forks** consists of a light colored, pyritic, slightly oil stained dolostone (Figure 9). Samples from the **Three Forks** were described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

Lateral

Overview

Drilling fluid consisting of open system salt water brine (9.9 ppg) was used while drilling the lateral section up to the TD of 20,760'. One 6" PDC bits was used to drill the lateral.

Bit #5 (Reed Z613) was a PDC bit used to drill the start of the lateral from 11,185' to a YD of 20,760' and averaged an ROP of 54.71 over 175 hours of use.

Drilling operations were concluded at 4:30 on February 7, 2015 at a bottom hole location of 10,022.35' E & 599.11' S of surface location or approximately 198.11' FNL & 280.4' FEL of NE NE Sec. 20, T153N, R100W.

Lithology

To aid in communications while drilling the lateral, a type log was created using a common gamma signature from the original offset well for the pad. The cool gamma "C" marker (50 api – 60 api) as well as the warm gamma signature characteristic of the claystone appeared to be reliable and served as our main markers for steering.

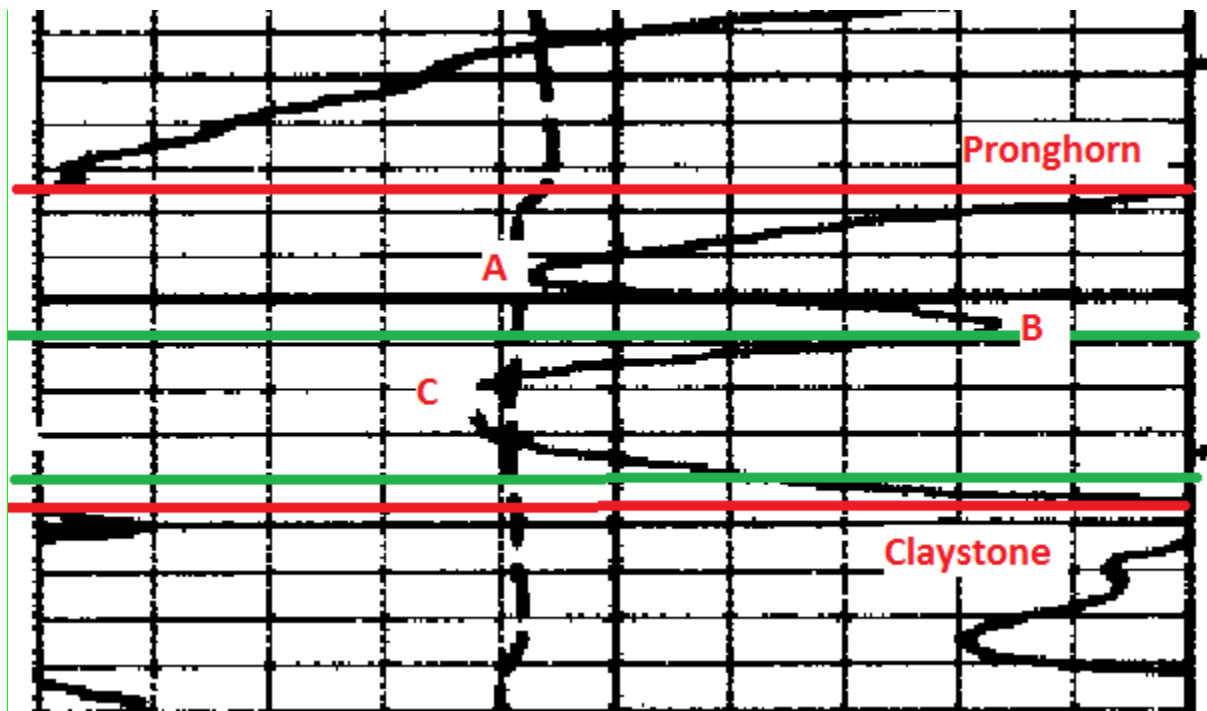


Diagram 1: Type log created from gamma log of the offset well

The target Zone of the **Three Forks** consisted of a dolostone with very apparent intergranular porosity, moderate gas, and moderate to abundant oil show (Figure 9). Samples from the target Zone were generally described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine and, trace intercrystalline porosity, trace disseminated pyrite, trace spotty light brown oil stain, fast white diffuse cut



Figure 9: Photograph of sandstone from "Target Zone" on Diagram 1

Gas and hydrocarbon Shows

It was found that the gas in the Three Forks 1st Bench averaged 40-90 units throughout the target zone with peaks of 160 units. In the claystone below the target zone gas levels averaged 30 units.

Hydrocarbon cuts using Entron solvent were moderate though the entire lateral. The target zone and above was characterized by moderate fast white to yellow diffuse cuts and common fast streaming cuts. No correlation was found between the strength of the hydrocarbon cuts and the position of the bit in the dolomite 1st Bench Three Forks.

Summary

- 1) The *Kline Federal 5300 41-18 13T2X* was initially spud for surface drilling on 17th of September 2014 in the Indian Hills field of McKenzie County, North Dakota. The well was spud by Nabors 486 and the vertical hole was drilled to a total depth (KOP) of 10,300'.
- 2) A mud program consisted of diesel invert (10.3-11.6 ppg) after surface casing through the curve build sections. The drilling mud was successful in maintaining stable hole conditions and minimizing washout through the salt intervals. Brine water with a weight of 9.8-9.9 ppg was used during lateral drilling operations to maintain hydrostatic balance until a TD of 20,760'.
- 3) One 8 3/4" Haliburton PDC bit was used to drill out of surface casing to a KOP of 10,300'. One 8 3/4" PDC bit was used to drill the curve to 11,185' MD. One 6" PDC bit was used to drill the lateral to 20,760' TD.
- 4) The lateral was targeting the **Three Forks** member ranging from about 8' below the base of the **Pronghorn** down to approximately 1' above the **Claystone**.
- 5) Gas levels between 40 and 100 units and peaks as high as 160 units were maintained through the lateral drilling operations while in the **Three Forks**.
- 6) The **Three Forks** target consisted of one observed gross lithology. The **Three Forks** was observed to be a dolostone with lots of disseminated pyrite and apparent intergranular porosity.
- 7) Trace to moderate oil stain was observed in zone associated with a fast, white, diffuse cut. Trace oil washed out of the samples into the wash water when they were cleaned.
- 8) Drilling operations were concluded at 14:00 on February 7th, 2015 at a bottom hole location of 10,022.35' E & 599.11' S of surface location or approximately 198.11' FNL & 280.4' FEL of NE NE Sec. 20, T153N, R100W.

Respectfully submitted,
Annika Tostengard
RPM Geologic LLC

Well Information

Operator: Oasis Petroleum North America

Address: 1001 Fannin Suite 202
Houston, TX 77002

Well Name: Kline Federal 5300 41-18 13T2X

Field/ Prospect: Indian Hill

Elevation: GL: 2,056'
KB: 2,082'

Spud Date: 17-Sep-14

API #: 33-053-06028

NDIC Well File #: 28654

Surface Location: SW SW Sec. 18 T153N, R100W

Footages: 401' FSL & 237' FWL

County, State: McKenzie Co., ND

Basin: Williston

Well Type: Horizontal Bakken

Contractor: Nabors 486

Toolpushers: Logan Penhollow, Tyler Schultz

Field Supervisors: Mark Lawlar/David Pawelek

Directional Drilling RPM Consulting
Ryan Romans
Ty Tschacher

Chemical Company Halliburton

Mud Engineer Tim Vaira

H₂S MONITORING: NOV WellSight Gas Watch

MWD Gyro Data
Beau Watt, Josh Elliott

Wellsite Geologist Annika Tostengard
Bob Witmer

Rock Sampling: 30' from 8,350' to 11,185' MD
50' 10,1185' MD to 20,760' (TD)

Prospect Geologist Brendan Hargrove

Gas Detector NOV WellSight Gas Watch

Sample Examination: Binocular microscope & fluoroscope

Sample Cuts: Acetone

Key Offset Wells:

Oasis Petroleum.	Kline Federal 41-18 11-18h	SE SE Sec. 10 T153N R100W	McKenzie Co., ND
Oasis Petroleum	Kline Federal 41-18 14BX	SE SE Sec. 10 T153N R100W	McKenzie Co., ND

Pumps: #1 & #2: HHF 1600 Triplex - 12" Stroke length Output: 0.0838 bbl/stk

Mud Type: Diesel invert mud 2,159-11,1850' (Curve TD)
Salt Water Brine, 11,185' - 20,760' (TD)

Casing: Surface: 13 3/8" 54# J-55 @ 2,159'
Intermediate: 9 5/8" 40# HCL80 @ 6,123', 7" 32# P-110 @ 11,185'

Hole Size: 17 1/2" to 2,159' MD, 12 1/2" to 6,135', 8 3/4" to 11,185' MD 6" to 20,760'

Total Drilling Days: 22

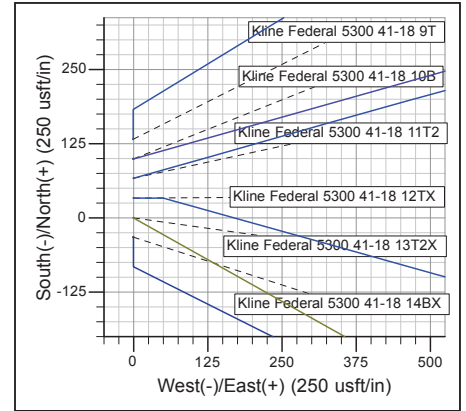
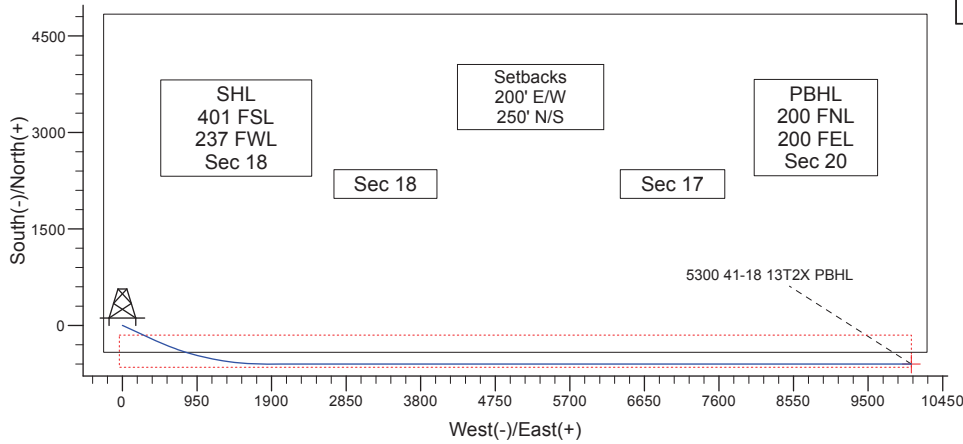
<u>Horizontal Target:</u>	<i>Three Forks</i>	<u>BOTTOM HOLE LOCATION:</u>
		10,022.35' E & 599.11' S
<u>Kick-Off Point / Date:</u>	<i>10,300' on October 12, 2014</i>	of surface location or approx.
		198.11' FNL & 280.4' FEL
<u>Total Depth/ Date:</u>	<i>20,760' Febuary 7, 2015</i>	NE NE Sec. 20, T153N, R100W
<u>Ending Vertical Section</u>	<i>10,022.35'</i>	
<u>Ending Azimuth</u>	<i>90.8°</i>	
<u>Status of Well:</u>	<i>Complete</i>	

Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 13T2X
Wellbore: Kline Federal 5300 41-18 13T2X
Design: Design #1



WELL DETAILS: Kline Federal 5300 41-18 13T2X

Ground Level: 2057.0
Northing 405103.22 Easting 1209966.87 Latitude 48° 4' 7.220 N Longitude 103° 36' 11.950 W

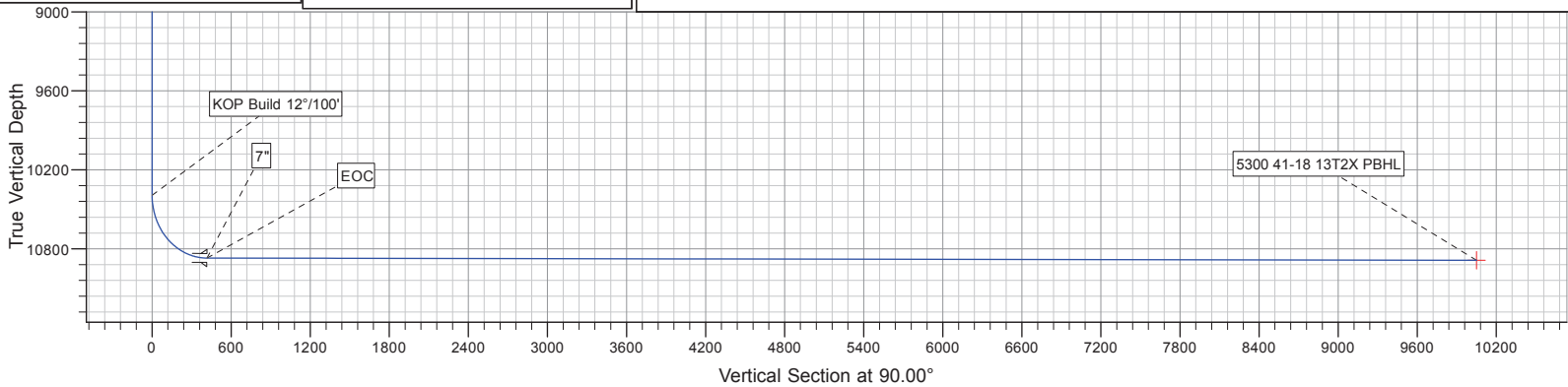


CASING DETAILS			
TVD	MD	Name	Size
10870.0	11142.0	7"	7



Azimuths to True North
Magnetic North: 8.27°
Magnetic Field
Strength: 56437.0snT
Dip Angle: 72.99°
Date: 5/23/2014
Model: IGRF2010

SECTION DETAILS							
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
10392.5	0.00	0.00	10392.5	0.0	0.0	0.00	
11141.7	89.90	119.35	10870.0	-233.6	415.5	12.00	
12609.2	89.90	90.00	10872.6	-601.3	1819.6	2.00	
20839.6	89.90	90.00	10886.9	-601.3	10050.0	0.00	5300 41-18 13T2X PBHL



Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
0	16-Sep	105'											
1	17-Sep	2,057'	1,977'	1	2	10	60				375	Rig up, F/drill 105'-2,057	
2	18-Sep	2,159'	102'	1	2	5	50				375	Drill F/2,057'-2,159' TOO, run 13 3/8" casing to 2,159', run cement	
3	19-Sep	2,159'		1	-	-	-				-	wait on cement, cut casing, release rig for next surface	
4	5-Oct	2,159'		2	-	-	-				-	NU BOP, P/U BHA, TIH	
5	6-Oct	4,411'	2,252'	2	10	20	45				-	TIH, D/) shoe, drill F/2,159' - 4,411'	
6	7-Oct	6,135'	1,724'	2	20	35	55				-	Drill F/ 4,411' - 6,135', TOH, rig up for casing	
7	8-Oct	6,135'	0'	3	-	-	-				-	Run 9 5/8" casing to 6,123', cement casing, Pick up BHA	
8	9-Oct	6,135'	0'	3	-	-	-				-	TIH, tag D/O shoe,	
9	10-Oct	7,442'	1,307'	3	5	20	60				600	Drill 6,135' to 7,442'	
10	10-Oct	8,840'	1,461'	3	8	0	45	3515	69	69	471	Drill F/ 7442-8226, Rig service, Drill F/8226-8840	Charles salt
11	11-Oct	9,937'	1,097'	3	37	0	46	3290	63	63	445	Drill F/8857-9421, Rig service, Drill F/9421-9519, BOP test, Drill F/9519-9937	Mission canyom

Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
12	12-Oct	10,300'	363'	3	37	0	46	3290	63	63	445	Dril F* 9923-10300, Rig service, Circulate and condition, TOO, Lay down BHA, Cased hole logs, Pick up BHA, TIH, Service top drive.	Mission Canyon
13	13-Oct	10,796'	496'	4	16	23	25	3500	61	63	440	TIH, Service rig, Drill F/10300-10796	Mission canyon
14	14-Oct	11,185'	389'	4	18	30	25	3125	61	63	440	Drill F/10798-11080, Rig Service, Drill F/11080-11185, Circ and condition, Short trip wet, Circ and condition, Rig service, TOH, Change rotating head, TOO, l/d tools	2nd Bench TF
15	15-Oct	11,185'		4	-	-	-	-	-	-	-	P/U casing tools, run 7" casing to 11,185', cement, move rig	2nd Bench TF
16	30-Jan	11,531'	346'	5	8	27	45	2000	75	0	297	Test BOPS // Change out swivel packing // Service Rig // Trip in hole // Drill actual F/ 11,185' - 11531'	2nd Bench TF
17	31-Jan	12,955'	1,424'	5	14	42	45	3150	0	73	297	Drill actual F/11,531 - 12,955' // Service Rig	2nd Bench TF
18	1-Feb	14,730'	1,775'	5	14	59	45	2900	74	0	297	Drill actual F/12,955' - 14,730' // Service Rig	2nd Bench TF
19	2-Feb	15,834'	1,104'	5	24	55	45	3150	70	0	227	Drill actual F/14,742' - 15,834' // Service Rig // Trip out for MWD tool	2nd Bench TF
20	3-Feb	16,380'	546'	5	17	55	40	3150	0	75	297	Replace MDW rool // TIH // Drill actual F/15,834' - 16,380'	2nd Bench TF

Daily Activity

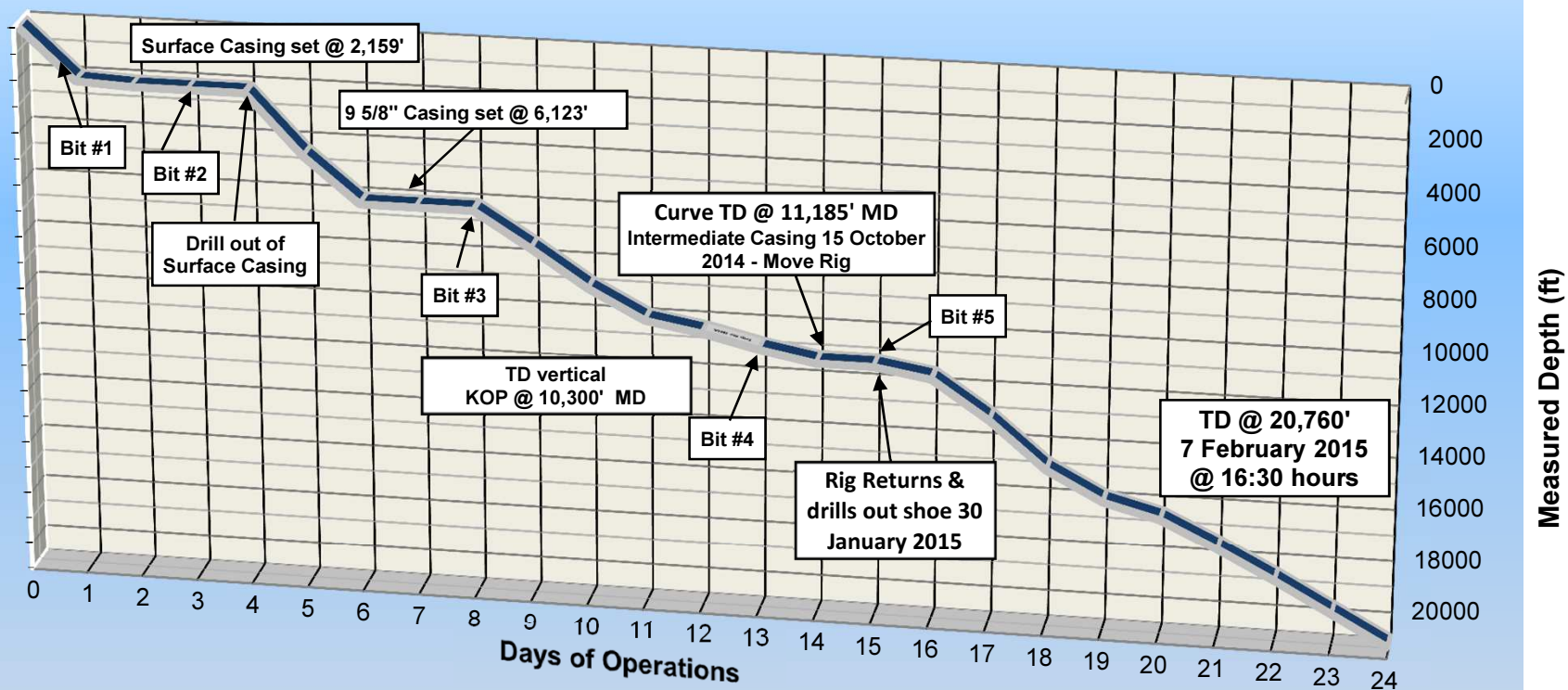
Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
21	4-Feb	17,380'	1,000'	5	17	46	40	3150	70	0	227	Drill actual F/16,380' - 17,380' // Service Rig	2nd Bench TF
22	5-Feb	18,460'	1,080'	5	18	42	40	2880	0	75	297	Drill actual F/ 17,380'-18460 // Service Rig	2nd Bench TF
23	6-Feb	19,615'	1,155'	5	18	42	40	2880	0	75	297	Drill actual F/ 18,460- 19,615' // Service Rig	2nd Bench TF
24	7-Feb	20,766'	1,151'	5	20	60	40	3400	0	75	297	Drill actual F/ 19,615 - 20,760 // Circulate and condition	2nd Bench TF



Daily Progress

Oasis Petroleum
Kline federal 5300 41-18 13T2X

Spud 17 September 2014



Bit Record

<i>Bit #</i>	<i>Size</i>	<i>Make</i>	<i>Model</i>	<i>Serial #</i>	<i>Jets</i>	<i>Depth In</i>	<i>Depth Out</i>	<i>Footage</i>	<i>Hours</i>	<i>Mean ROP (ft/hr)</i>	<i>Accum. Hours</i>
1	17.5	NOV	TFR6195	E163630	12x18	80'	2,159'	2,079'	20.5	101.4	20.5
2	12 1/4	NOV	TFR6195	TFR6195	7x18	2,159'	6,135'	3,976'	16	248.50	36.50
3	8 3/4	Haliburton	MM65D	1254843	6x18	6,135'	10,300'	4,165'	67.5	61.70	104.00
4	8 3/4	SEC	MMD55C	12553854	6x18	10,300'	11,185'	885'	32	27.66	136.00
5	6	Reed	Z613	JK2549	6Xx20	11,185'	20,760'	9,575'	175	54.71	311.00

Daily Mud Data

Day	Date 2014	Depth (0600 Hrs)	Mud WT (ppg)	VIS (sec)	PV (cP)	YP (lbs/ 100 ft²)	Gels (lbs/ 100 ft²)	600/ 300	HTHP (cc/30min)	NAP/ H₂O (ratio)	Cake (APT/ HT)	Solid s (%)	pH	Alk	Cl- (mg/l)	CaCl₂	ES (v)	Loss (Bbls)
1	8-Oct	6,135'	11.60	51	21	17	12/15/18	59/38	8	77.2/22.8	2	14.9	-	-	27K	-	775	-
2	9-Oct	7,379'	11.60	51	21	17	12/15/18	59/38	8	77.2/22.8	2	14.9	-	-	27K	-	775	-
3	10-Oct	8,840'	10.35	45	16	14	11/14/17	50/32	8	78.4/21.6	2	10.5	-	-	25K	-	885	-
4	11-Oct	9,937'	10.10	45	14	15	10/13/15	43/29	7	79.7/20.3	2	9.90	-	-	27K	-	1086	-
5	12-Oct	10,300'	10.30	45	15	13	9/11/12	43/28	7	81.2/18.8	2	7.00	-	-	28K	-	790	-
6	13-Oct	10,732'	10.50	51	19	14	10/12/13	52/33	7	81.2	2	7.00	-	-	29K	-	773	-
7	14-Oct	11,185'	10.50	51	19	14	10/12/13	52/33	7	81.2	2	7.00	-	-	29K	-	773	-
8	30-Jan	11,595'	9.85	28	1	1	0/1/0	3/2	-	0/89.7	-	0.40	9	-	167k	-	-	-
9	31-Jan	11,595'	9.85	28	1	1	0/1/0	3/2	-	0/89.7	-	0.40	9	-	167k	-	-	-
10	1-Feb	15,006'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.30	10	-	174	-	-	-
11	2-Feb	15,006'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.30	10	-	174	-	-	-
12	3-Feb	16,620'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.20	9	-	175k	-	-	-
13	4-Feb	16,620'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.20	9	-	175k	-	-	-
14	5-Feb	18,809'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.30	10	-	174.5k	-	-	-
15	6-Feb	18,809'	9.90	28	1	1	0/1/0	3/2	-	0/89.3	-	0.30	10	-	174.5k	-	-	-
16	7-Feb	20,105'	10.90	28	1	1	0/1/1	3/3	-	0/89.4	-	1.30	11	-	174.5k	-	-	-

DRILLING PLAN																																																
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND																																									
WELL NAME		Kline Federal 5300 41-18 13T2X			RIG		0																																									
WELL TYPE		Horizontal Three Forks																																														
LOCATION		SWSW 18-153N-100W			Surface Location (survey plat):		401' fsl 237' fwl																																									
EST. T.D.		20,840'			GROUND ELEV:		2057 Finished Pad Elev.		Sub Height: 25																																							
TOTAL LATERAL:		9,698' (est)			KB ELEV:		2082																																									
PROGNOSIS:				Based on 2,082' KB(est)																																												
LOGS:				Type OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD																																												
DEVIATION:				Interval Surf: 3 deg. max., 1 deg / 100'; srvy every 500' Prod: 5 deg. max., 1 deg / 100'; srvy every 100'																																												
DST'S:				None planned																																												
CORES:				None planned																																												
MUDLOGGING:				Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land																																												
BOP:				11" 5000 psi blind, pipe & annular																																												
Dip Rate: -0.1 Max. Anticipated BHP: 4714 Surface Formation: Glacial till																																																
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Comments: Request Sundry to Waive Open Hole Logs Exception well: Oasis Petroleum's Kline 5300 11-18H																																																
35 packers, 35 sleeves, no frac string																																																
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations. 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2) 68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)																																																
 Geology: M.Steed 4/23/14 Engineering: hbadler rpm																																																

Formation Tops

Operator: Well Name: Location: Elevation:	Subject Well:						
	Oasis Petroleum North America						
	<i>Kline Federal 5300 41-18 13T2X</i>						
	401' FSL & 237' FWL SW SW Sec. 18 T153N R100W 2,057 Sub: 25' KB: 2,082'						
Formation/ Zone	Prog. Top	Prog. MSL Datum	Est. MD Top (Gamma)	TVD Top (Gamma)	Est. MLS Datum	Thickness	Dip To <i>Kline Federal 5300 41-18 14BX</i>
Kibbey Lime	8,380	-6,298'	8,382'	8,382'	-6,300'	178'	Flat
Charles Salt	8,535	-6,453'	8,560'	8,560'	-6,478'	646'	4' High
Base Last Salt	9,205	-7,123'	9,206'	9,206'	-7,124'	222'	1' Low
Mission Canyon	9,428	-7,346'	9,428'	9,428'	-7,346'	547'	Flat
Lodgepole	9,975	-7,893'	9,975'	9,975'	-7,893'	730'	1' High
False Bakken	10,703	-8,621'	10,756'	10,705'	-8,623'	7'	1' Low
Upper Bakken Shale	10,712	-8,630'	10,764'	10,712'	-8,630'	18'	Flat
Middle Bakken	10,728	-8,646'	10,789'	10,730'	-8,648'	40'	2' Low
Lower Bakken	10,764	-8,682'	10,852'	10,770'	-8,688'	12'	-
Pronghorn	10,772	-8,690'	10,874'	10,782'	-8,700'	17'	-
Three Forks	10,793	-8,711'	10,909'	10,799'	-8,717'	25'	-
Claystone	10,818	-8,736'	10,974'	10,824'	-8,742'	9'	-
Three Forks 2nd Bench	10,830	-8,748'	11,004'	10,833'	-8,751'	-	-

Control Well

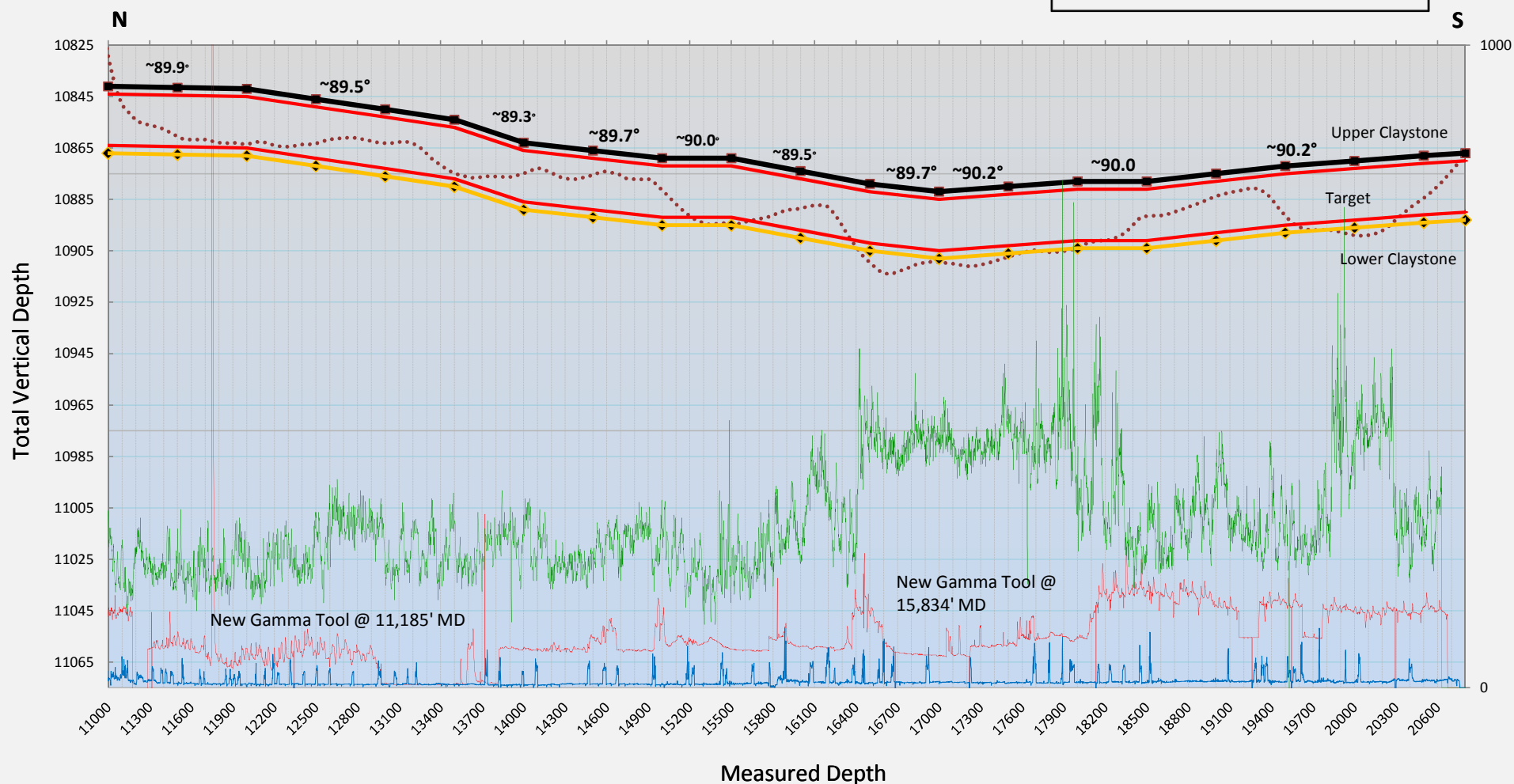
Operator:	Oasis Petroleum				Oasis Petroleum			
Well Name:	<i>Kline Federal 5300 41-18 14BX</i>				<i>Kline Federal 5300 11-18H</i>			
Location:	SW SW Sec. 18 T153N R100W McKenzie Co., ND Approximately 33 ft South				SW NW Sec. 18 T153N R100W McKenzie Co., ND Approximately .5 miles North			
Elevation:	KB: 2,082'				KB: 2,078'			
Formation/ Zone	Est. TVD	MSL Datum	Thickness	Dist to target	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382'	-6,300'	182'	2,471'	8,369	-6,291'	147'	2,467'
Charles Salt	8,564'	-6,482'	641'	2,289'	8,516	-6,438'	700'	2,320'
Base Last Salt	9,205'	-7,123'	223'	1,648'	9,216	-7,138'	224'	1,620'
Mission Canyon	9,428'	-7,346'	548'	1,425'	9,440	-7,362'	556'	1,396'
Lodgepole	9,976'	-7,894'	728'	877'	9,996	-7,918'	684'	840'
False Bakken	10,704'	-8,622'	8'	149'	10,680	-8,602'	13'	156'
Upper Bakken Shale	10,712'	-8,630'	16'	141'	10,693	-8,615'	18'	143'
Middle Bakken	10,728'	-8,646'	-	125'	10,711	-8,633'	-	125'



Surface Location:
4017' FSL & 240' FEL
SW SW Sec. 18 T153N, R100W
McKenzie County, ND

Kline Federal 5300 41-18 13T2X Cross Section

Bottom Hole Location:
10,022.35' E & 599.11' S
of surface location or approx.
19811' FNL & 280.4' FEL
NE NE Sec. 20, T153N, R100W



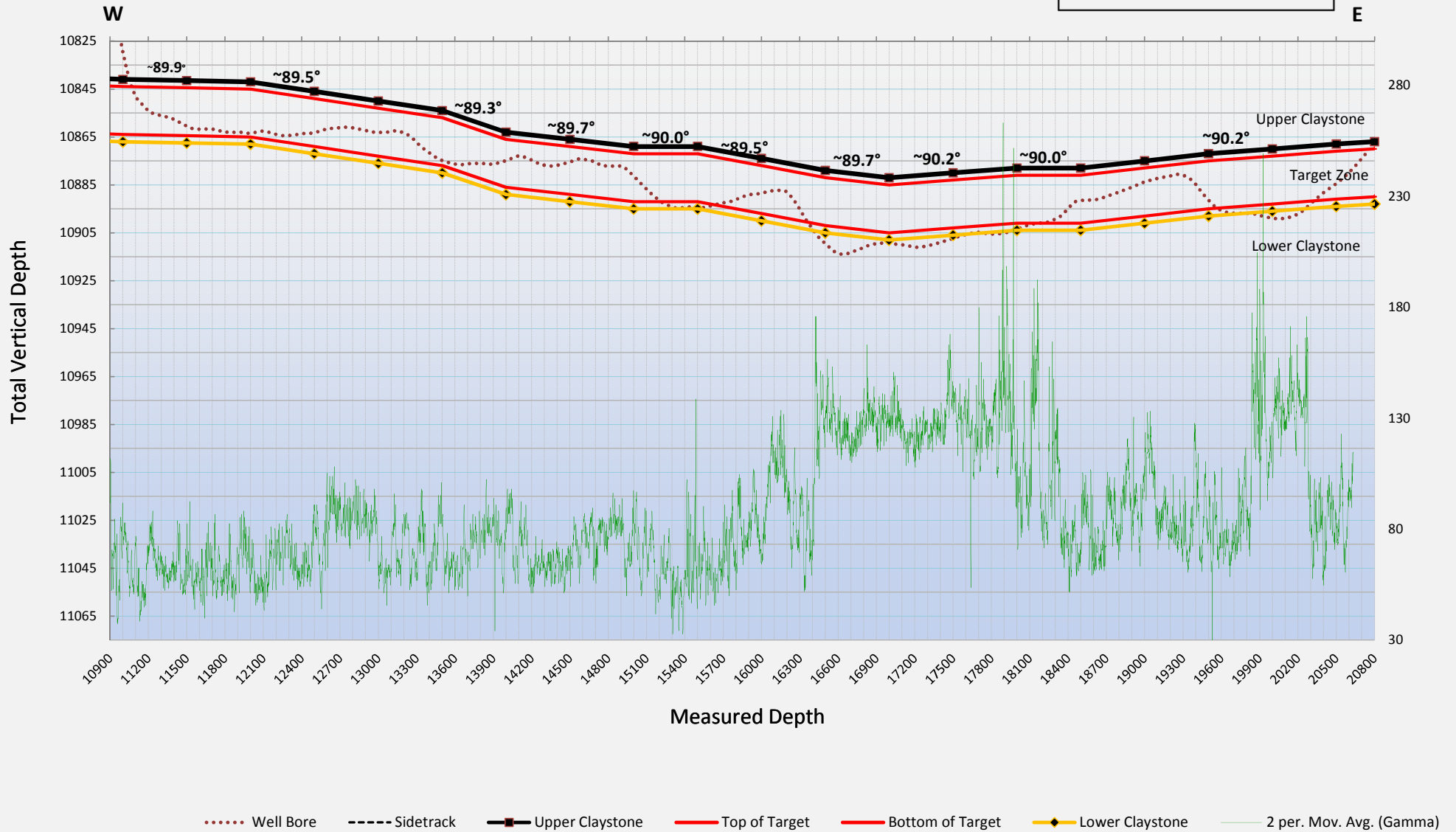
..... Well Bore —■— Upper Claystone — Top of Target — Bottom of Target —◆— Lower Claystone
----- Sidetrack — 2 per. Mov. Avg. (TG x 2) — 2 per. Mov. Avg. (ROP x 5) — 2 per. Mov. Avg. (GR x 2)



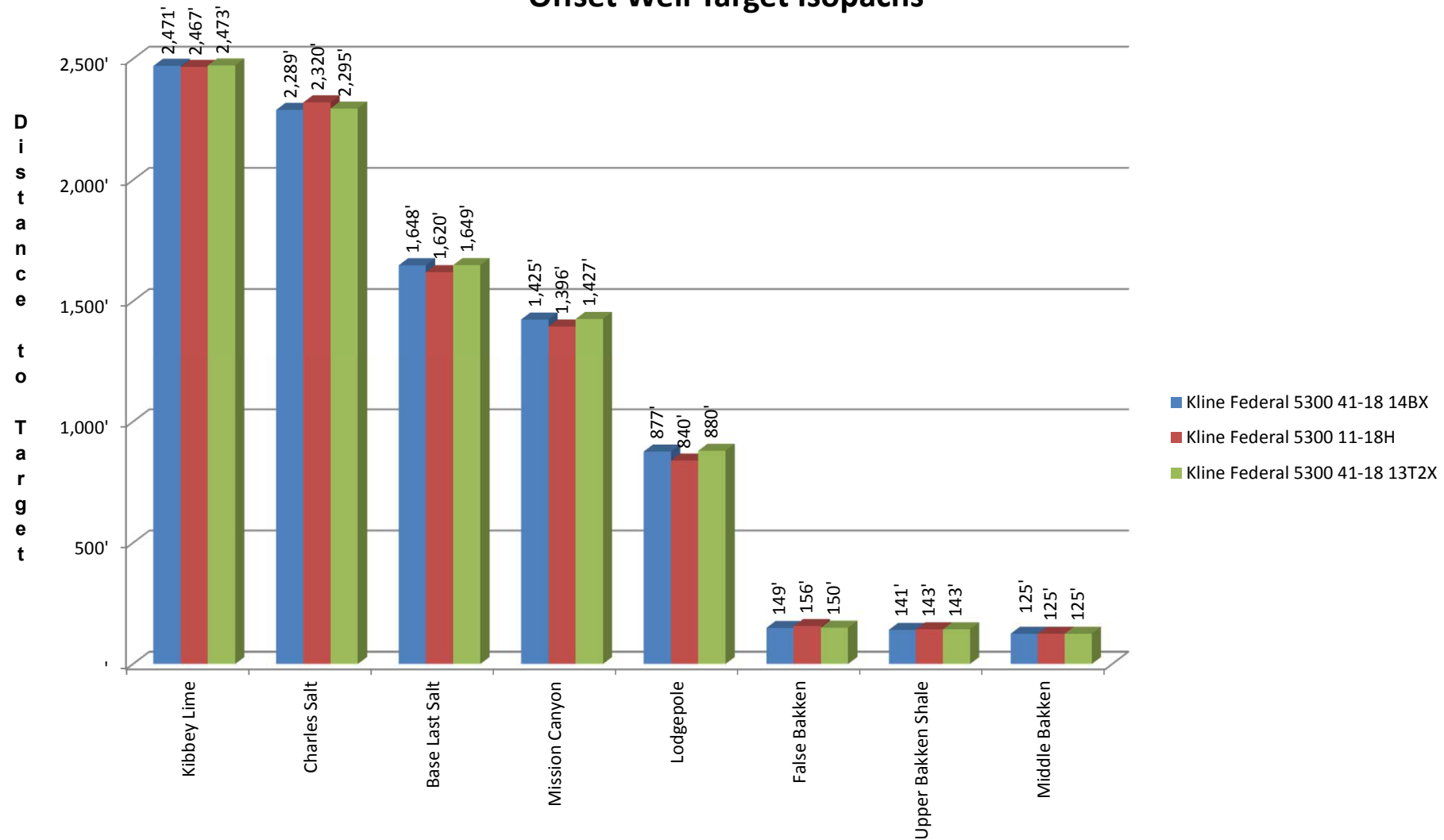
Surface: 4017' FSL & 240' FEL
SW SW Sec. 18 T153N, R100W
McKenzie County, ND

Kline Federal 5300 41-18 13T2X Gamma Cross Section

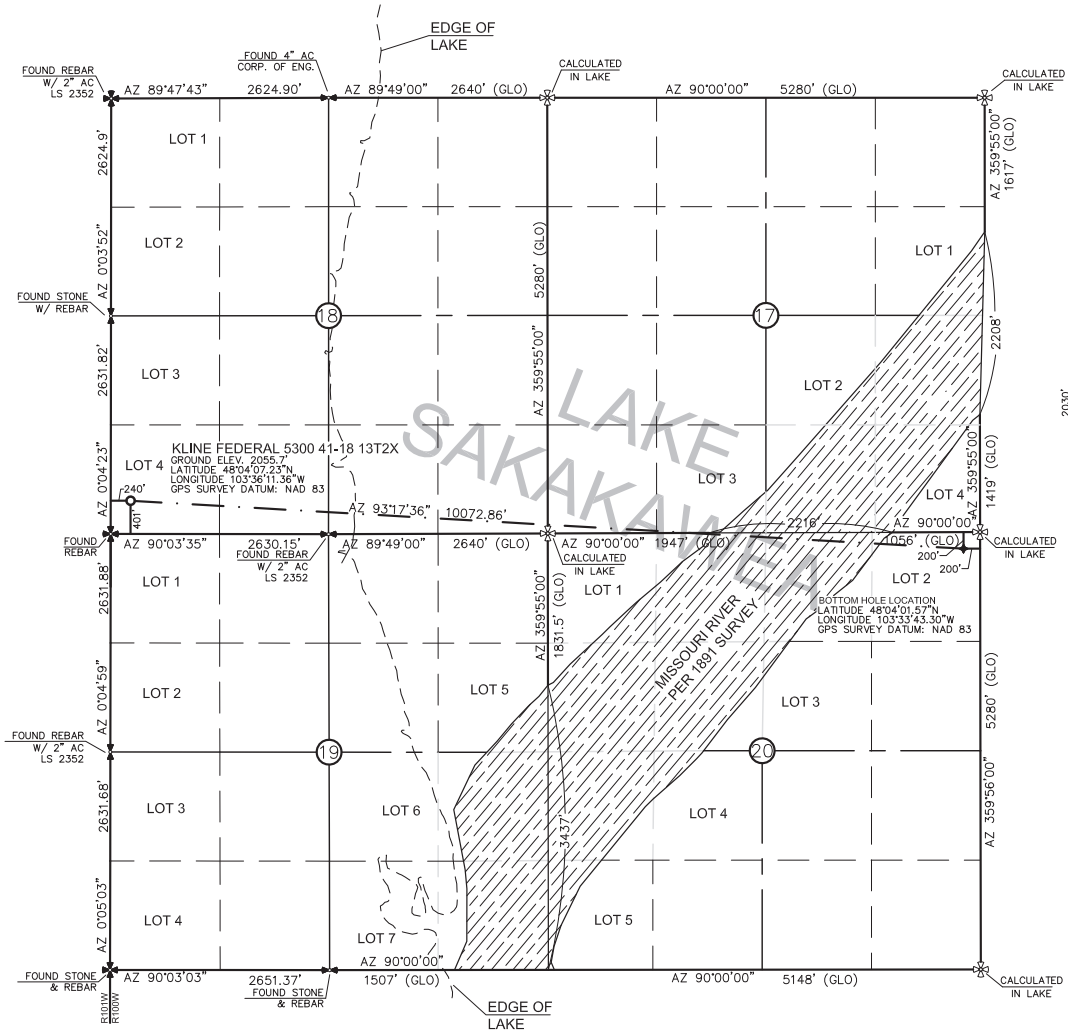
Bottom Hole Location:
10,022.35' E & 599.11' S
of surface location or approx.
19811' FNL & 280.4' FEL
NE NE Sec. 20, T153N, R100W



Offset Well Target Isopachs



WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 13T2X"
 401 FEET FROM SOUTH LINE AND 240 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

VICINITY MAP



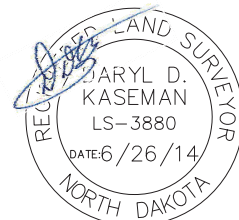
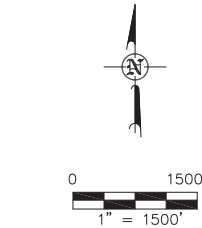
THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY DARYL D. KASEMAN,
 PLS, REGISTRATION NUMBER 3880 ON
 6/26/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.

STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC RAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880

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1/8

SHEET NO.



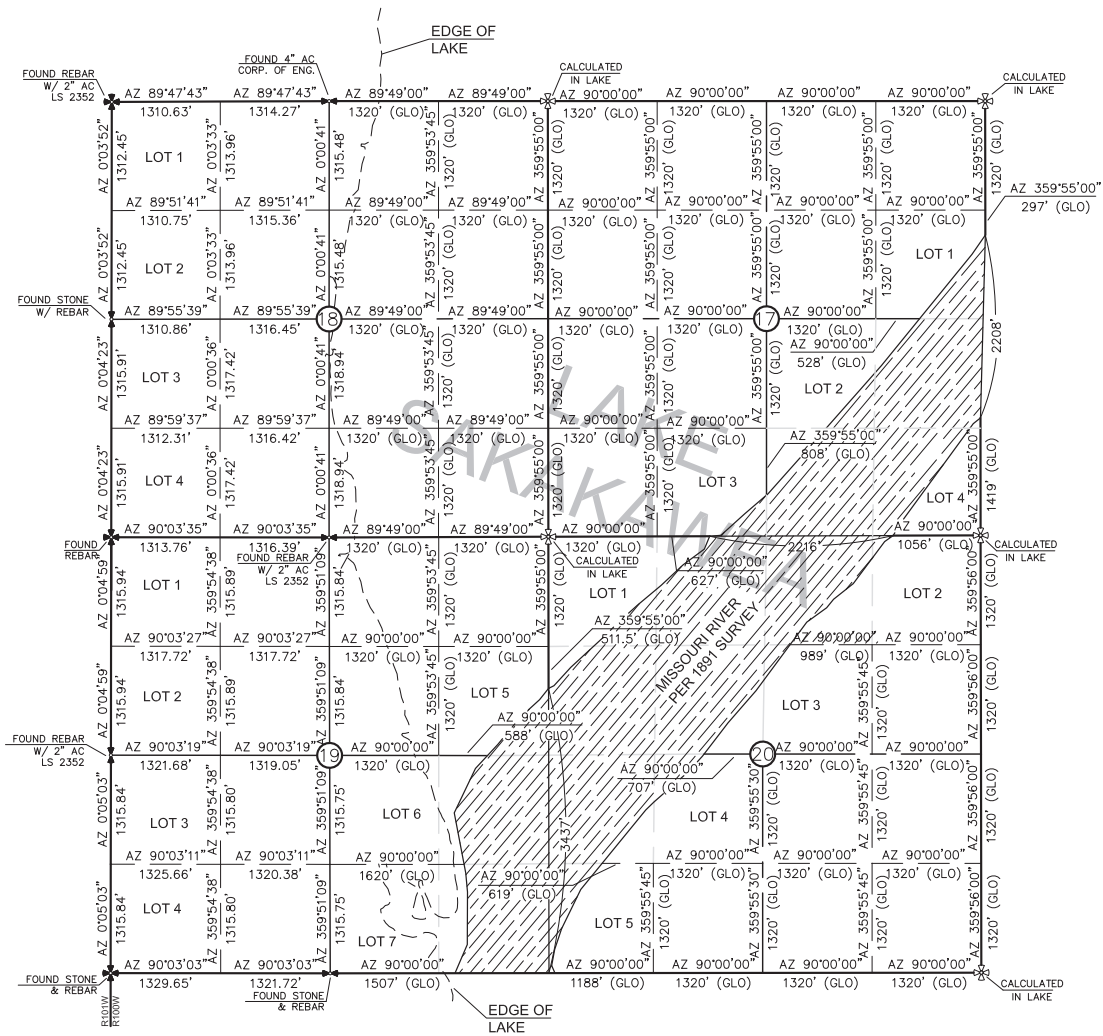
Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.01
 Checked By: D.D.K. Date: APRIL 2014

Revision	No.	Date	By	Description
REV 1	5/29/14	JJS		CORRECTED WELL CALLS
REV 2	6/25/14	BHH		MOVED WELLS ON PAD

SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 13T2X"
 401 FEET FROM SOUTH LINE AND 240 FEET FROM WEST LINE
 SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

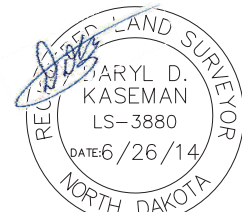
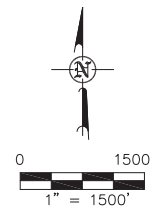
VICINITY MAP



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 AND SEALED BY DARYL D. KASEMAN,
 PLS, REGISTRATION NUMBER 3880 ON
 6/26/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.

ALL AZIMUTHS ARE BASED ON G.P.S.
 OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
 AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
 WAS 1891. THE CORNERS FOUND ARE AS
 INDICATED AND ALL OTHERS ARE COMPUTED FROM
 THOSE CORNERS FOUND AND BASED ON G.L.O.
 DATA. THE MAPPING ANGLE FOR THIS AREA IS
 APPROXIMATELY 0°03'.

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2/8
 SHEET NO.



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
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 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

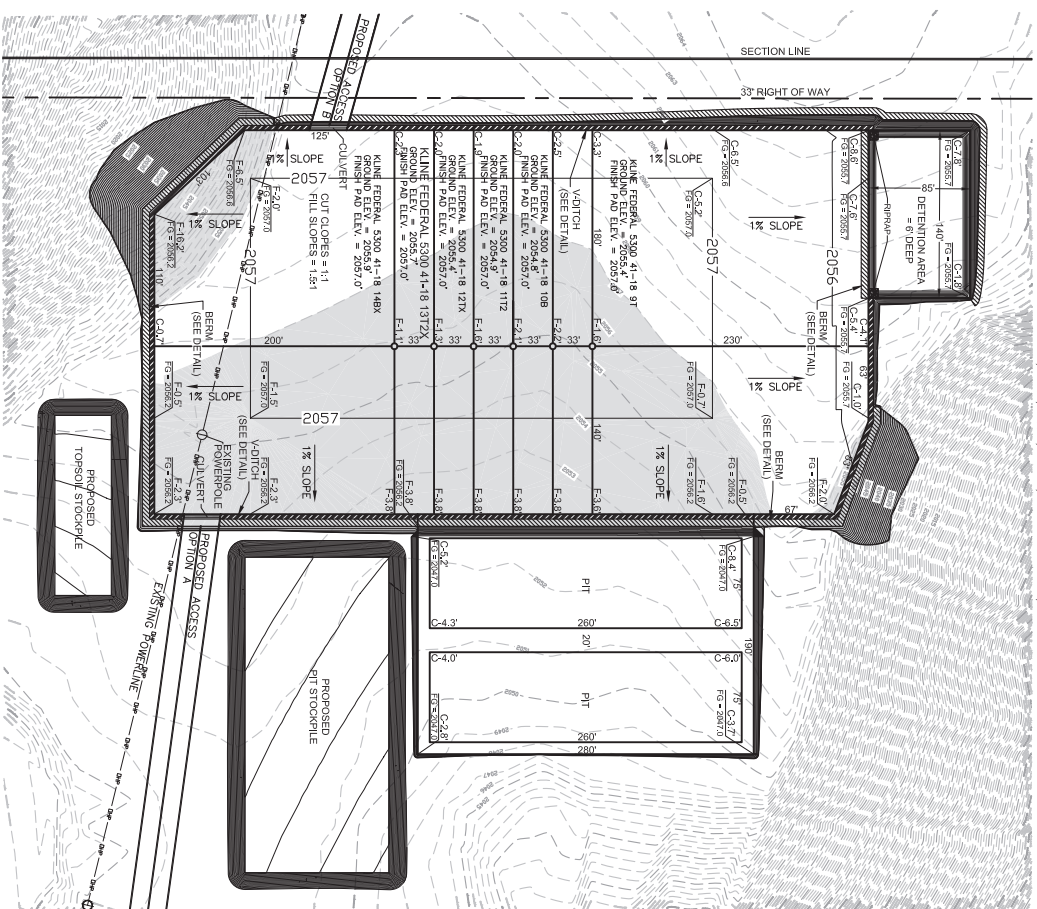
OASIS PETROLEUM NORTH AMERICA, LLC
 SECTION BREAKDOWN
 SECTIONS 17, 18, 19, & 20, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H. Project No.: S14-09-081.01
 Checked By: D.D.K. Date: APRIL 2014

Revision	No.	Date	By	Description
REV 1	5/29/14	JWS		CORRECTED WELL CALLS
REV 2	6/25/14	BHH		MOVED WELLS ON PAD

Updated/Verified: Federal 5300 41-18 13T2X Revised 6-25-14 dwg - 6/29/2014
 1:50 00 100 0000

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

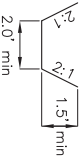
401 FEET FROM SOUTH LINE AND 240 FEET FROM WEST LINE



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NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

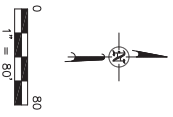
V-DITCH DETAIL



- Proposed Contours
- Original Contours

— BERM
— DITCH

NOTE: All utilities shown are preliminary only, a complete location is recommended before construction.

 $\frac{3}{8}$ 

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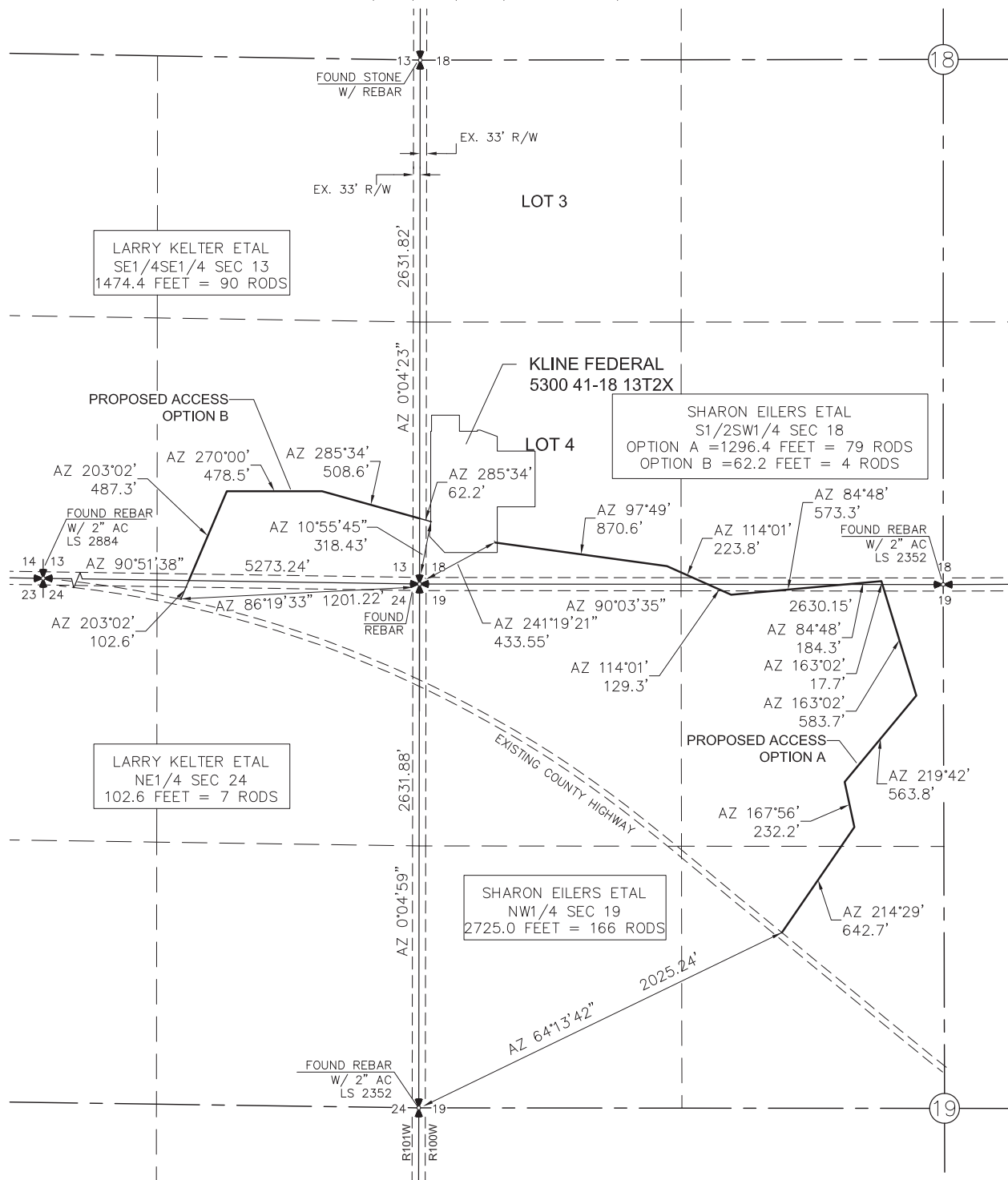
Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-6617
Fax (406) 433-6618
www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

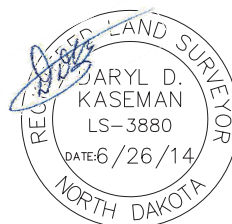
Drawn By: <u>BAH</u>	Project No.: <u>SL10-0000108</u>
Checked By: <u>D.D.K.</u>	Date: <u>APRIL 2014</u>

Revision No.	Date	By	Description
REV 1	5/29/14	JSB	CONNECTED WITH CALLS
REV 2	6/25/14	BHH	MOVED METS ON PAD

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 13T2X"
 401 FEET FROM SOUTH LINE AND 240 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.01
 Checked By: D.D.K. Date: APRIL 2014

Revision No.	Date	By	Description
REV 1	5/29/14	JJS	CORRECTED WELL CALLS
REV 2	6/25/14	BHH	MOVED WELLS ON PAD

Location: OASIS Petrochem Federal 5300 41-18 13T2X Revised: 6-25-14.dwg - 6/26/2014
 11:31 AM - 11:31 AM

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 41-18 13T2X
 401' FSL/240' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

WILLBUR
 ASH FEDERAL 5300 11-18T
 &
 KLINE FEDERAL 5300 11-18H

18
 KLINE FEDERAL
 5300 41-18 13T2X

PROPOSED
 ACCESS
 OPTION B

PROPOSED
 ACCESS
 OPTION A

CHALMERS 5300 31-19H

Indian Creek Bay
 Recreation Area

M I S S O U R I



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 Fax (406) 433-5618
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OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081_01
 Checked By: D.D.K. Date: APRIL 2014

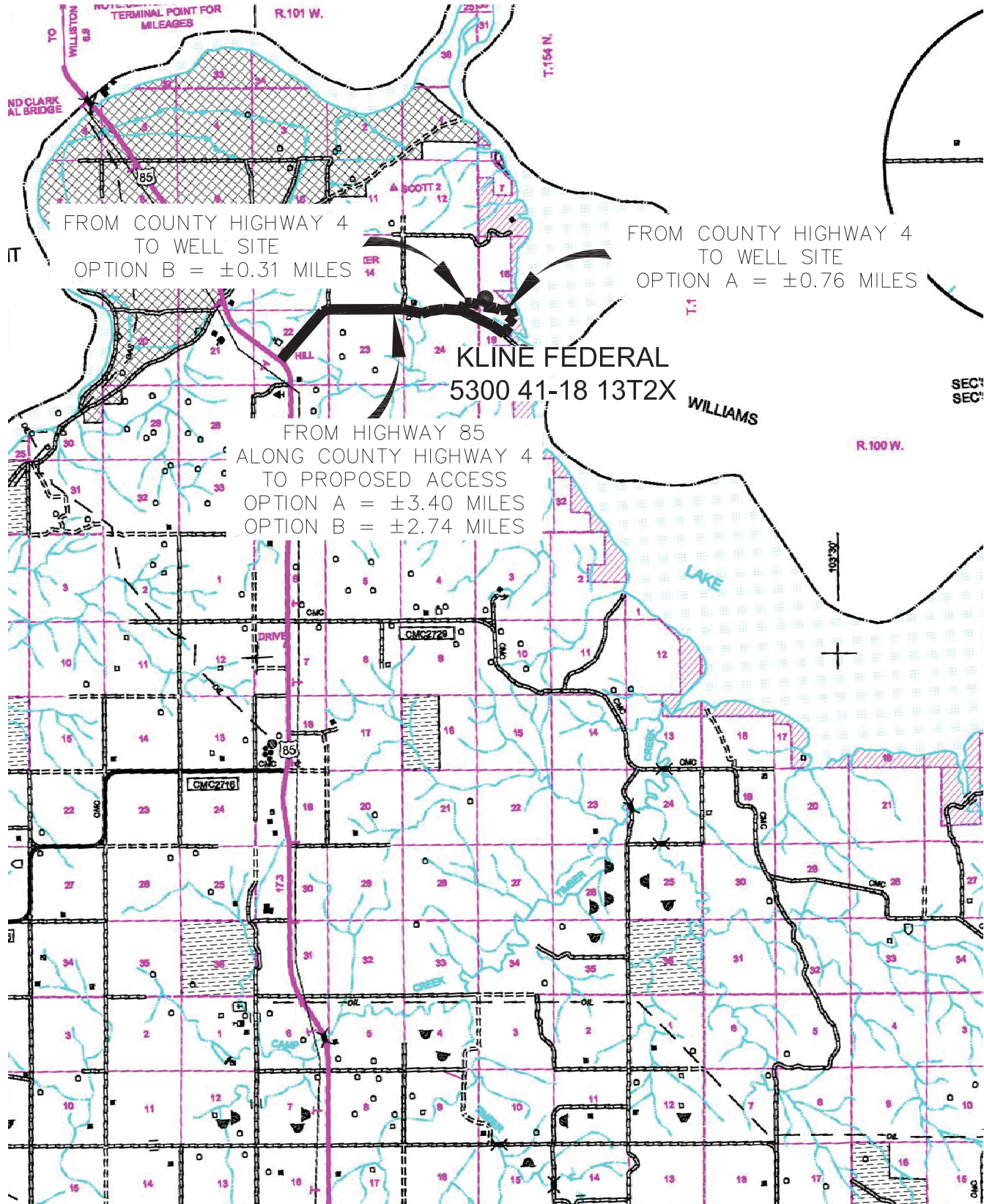
Revision No.	Date	By	Description
REV 1	5/29/14	JJS	CORRECTED WELL CALLS
REV 2	6/25/14	BHH	MOVED WELLS ON PAD

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 41-18 13T2X"

401 FEET FROM SOUTH LINE AND 240 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



FROM COUNTY HIGHWAY 4
TO WELL SITE
OPTION B = ± 0.31 MILES

FROM COUNTY HIGHWAY 4
TO WELL SITE
OPTION A = ± 0.76 MILES

FROM HIGHWAY 85
ALONG COUNTY HIGHWAY 4
TO PROPOSED ACCESS
OPTION A = ± 3.40 MILES
OPTION B = ± 2.74 MILES

KLINE FEDERAL
5300 41-18 13T2X

Co:	RPM Consulting Inc	Units:	Feet, °, °/100ft	VS Az:	90.00	Method:	Minimum
Drillers:	Ceynar	Elevation:	2082.00	Map System:			Curvature
Well Name:	Kline Federal 5300 41-18 13T2X	Northing:	405104.04	Latitude:	48.068672		
Location:	153N-100W-17/18	Easting:	1209969.29	Longitude:	-103.603319		

Indian Hills: 153N-100W-17/18: : Kline Fed 13T2X Surveys

No.	MD	CL	Inc.	Azi.	TVD	VS	+N/S-	+E/W-	BR	WR	DLS
1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
2	140.00	140.00	0.75	30.86	140.00	0.47	0.79	0.47	0.54	22.04	0.54
3	234.00	94.00	0.53	38.77	233.99	1.06	1.65	1.06	-0.23	8.41	0.25
4	327.00	93.00	0.66	24.71	326.99	1.55	2.48	1.55	0.14	-15.12	0.21
5	421.00	94.00	0.62	17.85	420.98	1.93	3.45	1.93	-0.04	-7.30	0.09
6	516.00	95.00	0.75	49.23	515.97	2.56	4.35	2.56	0.14	33.03	0.41
7	607.00	91.00	0.53	30.07	606.97	3.22	5.10	3.22	-0.24	-21.06	0.33
8	699.00	92.00	0.18	68.83	698.97	3.57	5.52	3.57	-0.38	42.13	0.44
9	791.00	92.00	0.22	289.26	790.97	3.54	5.63	3.54	0.04	-151.71	0.41
10	882.00	91.00	0.26	124.20	881.96	3.55	5.57	3.55	0.04	-181.38	0.52
11	974.00	92.00	0.62	112.33	973.96	4.18	5.27	4.18	0.39	-12.90	0.40
12	1066.00	92.00	0.79	109.34	1065.95	5.24	4.87	5.24	0.18	-3.25	0.19
13	1158.00	92.00	0.75	136.85	1157.95	6.25	4.22	6.25	-0.04	29.90	0.40
14	1250.00	92.00	0.57	121.30	1249.94	7.05	3.54	7.05	-0.20	-16.90	0.27
15	1342.00	92.00	0.26	132.72	1341.94	7.60	3.16	7.60	-0.34	12.41	0.35
16	1433.00	91.00	0.31	148.90	1432.94	7.87	2.81	7.87	0.05	17.78	0.10
17	1528.00	95.00	0.44	78.76	1527.94	8.37	2.66	8.37	0.14	-73.83	0.47
18	1623.00	95.00	0.35	227.82	1622.93	8.51	2.54	8.51	-0.09	156.91	0.80
19	1719.00	96.00	0.75	262.10	1718.93	7.67	2.25	7.67	0.42	35.71	0.52
20	1813.00	94.00	0.79	257.35	1812.92	6.43	2.03	6.43	0.04	-5.05	0.08
21	1908.00	95.00	0.84	247.60	1907.91	5.14	1.62	5.14	0.05	-10.26	0.15
22	2003.00	95.00	1.23	258.58	2002.90	3.50	1.15	3.50	0.41	11.56	0.46
23	2097.00	94.00	0.97	245.84	2096.88	1.79	0.63	1.79	-0.28	-13.55	0.38
24	2105.00	8.00	1.36	255.42	2104.88	1.63	0.58	1.63	4.87	119.75	5.43
25	2190.00	85.00	1.14	243.64	2189.86	-0.10	-0.05	-0.10	-0.26	-13.86	0.40
26	2284.00	94.00	1.27	263.59	2283.84	-1.98	-0.59	-1.98	0.14	21.22	0.46
27	2378.00	94.00	1.19	266.58	2377.82	-3.99	-0.76	-3.99	-0.09	3.18	0.11
28	2472.00	94.00	1.14	263.24	2471.80	-5.89	-0.93	-5.89	-0.05	-3.55	0.09

29	2567.00	95.00	0.44	158.12	2566.79	-6.69	-1.38	-6.69	-0.74	-110.65	1.39
30	2661.00	94.00	0.13	176.49	2660.79	-6.55	-1.82	-6.55	-0.33	19.54	0.34
31	2757.00	96.00	0.48	106.62	2756.79	-6.16	-2.04	-6.16	0.36	-72.78	0.47
32	2852.00	95.00	0.40	158.83	2851.78	-5.66	-2.47	-5.66	-0.08	54.96	0.41
33	2945.00	93.00	0.57	159.27	2944.78	-5.38	-3.20	-5.38	0.18	0.47	0.18
34	3039.00	94.00	0.35	167.88	3038.78	-5.15	-3.92	-5.15	-0.23	9.16	0.24
35	3134.00	95.00	0.35	170.25	3133.78	-5.04	-4.49	-5.04	0.00	2.49	0.02
36	3227.00	93.00	0.31	113.39	3226.78	-4.76	-4.87	-4.76	-0.04	-61.14	0.34
37	3322.00	95.00	0.48	124.37	3321.77	-4.20	-5.20	-4.20	0.18	11.56	0.20
38	3417.00	95.00	0.48	129.38	3416.77	-3.56	-5.67	-3.56	0.00	5.27	0.04
39	3512.00	95.00	0.31	162.61	3511.77	-3.18	-6.17	-3.18	-0.18	34.98	0.29
40	3607.00	95.00	0.57	172.10	3606.76	-3.03	-6.88	-3.03	0.27	9.99	0.28
41	3701.00	94.00	0.35	158.48	3700.76	-2.86	-7.62	-2.86	-0.23	-14.49	0.26
42	3796.00	95.00	0.31	71.20	3795.76	-2.51	-7.80	-2.51	-0.04	-91.87	0.48
43	3890.00	94.00	0.31	47.47	3889.76	-2.09	-7.55	-2.09	0.00	-25.24	0.14
44	3984.00	94.00	0.13	166.65	3983.76	-1.87	-7.48	-1.87	-0.19	126.79	0.42
45	4079.00	95.00	0.31	57.75	4078.76	-1.63	-7.45	-1.63	0.19	-114.63	0.39
46	4171.00	92.00	0.04	136.50	4170.76	-1.40	-7.34	-1.40	-0.29	85.60	0.33
47	4266.00	95.00	0.31	58.46	4265.76	-1.16	-7.23	-1.16	0.28	-82.15	0.32
48	4361.00	95.00	0.40	79.99	4360.76	-0.61	-7.04	-0.61	0.09	22.66	0.17
49	4456.00	95.00	0.31	121.39	4455.75	-0.07	-7.11	-0.07	-0.09	43.58	0.28
50	4550.00	94.00	0.35	76.03	4549.75	0.43	-7.18	0.43	0.04	-48.26	0.27
51	4645.00	95.00	0.09	161.55	4644.75	0.74	-7.18	0.74	-0.27	90.02	0.37
52	4740.00	95.00	0.04	85.44	4739.75	0.79	-7.24	0.79	-0.05	-80.12	0.09
53	4835.00	95.00	0.13	35.52	4834.75	0.89	-7.15	0.89	0.09	-52.55	0.11
54	4931.00	96.00	0.35	58.28	4930.75	1.20	-6.91	1.20	0.23	23.71	0.25
55	5025.00	94.00	0.44	350.60	5024.75	1.39	-6.40	1.39	0.10	-72.00	0.47
56	5119.00	94.00	0.57	10.91	5118.75	1.42	-5.59	1.42	0.14	21.61	0.23
57	5214.00	95.00	0.40	73.92	5213.74	1.82	-5.03	1.82	-0.18	66.33	0.56
58	5308.00	94.00	0.53	35.34	5307.74	2.39	-4.59	2.39	0.14	-41.04	0.35
59	5403.00	95.00	0.31	26.64	5402.74	2.76	-4.00	2.76	-0.23	-9.16	0.24
60	5497.00	94.00	0.35	98.45	5496.74	3.16	-3.81	3.16	0.04	76.39	0.41
61	5592.00	95.00	0.48	34.99	5591.73	3.67	-3.53	3.67	0.14	-66.80	0.47
62	5687.00	95.00	0.48	64.61	5686.73	4.26	-3.03	4.26	0.00	31.18	0.26

63	5782.00	95.00	0.44	78.32	5781.73	4.98	-2.79	4.98	-0.04	14.43	0.12
64	5876.00	94.00	0.53	65.75	5875.72	5.73	-2.54	5.73	0.10	-13.37	0.15
65	5971.00	95.00	0.26	24.49	5970.72	6.22	-2.16	6.22	-0.28	-43.43	0.40
66	6065.00	94.00	0.53	26.29	6064.72	6.50	-1.58	6.50	0.29	1.91	0.29
67	6085.00	20.00	0.44	79.02	6084.72	6.62	-1.48	6.62	-0.45	263.65	2.19
68	6161.00	76.00	0.53	35.87	6160.72	7.11	-1.14	7.11	0.12	-56.78	0.48
69	6255.00	94.00	0.66	26.81	6254.71	7.61	-0.30	7.61	0.14	-9.64	0.17
70	6350.00	95.00	0.66	13.28	6349.70	7.98	0.72	7.98	0.00	-14.24	0.16
71	6444.00	94.00	0.75	350.78	6443.70	8.01	1.85	8.01	0.10	-23.94	0.31
72	6539.00	95.00	1.01	353.33	6538.69	7.81	3.30	7.81	0.27	2.68	0.28
73	6634.00	95.00	1.19	353.50	6633.67	7.60	5.11	7.60	0.19	0.18	0.19
74	6727.00	93.00	0.70	351.57	6726.66	7.41	6.63	7.41	-0.53	-2.08	0.53
75	6822.00	95.00	0.40	329.69	6821.65	7.15	7.49	7.15	-0.32	-23.03	0.38
76	6916.00	94.00	0.75	336.01	6915.65	6.74	8.34	6.74	0.37	6.72	0.38
77	7011.00	95.00	0.04	64.08	7010.64	6.52	8.92	6.52	-0.75	92.71	0.79
78	7105.00	94.00	0.13	123.23	7104.64	6.63	8.88	6.63	0.10	62.93	0.12
79	7199.00	94.00	0.04	331.62	7198.64	6.71	8.85	6.71	-0.10	221.69	0.18
80	7294.00	95.00	0.44	263.15	7293.64	6.33	8.83	6.33	0.42	-72.07	0.45
81	7387.00	93.00	0.48	348.41	7386.64	5.90	9.17	5.90	0.04	91.68	0.67
82	7482.00	95.00	0.48	11.08	7481.64	5.89	9.95	5.89	0.00	23.86	0.20
83	7577.00	95.00	0.48	1.24	7576.63	5.98	10.74	5.98	0.00	-10.36	0.09
84	7671.00	94.00	0.57	11.87	7670.63	6.08	11.59	6.08	0.10	11.31	0.14
85	7765.00	94.00	0.62	76.21	7764.63	6.67	12.17	6.67	0.05	68.45	0.68
86	7858.00	93.00	0.79	107.67	7857.62	7.77	12.09	7.77	0.18	33.83	0.45
87	7952.00	94.00	0.35	189.41	7951.62	8.34	11.61	8.34	-0.47	86.96	0.87
88	8046.00	94.00	0.57	189.94	8045.61	8.22	10.87	8.22	0.23	0.56	0.23
89	8141.00	95.00	0.66	202.51	8140.61	7.93	9.90	7.93	0.09	13.23	0.17
90	8235.00	94.00	0.44	214.20	8234.60	7.52	9.10	7.52	-0.23	12.44	0.26
91	8329.00	94.00	0.48	209.72	8328.60	7.12	8.46	7.12	0.04	-4.77	0.06
92	8424.00	95.00	0.40	211.91	8423.60	6.74	7.83	6.74	-0.08	2.31	0.09
93	8518.00	94.00	0.66	263.94	8517.59	6.03	7.50	6.03	0.28	55.35	0.55
94	8612.00	94.00	0.84	254.10	8611.59	4.83	7.25	4.83	0.19	-10.47	0.23
95	8707.00	95.00	0.84	238.02	8706.58	3.57	6.69	3.57	0.00	-16.93	0.25
96	8802.00	95.00	0.53	221.67	8801.57	2.69	6.00	2.69	-0.33	-17.21	0.38

97	8896.00	94.00	0.57	257.62	8895.56	1.94	5.57	1.94	0.04	38.24	0.36
98	8990.00	94.00	0.35	242.76	8989.56	1.23	5.34	1.23	-0.23	-15.81	0.26
99	9084.00	94.00	0.44	255.86	9083.56	0.63	5.12	0.63	0.10	13.94	0.13
100	9178.00	94.00	0.35	282.84	9177.56	0.00	5.09	0.00	-0.10	28.70	0.22
101	9272.00	94.00	0.35	13.54	9271.56	-0.22	5.44	-0.22	0.00	96.49	0.53
102	9367.00	95.00	0.40	319.14	9366.55	-0.37	5.97	-0.37	0.05	-57.26	0.36
103	9461.00	94.00	0.53	334.17	9460.55	-0.77	6.61	-0.77	0.14	15.99	0.19
104	9555.00	94.00	0.66	11.43	9554.55	-0.85	7.53	-0.85	0.14	39.64	0.43
105	9649.00	94.00	0.53	10.64	9648.54	-0.67	8.49	-0.67	-0.14	-0.84	0.14
106	9742.00	93.00	0.66	92.12	9741.54	-0.05	8.89	-0.05	0.14	87.61	0.84
107	9836.00	94.00	0.70	115.23	9835.53	1.01	8.63	1.01	0.04	24.59	0.29
108	9930.00	94.00	0.53	119.45	9929.53	1.91	8.17	1.91	-0.18	4.49	0.19
109	10024.00	94.00	0.70	125.34	10023.52	2.75	7.62	2.75	0.18	6.27	0.19
110	10119.00	95.00	0.26	104.95	10118.52	3.44	7.23	3.44	-0.46	-21.46	0.49
111	10214.00	95.00	0.13	137.03	10213.52	3.72	7.10	3.72	-0.14	33.77	0.17
112	10246.00	32.00	0.40	155.40	10245.52	3.79	6.97	3.79	0.84	57.41	0.87



Operator: Oasis Petroleum	
Well : KLINE FEDERAL 5300 41-18 13T2X	
MWD Providers	Ryan Directional Services
Directional Supervision:	RPM Consulting

Section:	18	QQ:	SW SW	County:	McKenzie	State:	ND
Township:	153	N/S:	N	Footages:	401	FN/SL:	S
Range:	100	E/W:	W		237	FE/WL:	W

Vertical Section Plane:	90
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#	<i>MD</i>	<i>Inc.</i>	<i>Azm.</i>	<i>T.V.D.</i>	<i>Ver. Sect.</i>	<i>Coordinates</i>		<i>DLS</i>
						<i>+N / -S</i>	<i>+E / -W</i>	
Tie	10246.00	0.40	155.40	10245.52	3.78	6.98	3.78	0.87
1	10276.00	0.26	115.76	10275.52	3.88	6.86	3.88	0.87
2	10308.00	1.76	115.94	10307.51	4.39	6.61	4.39	4.69
3	10339.00	6.07	119.19	10338.43	6.25	5.60	6.25	13.92
4	10371.00	10.24	121.65	10370.10	10.15	3.28	10.15	13.07
5	10403.00	14.77	115.58	10401.34	16.26	0.03	16.26	14.72
6	10435.00	18.37	116.02	10432.00	24.47	-3.95	24.47	11.26
7	10466.00	21.01	116.55	10461.19	33.83	-8.58	33.83	8.54
8	10498.00	22.90	125.25	10490.87	44.05	-14.74	44.05	11.74
9	10530.00	26.42	130.53	10519.96	54.55	-22.96	54.55	12.97
10	10562.00	28.75	133.16	10548.32	65.58	-32.85	65.58	8.21
11	10593.00	29.45	132.46	10575.40	76.64	-43.10	76.64	2.51
12	10625.00	31.91	128.59	10602.92	89.06	-53.68	89.06	9.85
13	10656.00	35.60	127.19	10628.69	102.65	-64.25	102.65	12.16
14	10688.00	38.59	123.06	10654.22	118.44	-75.33	118.44	12.16
15	10719.00	41.45	121.21	10677.96	135.33	-85.92	135.33	9.99
16	10751.00	43.47	117.25	10701.57	154.18	-96.46	154.18	10.47
17	10782.00	44.66	118.57	10723.85	173.22	-106.55	173.22	4.85
18	10814.00	49.10	120.95	10745.71	193.49	-118.15	193.49	14.90
19	10845.00	53.93	120.68	10765.00	214.32	-130.58	214.32	15.60
20	10876.00	58.11	120.59	10782.32	236.43	-143.67	236.43	13.49
21	10908.00	63.60	120.51	10797.90	260.49	-157.87	260.49	17.16
22	10939.00	68.35	117.34	10810.52	285.27	-171.55	285.27	17.94
23	10971.00	72.57	115.76	10821.22	312.24	-185.02	312.24	13.98
24	11002.00	75.03	115.94	10829.87	339.03	-198.00	339.03	7.96
25	11034.00	77.98	115.23	10837.34	367.09	-211.44	367.09	9.47
26	11066.00	80.26	116.02	10843.38	395.43	-225.03	395.43	7.53
27	11097.00	83.52	116.20	10847.75	422.98	-238.53	422.98	10.53
28	11107.00	84.88	115.85	10848.76	431.92	-242.90	431.92	14.04
29	11207.00	88.57	116.81	10854.47	521.38	-287.17	521.38	3.81
30	11238.00	88.97	116.55	10855.14	549.08	-301.09	549.08	1.54
31	11269.00	89.01	116.73	10855.68	576.78	-314.98	576.78	0.59

32	11300.00	89.36	116.55	10856.13	604.49	-328.88	604.49	1.27
33	11332.00	89.23	115.23	10856.52	633.27	-342.85	633.27	4.14
34	11362.00	89.27	115.06	10856.91	660.43	-355.60	660.43	0.58
35	11393.00	88.48	113.12	10857.52	688.72	-368.25	688.72	6.76
36	11424.00	88.53	111.28	10858.33	717.41	-379.96	717.41	5.94
37	11455.00	88.26	106.52	10859.20	746.72	-389.99	746.72	15.37
38	11486.00	88.35	108.64	10860.12	776.26	-399.35	776.26	6.84
39	11517.00	88.35	108.73	10861.01	805.61	-409.28	805.61	0.29
40	11548.00	89.58	108.90	10861.57	834.95	-419.28	834.95	4.01
41	11580.00	89.80	108.90	10861.74	865.22	-429.64	865.22	0.69
42	11611.00	90.07	108.29	10861.78	894.60	-439.53	894.60	2.15
43	11642.00	90.11	108.20	10861.73	924.05	-449.23	924.05	0.32
44	11673.00	90.24	107.15	10861.63	953.58	-458.64	953.58	3.41
45	11704.00	89.32	106.09	10861.75	983.29	-467.51	983.29	4.53
46	11736.00	89.19	106.36	10862.17	1014.01	-476.45	1014.01	0.94
47	11767.00	89.32	105.74	10862.57	1043.80	-485.02	1043.80	2.04
48	11798.00	89.45	106.18	10862.90	1073.60	-493.54	1073.60	1.48
49	11829.00	89.89	105.83	10863.08	1103.40	-502.09	1103.40	1.81
50	11860.00	90.02	105.21	10863.11	1133.27	-510.39	1133.27	2.04
51	11891.00	90.37	103.46	10863.00	1163.30	-518.06	1163.30	5.76
52	11923.00	89.49	101.52	10863.04	1194.54	-524.98	1194.54	6.66
53	11954.00	89.49	100.12	10863.32	1224.99	-530.80	1224.99	4.52
54	11985.00	89.85	99.59	10863.49	1255.53	-536.11	1255.53	2.07
55	12016.00	90.59	99.85	10863.38	1286.09	-541.34	1286.09	2.53
56	12047.00	91.03	100.12	10862.94	1316.61	-546.71	1316.61	1.67
57	12078.00	90.07	99.15	10862.64	1347.17	-551.90	1347.17	4.40
58	12110.00	90.07	99.16	10862.60	1378.77	-556.99	1378.77	0.03
59	12141.00	88.44	99.76	10863.00	1409.34	-562.09	1409.34	5.60
60	12172.00	89.41	99.59	10863.59	1439.89	-567.30	1439.89	3.18
61	12203.00	89.19	96.95	10863.96	1470.57	-571.75	1470.57	8.55
62	12234.00	89.54	95.90	10864.31	1501.37	-575.22	1501.37	3.57
63	12266.00	89.80	95.15	10864.49	1533.22	-578.30	1533.22	2.48
64	12297.00	90.37	94.31	10864.45	1564.11	-580.86	1564.11	3.27
65	12328.00	90.29	93.17	10864.27	1595.05	-582.88	1595.05	3.69
66	12358.00	90.51	92.82	10864.06	1625.01	-584.45	1625.01	1.38
67	12390.00	90.81	93.26	10863.69	1656.96	-586.15	1656.96	1.66
68	12420.00	89.76	92.38	10863.54	1686.92	-587.62	1686.92	4.57
69	12451.00	90.11	91.50	10863.58	1717.90	-588.67	1717.90	3.05
70	12482.00	90.59	91.50	10863.39	1748.89	-589.48	1748.89	1.55
71	12514.00	90.64	90.36	10863.04	1780.88	-590.00	1780.88	3.57
72	12577.00	91.30	90.18	10861.98	1843.87	-590.30	1843.87	1.09
73	12608.00	90.02	90.36	10861.62	1874.87	-590.45	1874.87	4.17
74	12671.00	90.64	89.74	10861.26	1937.87	-590.50	1937.87	1.39
75	12766.00	89.71	89.83	10860.97	2032.87	-590.14	2032.87	0.98

76	12860.00	89.19	88.51	10861.87	2126.85	-588.78	2126.85	1.51
77	12955.00	89.54	88.51	10862.92	2221.81	-586.31	2221.81	0.37
78	12986.00	89.67	89.48	10863.13	2252.81	-585.77	2252.81	3.16
79	13049.00	90.46	89.39	10863.06	2315.80	-585.15	2315.80	1.26
80	13081.00	90.02	91.77	10862.93	2347.80	-585.47	2347.80	7.56
81	13144.00	90.77	90.71	10862.49	2410.78	-586.84	2410.78	2.06
82	13238.00	87.03	91.41	10864.30	2504.73	-588.57	2504.73	4.05
83	13333.00	87.34	90.36	10868.96	2599.60	-590.04	2599.60	1.15
84	13364.00	87.87	90.45	10870.26	2630.58	-590.26	2630.58	1.73
85	13427.00	87.82	90.98	10872.63	2693.53	-591.04	2693.53	0.84
86	13459.00	88.18	90.10	10873.75	2725.50	-591.34	2725.50	2.97
87	13522.00	89.05	89.92	10875.27	2788.49	-591.36	2788.49	1.41
88	13616.00	89.49	89.48	10876.47	2882.48	-590.86	2882.48	0.66
89	13647.00	90.02	89.31	10876.60	2913.47	-590.54	2913.47	1.80
90	13710.00	90.73	88.60	10876.19	2976.46	-589.39	2976.46	1.59
91	13741.00	89.85	90.54	10876.03	3007.46	-589.15	3007.46	6.87
92	13804.00	90.07	89.74	10876.07	3070.46	-589.31	3070.46	1.32
93	13836.00	89.63	91.41	10876.16	3102.45	-589.63	3102.45	5.40
94	13899.00	90.11	90.98	10876.30	3165.44	-590.94	3165.44	1.02
95	13994.00	91.47	91.24	10874.99	3260.41	-592.78	3260.41	1.46
96	14088.00	90.90	91.41	10873.05	3354.37	-594.96	3354.37	0.63
97	14120.00	89.27	90.36	10873.00	3386.36	-595.45	3386.36	6.06
98	14183.00	88.48	90.71	10874.24	3449.34	-596.04	3449.34	1.37
99	14277.00	88.75	91.06	10876.51	3543.31	-597.49	3543.31	0.47
100	14371.00	90.51	90.27	10877.12	3637.29	-598.58	3637.29	2.05
101	14466.00	90.86	89.74	10875.98	3732.29	-598.59	3732.29	0.67
102	14497.00	90.81	90.01	10875.53	3763.28	-598.52	3763.28	0.89
103	14561.00	91.38	89.13	10874.30	3827.27	-598.04	3827.27	1.64
104	14592.00	89.01	89.13	10874.20	3858.26	-597.57	3858.26	7.65
105	14655.00	89.85	90.80	10874.83	3921.26	-597.53	3921.26	2.97
106	14687.00	88.66	90.89	10875.24	3953.25	-598.00	3953.25	3.73
107	14718.00	88.35	90.27	10876.05	3984.24	-598.32	3984.24	2.24
108	14750.00	88.44	90.45	10876.95	4016.22	-598.52	4016.22	0.63
109	14844.00	91.25	90.98	10877.20	4110.21	-599.69	4110.21	3.04
110	14907.00	88.79	88.95	10877.18	4173.20	-599.65	4173.20	5.06
111	14939.00	87.38	89.57	10878.25	4205.18	-599.24	4205.18	4.81
112	15033.00	86.68	90.10	10883.12	4299.05	-598.97	4299.05	0.93
113	15065.00	86.86	89.57	10884.92	4331.00	-598.88	4331.00	1.75
114	15096.00	87.03	89.57	10886.57	4361.95	-598.65	4361.95	0.55
115	15127.00	86.95	89.22	10888.20	4392.91	-598.32	4392.91	1.16
116	15222.00	88.13	89.31	10892.28	4487.81	-597.10	4487.81	1.25
117	15316.00	89.19	89.31	10894.48	4581.78	-595.97	4581.78	1.13
118	15347.00	90.07	88.16	10894.68	4612.77	-595.29	4612.77	4.67
119	15378.00	90.29	87.90	10894.58	4643.75	-594.22	4643.75	1.10

120	15410.00	90.73	87.81	10894.30	4675.73	-593.02	4675.73	1.40
121	15442.00	89.49	89.31	10894.23	4707.71	-592.22	4707.71	6.08
122	15505.00	89.80	89.83	10894.62	4770.71	-591.75	4770.71	0.96
123	15568.00	90.73	89.66	10894.33	4833.71	-591.46	4833.71	1.50
124	15599.00	91.21	89.83	10893.81	4864.70	-591.33	4864.70	1.64
125	15694.00	90.29	90.18	10892.56	4959.70	-591.34	4959.70	1.04
126	15725.00	90.51	89.83	10892.35	4990.69	-591.34	4990.69	1.33
127	15757.00	90.81	91.41	10891.98	5022.69	-591.68	5022.69	5.03
128	15788.00	91.38	91.06	10891.39	5053.68	-592.35	5053.68	2.16
129	15883.00	91.03	90.62	10889.39	5148.64	-593.74	5148.64	0.59
130	15914.00	90.24	91.24	10889.05	5179.64	-594.25	5179.64	3.24
131	15977.00	90.37	91.68	10888.71	5242.62	-595.85	5242.62	0.73
132	16072.00	90.99	90.62	10887.58	5337.59	-597.76	5337.59	1.29
133	16103.00	89.85	91.41	10887.36	5368.58	-598.31	5368.58	4.47
134	16166.00	90.20	91.15	10887.33	5431.57	-599.72	5431.57	0.69
135	16197.00	87.56	90.18	10887.93	5462.56	-600.08	5462.56	9.07
136	16260.00	85.32	90.10	10891.85	5525.43	-600.23	5525.43	3.56
137	16355.00	84.84	89.92	10899.99	5620.08	-600.25	5620.08	0.54
138	16449.00	86.73	89.39	10906.90	5713.82	-599.68	5713.82	2.09
139	16481.00	87.21	89.92	10908.59	5745.77	-599.49	5745.77	2.23
140	16513.00	87.21	89.74	10910.15	5777.74	-599.39	5777.74	0.56
141	16544.00	87.21	89.83	10911.66	5808.70	-599.28	5808.70	0.29
142	16575.00	87.96	90.10	10912.97	5839.67	-599.26	5839.67	2.57
143	16607.00	89.36	89.92	10913.71	5871.66	-599.26	5871.66	4.41
144	16638.00	89.76	89.39	10913.95	5902.66	-599.08	5902.66	2.14
145	16670.00	91.03	89.22	10913.73	5934.66	-598.69	5934.66	4.00
146	16733.00	91.16	88.95	10912.53	5997.64	-597.68	5997.64	0.48
147	16796.00	91.21	88.51	10911.23	6060.61	-596.29	6060.61	0.70
148	16827.00	91.56	88.51	10910.48	6091.59	-595.48	6091.59	1.13
149	16923.00	89.76	89.04	10909.37	6187.55	-593.43	6187.55	1.95
150	17017.00	89.98	89.57	10909.58	6281.55	-592.29	6281.55	0.61
151	17111.00	89.41	89.57	10910.08	6375.54	-591.58	6375.54	0.61
152	17143.00	89.41	89.22	10910.41	6407.54	-591.25	6407.54	1.09
153	17206.00	89.58	89.22	10910.97	6470.53	-590.39	6470.53	0.27
154	17237.00	90.29	89.39	10911.00	6501.53	-590.01	6501.53	2.36
155	17300.00	90.68	89.39	10910.47	6564.52	-589.34	6564.52	0.62
156	17395.00	90.95	89.31	10909.12	6659.51	-588.26	6659.51	0.30
157	17489.00	90.99	88.43	10907.53	6753.47	-586.41	6753.47	0.94
158	17584.00	90.90	87.37	10905.96	6848.40	-582.93	6848.40	1.12
159	17678.00	90.46	88.07	10904.85	6942.31	-579.19	6942.31	0.88
160	17710.00	89.49	89.13	10904.86	6974.30	-578.41	6974.30	4.49
161	17773.00	89.49	89.13	10905.42	7037.29	-577.45	7037.29	0.00
162	17804.00	90.11	89.13	10905.53	7068.29	-576.98	7068.29	2.00
163	17867.00	90.20	88.51	10905.36	7131.28	-575.68	7131.28	0.99

164	17960.00	90.90	90.62	10904.47	7224.26	-574.98	7224.26	2.39
165	17992.00	91.47	90.89	10903.80	7256.25	-575.40	7256.25	1.97
166	18055.00	91.21	90.36	10902.33	7319.23	-576.09	7319.23	0.94
167	18148.00	90.29	90.54	10901.11	7412.22	-576.82	7412.22	1.01
168	18243.00	90.37	92.12	10900.57	7507.19	-579.02	7507.19	1.67
169	18337.00	92.48	92.82	10898.23	7601.07	-583.07	7601.07	2.36
170	18368.00	93.05	92.56	10896.73	7632.00	-584.52	7632.00	2.02
171	18431.00	93.36	92.21	10893.21	7694.84	-587.14	7694.84	0.74
172	18462.00	91.38	92.12	10891.93	7725.79	-588.31	7725.79	6.39
173	18526.00	89.58	91.68	10891.39	7789.75	-590.43	7789.75	2.90
174	18620.00	90.73	90.89	10891.14	7883.73	-592.54	7883.73	1.48
175	18714.00	91.12	90.54	10889.62	7977.71	-593.71	7977.71	0.56
176	18808.00	91.08	90.62	10887.82	8071.69	-594.67	8071.69	0.10
177	18902.00	91.21	90.62	10885.94	8165.66	-595.68	8165.66	0.14
178	18996.00	91.34	90.36	10883.85	8259.63	-596.49	8259.63	0.31
179	19090.00	90.37	90.18	10882.44	8353.62	-596.93	8353.62	1.05
180	19184.00	90.81	90.62	10881.48	8447.61	-597.59	8447.61	0.66
181	19278.00	90.15	90.98	10880.69	8541.60	-598.90	8541.60	0.80
182	19371.00	86.81	90.27	10883.15	8634.55	-599.91	8634.55	3.67
183	19466.00	85.76	90.54	10889.31	8729.35	-600.58	8729.35	1.14
184	19560.00	87.96	90.62	10894.46	8823.19	-601.53	8823.19	2.34
185	19654.00	89.41	89.92	10896.62	8917.17	-601.98	8917.17	1.71
186	19748.00	90.24	90.54	10896.90	9011.16	-602.35	9011.16	1.10
187	19843.00	89.63	90.89	10897.01	9106.15	-603.54	9106.15	0.74
188	19937.00	88.88	90.01	10898.23	9200.14	-604.28	9200.14	1.23
189	20031.00	90.07	89.57	10899.09	9294.14	-603.93	9294.14	1.35
190	20125.00	90.46	89.39	10898.66	9388.13	-603.08	9388.13	0.46
191	20220.00	92.09	88.69	10896.55	9483.09	-601.49	9483.09	1.87
192	20314.00	93.19	88.34	10892.22	9576.96	-599.05	9576.96	1.23
193	20409.00	92.00	89.48	10887.91	9671.84	-597.25	9671.84	1.73
194	20503.00	92.31	89.92	10884.38	9765.77	-596.76	9765.77	0.57
195	20597.00	92.97	90.45	10880.05	9859.67	-597.06	9859.67	0.90
196	20682.00	93.67	90.80	10875.13	9944.52	-597.99	9944.52	0.92
197	20760.00	93.67	90.85	10870.14	10022.35	-599.11	10022.35	0.06

Kline Federal 41-18 13T2X LITHOLOGY

Geologists caught lagged samples in 30' intervals in the vertical hole, from 8,350' to 10,300' MD; 30' intervals in the curve build from 10,300' to 11,190' MD; and 50' samples in the lateral from 11,190' to TD at 20,760'

Electric geophysical log, sample and/or MWD gamma ray markers and tops are included in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope in approximately 30' to 50' intervals. Sample descriptions begin in the **Kibbey** formation. The drilling fluid was diesel invert in the vertical and curve, and salt water brine in the lateral.

Drilling vertical hole in Dakota Formation

8350-8382 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

Kibbey Lime

8,382' MD, 8,382' TVD (-6,300')

8382-8410 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8410-8440 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8440-8470 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8470-8500 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8500-8530 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8530-8560 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8560-8590 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8590-8620 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8620-8650 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8650-8680 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8680-8710 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8710-8740 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8740-8770 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8770-8800 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8800-8830 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8830-8860 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8860-8890 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8890-8920 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8920-8950 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8950-8980 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8980-9010 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9010-9040 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline; ANHYDRITE: light to medium gray, amorphous, very soft to soft, earthy texture, no visible porosity or oil stain

9040-9070 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9070-9100 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9100-9130 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9130-9160 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9160-9206 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Base Charles Salt ***9,206' MD, 9,206' TVD (-7,124')***

9206-9220 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9220-9250 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9250-9280 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9280-9310 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9310-9340 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9340-9370 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9370-9400 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9400-9428 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9428-9460 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9460-9490 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9490-9520 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9520-9550 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9550-9580 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9580-9610 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9610-9640 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9640-9670 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9670-9700 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9700-9730 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9730-9760 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9760-9790 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9790-9820 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9820-9850 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9850-9880 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9880-9910 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9910-9940 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9940-9975 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Lodgepole

9,975' MD, 9,975' TVD (-7,893')

9975-10000 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10000-10030 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10030-10060 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10060-10090 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10090-10120 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10120-10150 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10150-10180 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10180-10210 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10210-10240 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10240-10270 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10270-10300 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10,300' Vertical TD reached @ 14:30 hours CST, 11 October 2014

Drilling curve build in the Lodgepole Formation

10300-10330 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10330-10360 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10360-10390 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10390-10420 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10420-10450 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10450-10480 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10480-10510 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10510-10540 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10540-10570 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10570-10600 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10600-10630 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10630-10660 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10660-10690 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10690-10720 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10720-10750 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

10750-10780 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, slight light brown spotty oil stain

False Bakken **10,756' MD, 10,705' TVD (-8,620')**

10756-10764 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Upper Bakken Shale **10,764' MD, 10,712' TVD (-8,627')**

10764-10789 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Middle Bakken **10,789' MD, 10,730' TVD (-8,645')**

10789-10810 SLTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10810-10852 SLTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

Lower Bakken Shale **10,852' MD, 10,770' TVD (-8,685')**

10852-10874 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Three Forks 1st Bench

10,874' MD, 10,782' TVD (-8,697')

10874-10900 SILTSTONE: medium gray, medium gray brown, friable to firm, sub-blocky to sbsplt, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity

10900-10930 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

10930-10974 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

10974-11004 CLAYSTONE: off white, cream, light gray, moderately soft, sub-blocky, very slightly calcareous, trace silt

Three Forks 2nd Bench

11,004' MD, 10,833' TVD (-8,751')

11004-11030 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11030-11060 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11060-11090 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11090-11120 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11120-11150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11150-11185 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

Curve TD of 11,185' @ 22:00 CST 13 October-2014

Intermediate 7" Casing Set @ 11,185' MD

11185-11250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

12400-12450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

14200-14250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

14850-14900 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

14900-14950 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

14950-15000 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15000-15050 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15050-15100 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15100-15150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15150-15200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15200-15250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15250-15300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15300-15350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15350-15400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15400-15450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft,

sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15450-15500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15500-15550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15550-15600 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15600-15650 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15650-15700 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15700-15750 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15750-15800 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15800-15850 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15850-15900 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15900-15950 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15950-16000 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16000-16050 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16050-16100 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16100-16150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16150-16200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16200-16250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16250-16300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16300-16350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16350-16400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16400-16450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

16450-16500 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

16500-16550 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

16550-16600 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17350-17400 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17400-17450 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17450-17500 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17500-17550 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17550-17600 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17600-17650 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17650-17700 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17700-17750 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17750-17800 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17800-17850 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17850-17900 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17900-17950 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

17950-18000 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

18000-18050 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

18050-18100 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

18650-18700 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19250-19300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19300-19350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19350-19400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19400-19450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19450-19500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19500-19550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19550-19600 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19600-19650 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19650-19700 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19700-19750 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19750-19800 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19800-19850 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19850-19900 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft,

sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19900-19950 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19950-20000 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20000-20050 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20050-20100 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20100-20150 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20150-20200 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20200-20250 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20250-20300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20300-20350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20350-20400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20400-20450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20450-20500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20500-20550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20550-20600 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

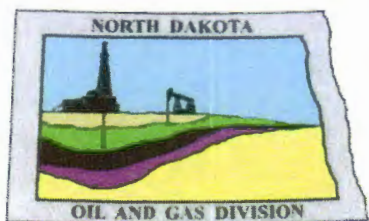
20600-20650 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20650-20700 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20700-20760 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

18,340' TD reached @ 15:00 hours CST, 29 August 2014





Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28654

January 8, 2016

Michael Kukuk
OASIS PETRO NO AMER
1001 FANNIN ST. SUITE 1500
HOUSTON, TX 77002

RE: See Attached List of Wells

List Reference WF# 28654

Dear Michael Kukuk:

We have not received the geological reports on the referenced well list. Please submit one paper copy and one digital pdf file of this report for each well.

If you have any questions, please contact Richard Suggs at (701) 328-8020.

Sincerely,

A handwritten signature in blue ink, which appears to read "Taylor Roth".

Taylor Roth
Engineering Technician



Gyrodata Incorporated
301 Thelma Dr. #413
Casper, Wyoming 82601

307-472-0182
Fax: 307-472-0204

DIRECTIONAL SURVEY CERTIFICATION

RE: Oasis Petroleum (Operator)
Kline Federal 5300 41-18 13T2X (Well Name & No.)
8.27 (Magnetic Declination)

Surveyor: Blake Hunter

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 140 feet to a measured depth of 20682 feet and is projected to 20760 feet, is true and correct as determined from all available records.


Signature

MWD Coordinator
Title

Gyrodata, Inc.
Company

*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: Nabors #486, McKenzie, North Dakota

Job Number: RD0814M179

Run Date: 03 Mar 2015

Surveyor: Blake Hunter

Calculation Method: MINIMUM CURVATURE

Survey surface coordinates obtained from: Company Man

Survey Latitude: 48.068701 deg. N Longitude: 103.602345 deg. W

Magnetic Declination: 8.2700 deg

Total Correction: 8.2700 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
0.00	0.00	0.00	0.00	0.00	0.0	0.0	0.00 N 0.00 E

NO TIE IN DATA PROVIDED. TIE INTO SURFACE.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. AT 25 FT.

140-20682 FT. MD ARE MAGNETIC MEASUREMENT WHILE DRILLING SURVEYS.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20760 FT. MD.

140.00	0.75	30.86	0.54	140.00	0.9	30.9	0.79 N	0.47 E
234.00	0.53	38.77	0.25	233.99	2.0	32.6	1.65 N	1.06 E
327.00	0.66	24.71	0.21	326.99	2.9	32.1	2.48 N	1.55 E
421.00	0.62	17.85	0.09	420.98	4.0	29.3	3.45 N	1.93 E
516.00	0.75	49.23	0.41	515.97	5.0	30.5	4.35 N	2.56 E
607.00	0.53	30.07	0.33	606.97	6.0	32.3	5.10 N	3.22 E
699.00	0.18	68.83	0.44	698.97	6.6	32.9	5.52 N	3.57 E
791.00	0.22	289.26	0.41	790.97	6.7	32.2	5.63 N	3.54 E
882.00	0.26	124.20	0.52	881.96	6.6	32.5	5.57 N	3.55 E
974.00	0.62	112.33	0.40	973.96	6.7	38.4	5.27 N	4.18 E
1066.00	0.79	109.34	0.19	1065.95	7.1	47.1	4.87 N	5.24 E
1158.00	0.75	136.85	0.40	1157.95	7.5	56.0	4.22 N	6.25 E
1250.00	0.57	121.30	0.27	1249.94	7.9	63.3	3.54 N	7.05 E
1342.00	0.26	132.72	0.35	1341.94	8.2	67.4	3.16 N	7.59 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
1433.00	0.31	148.90	0.10	1432.94	8.4	70.4	2.81 N 7.87 E
1528.00	0.44	78.76	0.47	1527.94	8.8	72.4	2.66 N 8.36 E
1623.00	0.35	227.82	0.80	1622.93	8.9	73.4	2.54 N 8.51 E
1719.00	0.75	262.10	0.52	1718.93	8.0	73.6	2.25 N 7.67 E
1813.00	0.79	257.35	0.08	1812.92	6.7	72.5	2.03 N 6.43 E
1908.00	0.84	247.60	0.15	1907.91	5.4	72.5	1.62 N 5.14 E
2003.00	1.23	258.58	0.46	2002.90	3.7	71.8	1.15 N 3.50 E
2097.00	0.97	245.84	0.38	2096.88	1.9	70.7	0.63 N 1.78 E
2105.00	1.36	255.42	5.43	2104.88	1.7	70.6	0.57 N 1.63 E
2190.00	1.14	243.64	0.40	2189.86	0.1	242.0	0.05 S 0.10 W
2284.00	1.27	263.59	0.46	2283.84	2.1	253.5	0.59 S 1.98 W
2378.00	1.19	266.58	0.11	2377.82	4.1	259.2	0.76 S 3.99 W
2472.00	1.14	263.24	0.09	2471.80	6.0	261.0	0.93 S 5.89 W
2567.00	0.44	158.12	1.39	2566.79	6.8	258.4	1.38 S 6.69 W
2661.00	0.13	176.49	0.34	2660.79	6.8	254.5	1.82 S 6.55 W
2757.00	0.48	106.62	0.47	2756.79	6.5	251.6	2.04 S 6.16 W
2852.00	0.40	158.83	0.41	2851.78	6.2	246.4	2.47 S 5.66 W
2945.00	0.57	159.27	0.18	2944.78	6.3	239.2	3.20 S 5.38 W
3039.00	0.35	167.88	0.24	3038.78	6.5	232.7	3.92 S 5.15 W
3134.00	0.35	170.25	0.02	3133.78	6.7	228.3	4.49 S 5.04 W
3227.00	0.31	113.39	0.34	3226.78	6.8	224.4	4.87 S 4.76 W
3322.00	0.48	124.37	0.20	3321.77	6.7	218.9	5.20 S 4.20 W

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
3417.00	0.48	129.38	0.04	3416.77	6.7 212.1	5.67 S 3.56 W
3512.00	0.31	162.61	0.29	3511.77	6.9 207.2	6.17 S 3.18 W
3607.00	0.57	172.10	0.28	3606.76	7.5 203.8	6.88 S 3.03 W
3701.00	0.35	158.48	0.26	3700.76	8.1 200.6	7.61 S 2.86 W
3796.00	0.31	71.20	0.48	3795.76	8.2 197.9	7.80 S 2.52 W
3890.00	0.31	47.47	0.14	3889.76	7.8 195.5	7.55 S 2.09 W
3984.00	0.13	166.65	0.42	3983.76	7.7 194.1	7.48 S 1.88 W
4079.00	0.31	57.75	0.39	4078.76	7.6 192.4	7.45 S 1.63 W
4171.00	0.04	136.50	0.33	4170.76	7.5 190.8	7.34 S 1.40 W
4266.00	0.31	58.46	0.32	4265.76	7.3 189.1	7.23 S 1.16 W
4361.00	0.40	79.99	0.17	4360.76	7.1 185.0	7.04 S 0.61 W
4456.00	0.31	121.39	0.28	4455.75	7.1 180.5	7.11 S 0.07 W
4550.00	0.35	76.03	0.27	4549.75	7.2 176.6	7.17 S 0.43 E
4645.00	0.09	161.55	0.37	4644.75	7.2 174.2	7.18 S 0.73 E
4740.00	0.04	85.44	0.09	4739.75	7.3 173.8	7.24 S 0.79 E
4835.00	0.13	35.52	0.11	4834.75	7.2 172.9	7.15 S 0.89 E
4931.00	0.35	58.28	0.25	4930.75	7.0 170.2	6.91 S 1.20 E
5025.00	0.44	350.60	0.47	5024.75	6.6 167.8	6.40 S 1.38 E
5119.00	0.57	10.91	0.23	5118.75	5.8 165.8	5.59 S 1.41 E
5214.00	0.40	73.92	0.55	5213.74	5.4 160.1	5.03 S 1.82 E
5308.00	0.53	35.34	0.35	5307.74	5.2 152.5	4.59 S 2.39 E
5403.00	0.31	26.64	0.24	5402.74	4.9 145.4	4.00 S 2.76 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
5497.00	0.35	98.45	0.41	5496.74	5.0 140.4	3.81 S 3.16 E
5592.00	0.48	34.99	0.47	5591.73	5.1 133.9	3.53 S 3.67 E
5687.00	0.48	64.61	0.26	5686.73	5.2 125.5	3.03 S 4.26 E
5782.00	0.44	78.32	0.12	5781.73	5.7 119.3	2.79 S 4.98 E
5876.00	0.53	65.75	0.15	5875.72	6.3 113.9	2.54 S 5.73 E
5971.00	0.26	24.79	0.39	5970.72	6.6 109.2	2.16 S 6.22 E
6065.00	0.53	26.29	0.29	6064.72	6.7 103.6	1.58 S 6.50 E
6085.00	0.44	79.02	2.19	6084.72	6.8 102.6	1.48 S 6.61 E
6161.00	0.53	35.87	0.48	6160.72	7.2 99.1	1.14 S 7.11 E
6255.00	0.66	26.81	0.17	6254.71	7.6 92.3	0.30 S 7.61 E
6350.00	0.66	13.28	0.16	6349.70	8.0 84.9	0.72 N 7.98 E
6444.00	0.75	350.78	0.31	6443.70	8.2 77.0	1.85 N 8.00 E
6539.00	1.01	353.33	0.28	6538.69	8.5 67.1	3.30 N 7.81 E
6634.00	1.19	353.50	0.19	6633.67	9.2 56.1	5.11 N 7.60 E
6727.00	0.70	351.57	0.53	6726.66	9.9 48.2	6.63 N 7.41 E
6822.00	0.40	329.69	0.38	6821.65	10.4 43.7	7.49 N 7.15 E
6916.00	0.75	336.01	0.38	6915.65	10.7 39.0	8.33 N 6.74 E
7011.00	0.04	64.08	0.79	7010.64	11.0 36.2	8.92 N 6.52 E
7105.00	0.13	123.23	0.12	7104.64	11.1 36.8	8.87 N 6.63 E
7199.00	0.04	331.62	0.18	7198.64	11.1 37.2	8.84 N 6.71 E
7294.00	0.44	263.15	0.45	7293.64	10.9 35.6	8.83 N 6.33 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
7387.00	0.48	348.41	0.67	7386.64	10.9 32.7	9.17 N 5.90 E
7482.00	0.48	11.08	0.20	7481.64	11.6 30.6	9.95 N 5.89 E
7577.00	0.48	1.24	0.09	7576.63	12.3 29.1	10.74 N 5.98 E
7671.00	0.57	11.87	0.14	7670.63	13.1 27.7	11.59 N 6.08 E
7765.00	0.62	76.21	0.68	7764.63	13.9 28.7	12.17 N 6.67 E
7858.00	0.79	107.67	0.45	7857.62	14.4 32.7	12.09 N 7.77 E
7952.00	0.35	189.41	0.87	7951.62	14.3 35.7	11.61 N 8.34 E
8046.00	0.57	189.94	0.23	8045.61	13.6 37.1	10.87 N 8.22 E
8141.00	0.66	202.51	0.17	8140.61	12.7 38.7	9.90 N 7.92 E
8235.00	0.44	214.20	0.26	8234.60	11.8 39.6	9.10 N 7.51 E
8329.00	0.48	209.72	0.06	8328.60	11.1 40.1	8.46 N 7.12 E
8424.00	0.40	211.91	0.09	8423.60	10.3 40.7	7.83 N 6.74 E
8518.00	0.66	263.94	0.55	8517.59	9.6 38.8	7.50 N 6.03 E
8612.00	0.84	254.10	0.23	8611.59	8.7 33.7	7.25 N 4.83 E
8707.00	0.84	238.02	0.25	8706.58	7.6 28.1	6.69 N 3.57 E
8802.00	0.53	221.67	0.38	8801.57	6.6 24.2	5.99 N 2.69 E
8896.00	0.57	257.62	0.36	8895.56	5.9 19.2	5.57 N 1.94 E
8990.00	0.35	242.76	0.26	8989.56	5.5 13.0	5.34 N 1.23 E
9084.00	0.44	255.86	0.14	9083.56	5.2 7.0	5.12 N 0.62 E
9178.00	0.35	282.84	0.22	9177.56	5.1 359.9	5.09 N 0.01 W
9272.00	0.35	13.54	0.53	9271.56	5.4 357.7	5.44 N 0.22 W
9367.00	0.40	319.14	0.36	9366.55	6.0 356.5	5.97 N 0.37 W

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
9461.00	0.53	334.17	0.19	9460.55	6.7 353.3	6.61 N 0.77 W
9555.00	0.66	11.43	0.43	9554.55	7.6 353.5	7.53 N 0.85 W
9649.00	0.53	10.64	0.14	9648.54	8.5 355.5	8.49 N 0.67 W
9742.00	0.66	92.12	0.84	9741.54	8.9 359.7	8.89 N 0.05 W
9836.00	0.70	115.23	0.29	9835.53	8.7 6.7	8.63 N 1.01 E
9930.00	0.53	119.45	0.19	9929.53	8.4 13.1	8.17 N 1.91 E
10024.00	0.70	125.34	0.19	10023.52	8.1 19.9	7.62 N 2.75 E
10119.00	0.26	104.95	0.49	10118.52	8.0 25.4	7.23 N 3.44 E
10214.00	0.13	137.03	0.17	10213.52	8.0 27.7	7.10 N 3.72 E
10246.00	0.40	155.40	0.87	10245.52	7.9 28.5	6.97 N 3.79 E
10276.00	0.26	115.76	0.87	10275.52	7.9 29.6	6.84 N 3.89 E
10308.00	1.76	115.94	4.69	10307.51	7.9 33.7	6.60 N 4.40 E
10339.00	6.07	119.19	13.92	10338.43	8.4 48.3	5.59 N 6.26 E
10371.00	10.24	121.65	13.07	10370.10	10.7 72.2	3.27 N 10.16 E
10403.00	14.77	115.58	14.72	10401.33	16.3 90.0	0.01 N 16.27 E
10435.00	18.37	116.02	11.26	10432.00	24.8 99.2	3.96 S 24.48 E
10466.00	21.01	116.55	8.54	10461.18	34.9 104.2	8.59 S 33.84 E
10498.00	22.90	125.25	11.74	10490.87	46.5 108.5	14.75 S 44.06 E
10530.00	26.42	130.53	12.97	10519.95	59.2 112.8	22.97 S 54.56 E
10562.00	28.75	133.16	8.21	10548.31	73.4 116.6	32.86 S 65.59 E
10593.00	29.45	132.46	2.51	10575.40	87.9 119.4	43.11 S 76.65 E
10625.00	31.91	128.59	9.85	10602.92	104.0 121.1	53.70 S 89.07 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
10656.00	35.60	127.19	12.16	10628.69	121.1 122.0	64.26 S 102.66 E
10688.00	38.59	123.06	12.16	10654.22	140.4 122.5	75.34 S 118.45 E
10719.00	41.45	121.21	9.99	10677.96	160.3 122.4	85.94 S 135.33 E
10751.00	43.47	117.25	10.47	10701.57	181.9 122.0	96.47 S 154.18 E
10782.00	44.66	118.57	4.85	10723.84	203.4 121.6	106.56 S 173.23 E
10814.00	49.10	120.95	14.90	10745.71	226.7 121.4	118.17 S 193.49 E
10845.00	53.93	120.68	15.60	10765.00	251.0 121.4	130.59 S 214.33 E
10876.00	58.11	120.59	13.49	10782.32	276.7 121.3	143.69 S 236.44 E
10908.00	63.60	120.51	17.16	10797.90	304.6 121.2	157.89 S 260.50 E
10939.00	68.35	117.34	17.94	10810.52	332.9 121.0	171.56 S 285.28 E
10971.00	72.57	115.76	13.98	10821.22	363.0 120.7	185.03 S 312.25 E
11002.00	75.03	115.94	7.96	10829.87	392.6 120.3	198.01 S 339.04 E
11034.00	77.98	115.23	9.47	10837.33	423.6 119.9	211.45 S 367.10 E
11066.00	80.26	116.02	7.53	10843.37	455.0 119.6	225.04 S 395.44 E
11097.00	83.52	116.20	10.53	10847.75	485.6 119.4	238.54 S 422.99 E
11107.00	84.88	115.85	14.04	10848.76	495.5 119.4	242.91 S 431.93 E
11207.00	88.57	116.81	3.81	10854.47	595.3 118.8	287.18 S 521.39 E
11238.00	88.97	116.55	1.54	10855.13	626.2 118.7	301.10 S 549.08 E
11269.00	89.01	116.73	0.59	10855.68	657.2 118.6	315.00 S 576.79 E
11300.00	89.36	116.55	1.27	10856.12	688.2 118.5	328.90 S 604.50 E
11332.00	89.23	115.23	4.14	10856.52	720.1 118.4	342.87 S 633.28 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
11362.00	89.27	115.06	0.58	10856.91	750.1 118.3	355.61 S 660.44 E
11393.00	88.48	113.12	6.76	10857.52	781.0 118.1	368.26 S 688.73 E
11424.00	88.53	111.28	5.94	10858.33	811.8 117.9	379.97 S 717.42 E
11455.00	88.26	109.52	5.74	10859.19	842.6 117.6	390.77 S 746.46 E
11486.00	88.35	108.64	2.85	10860.11	873.2 117.3	400.90 S 775.75 E
11517.00	88.35	108.73	0.29	10861.00	903.9 117.0	410.83 S 805.10 E
11548.00	89.58	108.90	4.01	10861.56	934.5 116.8	420.82 S 834.44 E
11580.00	89.80	108.90	0.69	10861.74	966.3 116.5	431.19 S 864.71 E
11611.00	90.07	108.29	2.15	10861.77	997.0 116.3	441.08 S 894.09 E
11642.00	90.11	108.20	0.32	10861.72	1027.7 116.0	450.78 S 923.54 E
11673.00	90.24	107.15	3.41	10861.63	1058.4 115.8	460.19 S 953.07 E
11704.00	89.32	106.09	4.53	10861.75	1089.0 115.5	469.06 S 982.78 E
11736.00	89.19	106.36	0.94	10862.16	1120.6 115.3	478.00 S 1013.50 E
11767.00	89.32	105.74	2.04	10862.57	1151.2 115.0	486.57 S 1043.29 E
11798.00	89.45	106.18	1.48	10862.90	1181.8 114.8	495.09 S 1073.09 E
11829.00	89.89	105.83	1.81	10863.08	1212.4 114.5	503.64 S 1102.89 E
11860.00	90.02	105.21	2.04	10863.10	1243.1 114.3	511.93 S 1132.76 E
11891.00	90.37	103.46	5.76	10863.00	1273.6 114.1	519.61 S 1162.79 E
11923.00	89.49	101.52	6.66	10863.04	1305.0 113.8	526.53 S 1194.03 E
11954.00	89.49	100.12	4.52	10863.31	1335.2 113.5	532.35 S 1224.48 E
11985.00	89.85	99.59	2.07	10863.49	1365.3 113.2	537.65 S 1255.02 E
12016.00	90.59	99.85	2.53	10863.37	1395.5 112.9	542.89 S 1285.58 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
12047.00	91.03	100.12	1.67	10862.93	1425.7 112.6	548.26 S 1316.10 E
12078.00	90.07	99.15	4.40	10862.64	1456.0 112.3	553.45 S 1346.66 E
12110.00	90.07	99.24	0.28	10862.60	1487.1 112.1	558.56 S 1378.25 E
12141.00	88.44	99.76	5.52	10863.00	1517.4 111.8	563.68 S 1408.82 E
12172.00	89.41	99.59	3.18	10863.58	1547.7 111.6	568.89 S 1439.38 E
12203.00	89.19	96.95	8.55	10863.96	1577.9 111.3	573.35 S 1470.05 E
12234.00	89.54	95.90	3.57	10864.30	1607.9 111.0	576.82 S 1500.85 E
12266.00	89.80	95.19	2.36	10864.49	1638.7 110.7	579.91 S 1532.70 E
12297.00	90.37	94.31	3.38	10864.44	1668.6 110.4	582.48 S 1563.59 E
12328.00	90.29	93.17	3.69	10864.26	1698.3 110.1	584.50 S 1594.53 E
12358.00	90.51	92.82	1.38	10864.05	1727.0 109.8	586.06 S 1624.49 E
12390.00	90.81	93.26	1.66	10863.68	1757.6 109.5	587.76 S 1656.44 E
12420.00	89.76	92.38	4.57	10863.54	1786.4 109.3	589.24 S 1686.40 E
12451.00	90.11	91.50	3.05	10863.57	1816.0 109.0	590.29 S 1717.38 E
12482.00	90.59	91.50	1.55	10863.38	1845.6 108.7	591.10 S 1748.37 E
12514.00	90.64	90.36	3.57	10863.04	1876.1 108.4	591.62 S 1780.37 E
12577.00	91.30	90.18	1.09	10861.97	1936.1 107.8	591.91 S 1843.36 E
12608.00	90.02	90.36	4.17	10861.61	1965.6 107.5	592.06 S 1874.35 E
12671.00	90.64	89.74	1.39	10861.25	2025.8 107.0	592.12 S 1937.35 E
12766.00	89.71	89.83	0.98	10860.96	2116.7 106.2	591.76 S 2032.35 E
12860.00	89.19	88.51	1.51	10861.86	2206.8 105.5	590.40 S 2126.33 E
12955.00	89.54	88.51	0.37	10862.92	2297.8 104.8	587.93 S 2221.29 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
12986.00	89.67	89.48	3.16	10863.13	2327.6 104.6	587.38 S 2252.29 E
13049.00	90.46	89.39	1.26	10863.06	2388.5 104.2	586.76 S 2315.28 E
13081.00	90.02	91.77	7.56	10862.92	2419.6 104.0	587.09 S 2347.28 E
13144.00	90.77	90.71	2.06	10862.49	2481.1 103.7	588.45 S 2410.26 E
13238.00	87.03	91.41	4.05	10864.29	2572.8 103.3	590.19 S 2504.21 E
13333.00	87.34	90.36	1.15	10868.96	2665.6 102.8	591.65 S 2599.08 E
13364.00	87.87	90.45	1.73	10870.26	2695.8 102.7	591.87 S 2630.06 E
13427.00	87.82	90.98	0.84	10872.62	2757.4 102.4	592.66 S 2693.01 E
13459.00	88.18	90.10	2.97	10873.74	2788.8 102.3	592.96 S 2724.99 E
13522.00	89.05	89.92	1.41	10875.26	2850.3 102.0	592.97 S 2787.97 E
13616.00	89.49	89.48	0.66	10876.46	2942.2 101.6	592.48 S 2881.96 E
13647.00	90.02	89.31	1.80	10876.59	2972.5 101.5	592.15 S 2912.95 E
13710.00	90.73	88.60	1.59	10876.18	3034.1 101.2	591.00 S 2975.94 E
13741.00	89.85	90.54	6.87	10876.03	3064.4 101.1	590.77 S 3006.94 E
13804.00	90.07	89.74	1.32	10876.07	3126.3 100.9	590.92 S 3069.94 E
13836.00	89.63	91.41	5.40	10876.15	3157.8 100.8	591.24 S 3101.94 E
13899.00	90.11	90.98	1.02	10876.30	3219.9 100.6	592.56 S 3164.92 E
13994.00	91.47	91.24	1.46	10874.99	3313.6 100.3	594.40 S 3259.89 E
14088.00	90.90	91.41	0.63	10873.04	3406.5 100.1	596.57 S 3353.85 E
14120.00	89.27	90.36	6.06	10872.99	3438.1 100.0	597.07 S 3385.84 E
14183.00	88.48	90.71	1.37	10874.23	3500.2 99.8	597.65 S 3448.83 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
14277.00	88.75	91.06	0.47	10876.50	3593.1 99.6	599.11 S 3542.79 E
14371.00	90.51	90.27	2.05	10877.11	3686.0 99.4	600.20 S 3636.77 E
14466.00	90.86	89.74	0.67	10875.97	3779.7 99.1	600.20 S 3731.77 E
14497.00	90.81	90.01	0.89	10875.52	3810.3 99.1	600.14 S 3762.76 E
14561.00	91.38	89.13	1.64	10874.30	3873.4 98.9	599.66 S 3826.75 E
14592.00	89.01	89.13	7.65	10874.19	3904.0 98.8	599.19 S 3857.74 E
14655.00	89.85	90.80	2.97	10874.82	3966.3 98.7	599.15 S 3920.74 E
14687.00	88.66	90.89	3.73	10875.24	3998.0 98.6	599.62 S 3952.73 E
14718.00	88.35	90.27	2.24	10876.05	4028.6 98.6	599.93 S 3983.72 E
14750.00	88.44	90.45	0.63	10876.94	4060.3 98.5	600.13 S 4015.70 E
14844.00	91.25	90.98	3.04	10877.20	4153.4 98.3	601.31 S 4109.69 E
14907.00	88.79	88.95	5.06	10877.17	4215.8 98.2	601.27 S 4172.68 E
14939.00	87.38	89.57	4.81	10878.24	4247.4 98.1	600.86 S 4204.66 E
15033.00	86.68	90.10	0.93	10883.11	4340.3 98.0	600.58 S 4298.53 E
15065.00	86.86	89.57	1.75	10884.92	4371.9 97.9	600.49 S 4330.48 E
15096.00	87.03	89.57	0.55	10886.57	4402.5 97.8	600.26 S 4361.43 E
15127.00	86.95	89.22	1.16	10888.20	4433.2 97.8	599.93 S 4392.39 E
15222.00	88.13	89.31	1.25	10892.27	4527.1 97.6	598.72 S 4487.29 E
15316.00	89.19	89.31	1.13	10894.47	4620.1 97.4	597.58 S 4581.26 E
15347.00	90.07	88.16	4.67	10894.67	4650.7 97.4	596.90 S 4612.25 E
15378.00	90.29	87.90	1.10	10894.58	4681.3 97.3	595.83 S 4643.23 E
15410.00	90.73	87.81	1.40	10894.29	4712.9 97.2	594.64 S 4675.21 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
15442.00	89.49	89.31	6.08	10894.23	4744.5 97.2	593.83 S 4707.20 E
15505.00	89.80	89.83	0.96	10894.62	4807.0 97.1	593.36 S 4770.19 E
15568.00	90.73	89.66	1.50	10894.33	4869.4 97.0	593.08 S 4833.19 E
15599.00	91.21	89.83	1.64	10893.80	4900.2 97.0	592.94 S 4864.19 E
15694.00	90.29	90.18	1.04	10892.56	4994.5 96.8	592.95 S 4959.18 E
15725.00	90.51	89.83	1.33	10892.34	5025.3 96.8	592.95 S 4990.18 E
15757.00	90.81	91.41	5.03	10891.98	5057.1 96.7	593.30 S 5022.17 E
15788.00	91.38	91.06	2.16	10891.38	5087.9 96.7	593.97 S 5053.16 E
15883.00	91.03	90.62	0.59	10889.39	5182.4 96.6	595.36 S 5148.13 E
15914.00	90.24	91.24	3.24	10889.04	5213.3 96.6	595.86 S 5179.12 E
15977.00	90.37	91.68	0.73	10888.71	5276.0 96.5	597.47 S 5242.10 E
16072.00	90.99	90.62	1.29	10887.58	5370.6 96.4	599.37 S 5337.07 E
16103.00	89.85	91.41	4.47	10887.35	5401.5 96.4	599.92 S 5368.06 E
16166.00	90.20	91.15	0.69	10887.32	5464.2 96.3	601.33 S 5431.05 E
16197.00	87.55	90.18	9.10	10887.93	5495.1 96.3	601.69 S 5462.04 E
16260.00	85.32	90.10	3.54	10891.85	5557.6 96.2	601.84 S 5524.91 E
16355.00	84.84	89.92	0.54	10900.00	5651.7 96.1	601.86 S 5619.56 E
16449.00	86.73	89.39	2.09	10906.91	5744.9 96.0	601.30 S 5713.30 E
16481.00	87.21	89.92	2.23	10908.60	5776.6 96.0	601.10 S 5745.25 E
16513.00	87.21	89.74	0.56	10910.16	5808.4 95.9	601.01 S 5777.22 E
16544.00	87.21	89.83	0.29	10911.66	5839.2 95.9	600.89 S 5808.18 E
16575.00	87.96	90.10	2.57	10912.97	5870.0 95.9	600.87 S 5839.15 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
16607.00	89.36	89.92	4.41	10913.72	5901.8 95.8	600.88 S 5871.14 E
16638.00	89.76	89.39	2.14	10913.96	5932.6 95.8	600.69 S 5902.14 E
16670.00	91.03	89.22	4.00	10913.74	5964.4 95.8	600.30 S 5934.14 E
16733.00	91.16	88.95	0.48	10912.53	6027.0 95.7	599.30 S 5997.12 E
16796.00	91.21	88.51	0.70	10911.23	6089.5 95.6	597.90 S 6060.09 E
16827.00	91.56	88.16	1.60	10910.48	6120.3 95.6	597.00 S 6091.07 E
16923.00	89.76	89.04	2.09	10909.37	6215.5 95.5	594.66 S 6187.03 E
17017.00	89.98	89.57	0.61	10909.59	6309.0 95.4	593.52 S 6281.02 E
17111.00	89.41	89.57	0.61	10910.09	6402.5 95.3	592.81 S 6375.01 E
17143.00	89.41	89.22	1.09	10910.42	6434.3 95.3	592.47 S 6407.01 E
17206.00	89.58	89.22	0.27	10910.97	6497.0 95.2	591.62 S 6470.00 E
17237.00	90.29	89.39	2.36	10911.01	6527.8 95.2	591.24 S 6501.00 E
17300.00	90.68	89.39	0.62	10910.47	6590.5 95.1	590.57 S 6563.99 E
17395.00	90.95	89.31	0.30	10909.12	6685.0 95.1	589.49 S 6658.98 E
17489.00	90.99	88.43	0.94	10907.53	6778.5 95.0	587.64 S 6752.94 E
17584.00	90.90	87.37	1.12	10905.97	6872.7 94.9	584.16 S 6847.87 E
17678.00	90.46	88.07	0.88	10904.85	6966.0 94.8	580.42 S 6941.78 E
17710.00	89.49	89.39	5.12	10904.86	6997.8 94.8	579.71 S 6973.78 E
17773.00	89.49	89.13	0.41	10905.42	7060.5 94.7	578.89 S 7036.77 E
17804.00	90.11	89.13	2.00	10905.53	7091.4 94.7	578.42 S 7067.76 E
17867.00	90.20	88.51	0.99	10905.36	7154.1 94.6	577.13 S 7130.75 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
17960.00	90.90	90.62	2.39	10904.47	7246.7 94.6	576.42 S 7223.74 E
17992.00	91.47	90.89	1.97	10903.81	7278.6 94.5	576.84 S 7255.73 E
18055.00	91.21	90.36	0.94	10902.33	7341.5 94.5	577.53 S 7318.71 E
18148.00	90.29	90.54	1.01	10901.12	7434.2 94.5	578.26 S 7411.69 E
18243.00	90.37	92.12	1.67	10900.57	7529.1 94.4	580.46 S 7506.66 E
18337.00	92.48	92.82	2.36	10898.23	7623.0 94.4	584.51 S 7600.54 E
18368.00	93.05	92.56	2.02	10896.74	7653.9 94.4	585.97 S 7631.47 E
18431.00	93.36	92.21	0.74	10893.21	7716.8 94.4	588.58 S 7694.32 E
18462.00	91.38	92.12	6.39	10891.93	7747.7 94.4	589.75 S 7725.27 E
18526.00	89.58	91.68	2.90	10891.40	7811.7 94.3	591.88 S 7789.23 E
18620.00	90.73	90.89	1.48	10891.14	7905.5 94.3	593.98 S 7883.20 E
18714.00	91.12	90.54	0.56	10889.62	7999.4 94.3	595.16 S 7977.18 E
18808.00	91.08	90.62	0.10	10887.82	8093.1 94.2	596.11 S 8071.16 E
18902.00	91.21	90.62	0.14	10885.94	8186.9 94.2	597.13 S 8165.14 E
18996.00	91.34	90.36	0.31	10883.85	8280.7 94.1	597.93 S 8259.11 E
19090.00	90.37	90.18	1.05	10882.45	8374.5 94.1	598.37 S 8353.10 E
19184.00	90.81	90.62	0.66	10881.48	8468.3 94.1	599.03 S 8447.09 E
19278.00	90.15	90.98	0.80	10880.69	8562.1 94.0	600.34 S 8541.07 E
19371.00	86.81	90.27	3.67	10883.16	8654.9 94.0	601.36 S 8634.02 E
19466.00	85.76	90.54	1.14	10889.31	8749.6 93.9	602.03 S 8728.82 E
19560.00	87.96	90.62	2.34	10894.46	8843.2 93.9	602.98 S 8822.67 E
19654.00	89.41	89.92	1.71	10896.62	8937.0 93.9	603.42 S 8916.64 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 13T2X, 8 3/4" Case, and 6" Hole

Location: NABORS #486, McKenzie, North Dakota

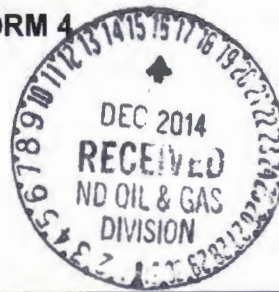
Job Number: RD0814M179

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
19748.00	90.24	90.54	1.10	10896.91	9030.8 93.8	603.80 S 9010.64 E
19843.00	89.63	90.89	0.74	10897.01	9125.7 93.8	604.98 S 9105.63 E
19937.00	88.88	90.01	1.23	10898.24	9219.5 93.8	605.72 S 9199.62 E
20031.00	90.07	89.57	1.35	10899.10	9313.3 93.7	605.37 S 9293.61 E
20125.00	90.46	89.39	0.46	10898.66	9407.0 93.7	604.52 S 9387.60 E
20220.00	92.09	88.69	1.87	10896.55	9501.7 93.6	602.93 S 9482.56 E
20314.00	93.19	88.34	1.23	10892.22	9595.2 93.6	600.50 S 9576.43 E
20409.00	92.00	89.48	1.73	10887.92	9689.8 93.5	598.69 S 9671.31 E
20503.00	92.31	89.92	0.57	10884.38	9783.5 93.5	598.20 S 9765.24 E
20597.00	92.97	90.45	0.90	10880.05	9877.3 93.5	598.50 S 9859.14 E
20682.00	93.67	90.80	0.92	10875.13	9962.0 93.4	599.43 S 9944.00 E
STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20760 FT. MD.						
20760.00	93.67	90.80	0.00	10870.14	10039.8 93.4	600.52 S 10021.83 E

Final Station Closure: Distance: 10039.80 ft Az: 93.43 deg.

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 15, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	<u>Waiver from tubing/packer requirement</u>

Well Name and Number Kline Federal 5300 41-18 13T2X					
Footages	Qtr-Qtr	Section	Township	Range	
401 F S L 237 F W L	LOT4	18	153 N	100 W	
Field BAKER	Pool Bakken	County MCKENZIE			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Assistant	Date December 15, 2014		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date December 22, 2014	
By 	
Title PETROLEUM ENGINEER	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 28, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	
Approximate Start Date	

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Change casing

Well Name and Number Kline Federal 5300 41-18 13T2X					
Footages 401 F S L 237 F W L		Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:

Surface Casing: 13-3/8, 54.5#, 17-1/2" Hole, 2,072' MD
Dakota Contingency: 9-5/8, 40#, 12-1/4" Hole, 6,101' MD

Attached are revised plats, drill plan, well summary, directional plan and plot

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563
Address 1001 Fannin, Suite 1500		
City Houston	State TX	Zip Code 77002
Signature 	Printed Name Heather McCowan	
Title Regulatory Assistant	Date July 28, 2014	
Email Address hmccowan@oasispetroleum.com		

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-5-14	
By 	
Title Petroleum Resource Specialist	



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 13T2X

Kline Federal 5300 41-18 13T2X

Plan: Design #1

Standard Planning Report

23 May, 2014



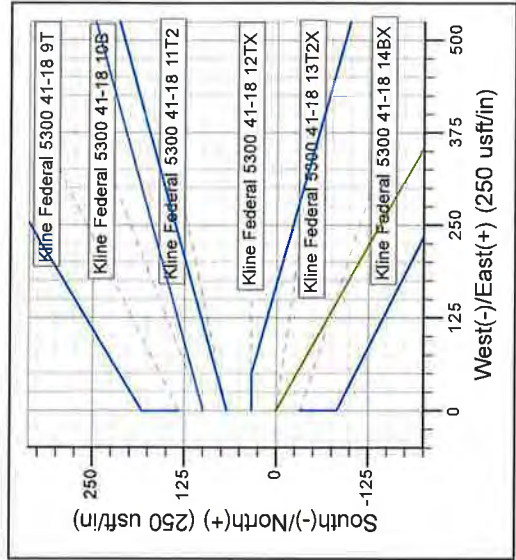
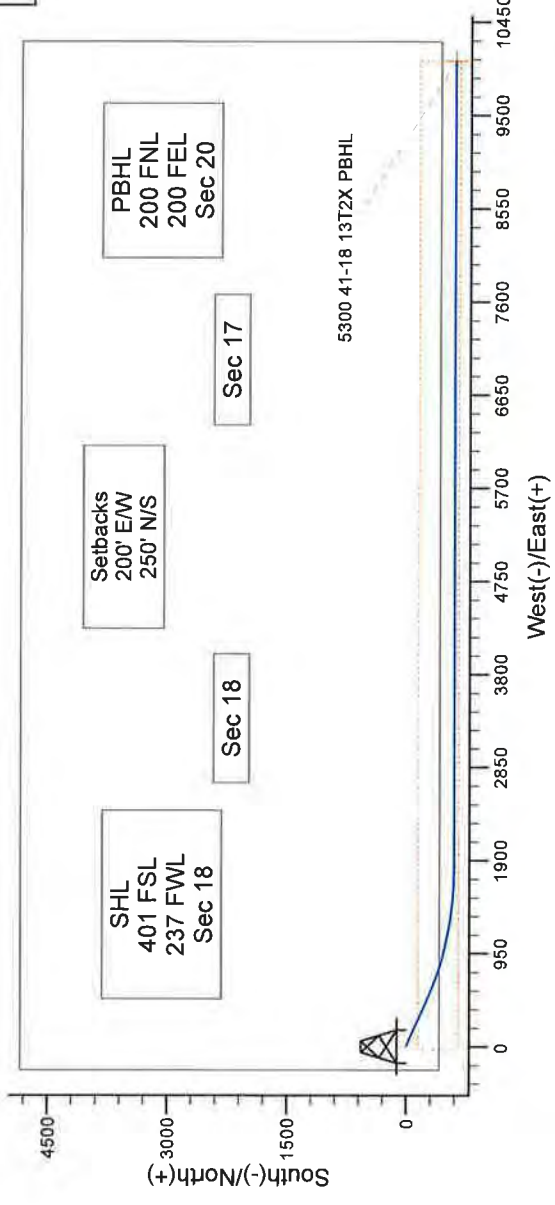
**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY



Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 13T2X
Wellbore: Kline Federal 5300 41-18 13T2X
Design: Design #1

WELL DETAILS: Kline Federal 5300 41-18 13T2X

Northing	Ground Level	2057.0	Longitude
405103.22	Easting	1209966.87	48° 4' 7.220 N 103° 36' 11.950 W



CASING DETAILS

TVD	MD	Name	Size
10870.0	11142.0	7"	7

SECTION DETAILS

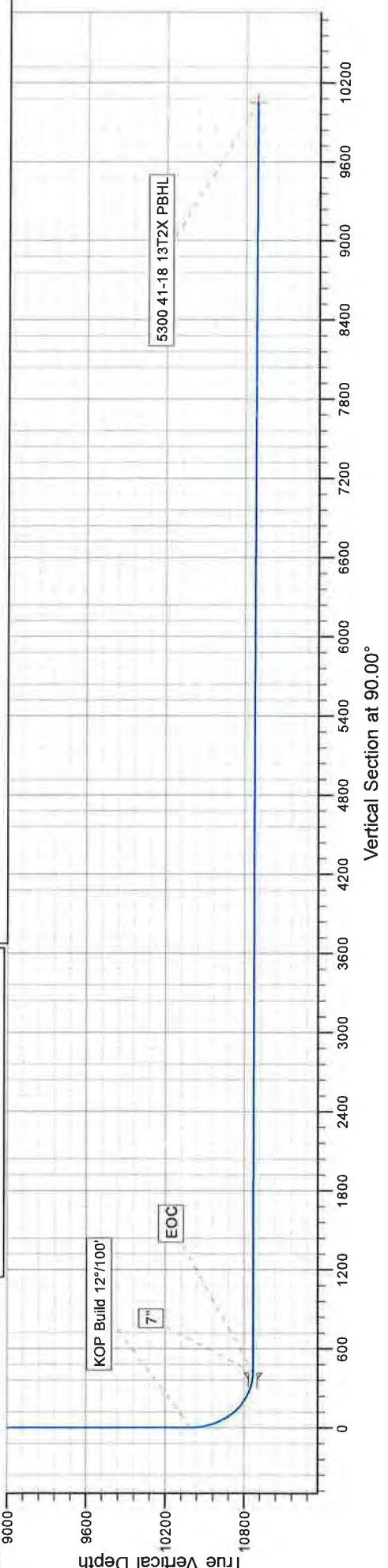
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00
10392.5	0.00	0.00	10392.5	0.0	0.0	0.00	0.00
11141.7	89.90	119.35	10870.0	-233.6	415.5	12.00	12.00
12609.2	89.90	90.00	10872.6	-601.3	1819.6	2.00	2.00
20839.6	89.90	90.00	10886.9	-601.3	10050.0	0.00	0.00

CASING DETAILS

TVD	MD	Name	Size
10870.0	11142.0	7"	7

SECTION DETAILS

MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00
10392.5	0.00	0.00	10392.5	0.0	0.0	0.00	0.00
11141.7	89.90	119.35	10870.0	-233.6	415.5	12.00	12.00
12609.2	89.90	90.00	10872.6	-601.3	1819.6	2.00	2.00
20839.6	89.90	90.00	10886.9	-601.3	10050.0	0.00	0.00





Ryan Directional Services
Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18		
Site Position:		Northing:	408,962.44 usft
From:	Lat/Long	Easting:	1,210,229.18 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 4' 45.380 N
		Longitude:	103° 36' 10.380 W
		Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 13T2X		
Well Position	+N/-S	-3,866.7 usft	Northing: 405,103.22 usft
	+E/-W	-106.6 usft	Easting: 1,209,966.86 usft
Position Uncertainty	2.0 usft	Wellhead Elevation:	Latitude: 48° 4' 7.220 N
			Longitude: 103° 36' 11.950 W
			Ground Level: 2,057.0 usft

Wellbore	Kline Federal 5300 41-18 13T2X				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,392.5	0.00	0.00	10,392.5	0.0	0.0	0.00	0.00	0.00	0.00	
11,141.7	89.90	119.35	10,870.0	-233.6	415.5	12.00	12.00	0.00	119.35	
12,609.2	89.90	90.00	10,872.6	-601.3	1,819.6	2.00	0.00	-2.00	-90.03	
20,839.6	89.90	90.00	10,886.9	-601.3	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 13T2X PE



Ryan Directional Services

Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 13T2X
Wellbore: Kline Federal 5300 41-18 13T2X
Design: Design #1

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 13T2X
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,392.5	0.00	0.00	10,392.5	0.0	0.0	0.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,400.0	0.90	119.35	10,400.0	0.0	0.1	0.1	12.00	12.00	0.00
10,425.0	3.90	119.35	10,425.0	-0.5	1.0	1.0	12.00	12.00	0.00
10,450.0	6.90	119.35	10,449.9	-1.7	3.0	3.0	12.00	12.00	0.00
10,475.0	9.90	119.35	10,474.6	-3.5	6.2	6.2	12.00	12.00	0.00
10,500.0	12.90	119.35	10,499.1	-5.9	10.5	10.5	12.00	12.00	0.00
10,525.0	15.90	119.35	10,523.3	-9.0	15.9	15.9	12.00	12.00	0.00
10,550.0	18.90	119.35	10,547.2	-12.6	22.4	22.4	12.00	12.00	0.00
10,575.0	21.90	119.35	10,570.6	-16.9	30.0	30.0	12.00	12.00	0.00
10,600.0	24.90	119.35	10,593.5	-21.8	38.7	38.7	12.00	12.00	0.00
10,625.0	27.90	119.35	10,615.9	-27.2	48.4	48.4	12.00	12.00	0.00
10,650.0	30.90	119.35	10,637.7	-33.2	59.1	59.1	12.00	12.00	0.00
10,675.0	33.90	119.35	10,658.8	-39.8	70.7	70.7	12.00	12.00	0.00
10,700.0	36.90	119.35	10,679.2	-46.9	83.4	83.4	12.00	12.00	0.00
10,725.0	39.90	119.35	10,698.8	-54.5	96.9	96.9	12.00	12.00	0.00
10,750.0	42.90	119.35	10,717.5	-62.6	111.3	111.3	12.00	12.00	0.00
10,761.7	44.31	119.35	10,726.0	-66.6	118.3	118.3	12.00	12.00	0.00
False Bakken									
10,774.5	45.83	119.35	10,735.0	-71.0	126.2	126.2	12.00	12.00	0.00
Upper Bakken									
10,775.0	45.90	119.35	10,735.4	-71.2	126.6	126.6	12.00	12.00	0.00
10,798.0	48.66	119.35	10,751.0	-79.5	141.3	141.3	12.00	12.00	0.00
Middle Bakken									
10,800.0	48.90	119.35	10,752.3	-80.2	142.6	142.6	12.00	12.00	0.00
10,825.0	51.90	119.35	10,768.2	-89.6	159.4	159.4	12.00	12.00	0.00
10,850.0	54.90	119.35	10,783.1	-99.5	176.9	176.9	12.00	12.00	0.00
10,853.3	55.29	119.35	10,785.0	-100.8	179.2	179.2	12.00	12.00	0.00
Lower Bakken									
10,864.0	56.58	119.35	10,791.0	-105.1	186.9	186.9	12.00	12.00	0.00
Pronghorn									
10,875.0	57.90	119.35	10,797.0	-109.7	195.0	195.0	12.00	12.00	0.00
10,900.0	60.90	119.35	10,809.7	-120.2	213.8	213.8	12.00	12.00	0.00
10,906.9	61.73	119.35	10,813.0	-123.2	219.0	219.0	12.00	12.00	0.00
Three Forks 1st Bench									
10,925.0	63.90	119.35	10,821.3	-131.1	233.1	233.1	12.00	12.00	0.00
10,950.0	66.90	119.35	10,831.7	-142.2	252.9	252.9	12.00	12.00	0.00
10,966.8	68.92	119.35	10,838.0	-149.8	266.5	266.5	12.00	12.00	0.00
Three Forks 1st Bench Claystone									
10,975.0	69.90	119.35	10,840.9	-153.6	273.2	273.2	12.00	12.00	0.00
11,000.0	72.90	119.35	10,848.9	-165.2	293.8	293.8	12.00	12.00	0.00
11,007.5	73.80	119.35	10,851.0	-168.7	300.0	300.0	12.00	12.00	0.00
Three Forks 2nd Bench									
11,025.0	75.90	119.35	10,855.6	-177.0	314.8	314.8	12.00	12.00	0.00
11,050.0	78.90	119.35	10,861.0	-189.0	336.1	336.1	12.00	12.00	0.00
11,075.0	81.90	119.35	10,865.2	-201.1	357.5	357.5	12.00	12.00	0.00
11,100.0	84.90	119.35	10,868.1	-213.2	379.2	379.2	12.00	12.00	0.00
11,125.0	87.90	119.35	10,869.6	-225.5	400.9	400.9	12.00	12.00	0.00
11,141.7	89.90	119.35	10,870.0	-233.6	415.5	415.5	11.98	11.98	0.00
EOC									
11,142.0	89.90	119.34	10,870.0	-233.8	415.7	415.7	2.22	0.00	-2.22
7"									
11,200.0	89.90	118.18	10,870.1	-261.7	466.6	466.6	2.00	0.00	-2.00



Ryan Directional Services
Planning Report



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Wellbore: Kline Federal 5300 41-18 13T2X
Design: Design #1

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 13T2X
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
11,300.0	89.90	116.18	10,870.2	-307.4	555.5	555.5	2.00	0.00	-2.00
11,400.0	89.90	114.18	10,870.4	-349.9	646.0	646.0	2.00	0.00	-2.00
11,500.0	89.90	112.18	10,870.6	-389.3	737.9	737.9	2.00	0.00	-2.00
11,600.0	89.90	110.18	10,870.8	-425.4	831.2	831.2	2.00	0.00	-2.00
11,700.0	89.90	108.18	10,871.0	-458.3	925.6	925.6	2.00	0.00	-2.00
11,800.0	89.90	106.18	10,871.1	-487.8	1,021.2	1,021.2	2.00	0.00	-2.00
11,900.0	89.90	104.18	10,871.3	-514.0	1,117.7	1,117.7	2.00	0.00	-2.00
12,000.0	89.90	102.18	10,871.5	-536.8	1,215.0	1,215.0	2.00	0.00	-2.00
12,100.0	89.90	100.18	10,871.7	-556.2	1,313.1	1,313.1	2.00	0.00	-2.00
12,200.0	89.90	98.18	10,871.9	-572.2	1,411.8	1,411.8	2.00	0.00	-2.00
12,300.0	89.90	96.18	10,872.0	-584.7	1,511.0	1,511.0	2.00	0.00	-2.00
12,400.0	89.90	94.18	10,872.2	-593.7	1,610.6	1,610.6	2.00	0.00	-2.00
12,500.0	89.90	92.18	10,872.4	-599.3	1,710.5	1,710.5	2.00	0.00	-2.00
12,609.2	89.90	90.00	10,872.6	-601.3	1,819.6	1,819.6	2.00	0.00	-2.00
12,700.0	89.90	90.00	10,872.7	-601.3	1,910.4	1,910.4	0.00	0.00	0.00
12,800.0	89.90	90.00	10,872.9	-601.3	2,010.4	2,010.4	0.00	0.00	0.00
12,900.0	89.90	90.00	10,873.1	-601.3	2,110.4	2,110.4	0.00	0.00	0.00
13,000.0	89.90	90.00	10,873.3	-601.3	2,210.4	2,210.4	0.00	0.00	0.00
13,100.0	89.90	90.00	10,873.4	-601.3	2,310.4	2,310.4	0.00	0.00	0.00
13,200.0	89.90	90.00	10,873.6	-601.3	2,410.4	2,410.4	0.00	0.00	0.00
13,300.0	89.90	90.00	10,873.8	-601.3	2,510.4	2,510.4	0.00	0.00	0.00
13,400.0	89.90	90.00	10,874.0	-601.3	2,610.4	2,610.4	0.00	0.00	0.00
13,500.0	89.90	90.00	10,874.1	-601.3	2,710.4	2,710.4	0.00	0.00	0.00
13,600.0	89.90	90.00	10,874.3	-601.3	2,810.4	2,810.4	0.00	0.00	0.00
13,700.0	89.90	90.00	10,874.5	-601.3	2,910.4	2,910.4	0.00	0.00	0.00
13,800.0	89.90	90.00	10,874.7	-601.3	3,010.4	3,010.4	0.00	0.00	0.00
13,900.0	89.90	90.00	10,874.8	-601.3	3,110.4	3,110.4	0.00	0.00	0.00
14,000.0	89.90	90.00	10,875.0	-601.3	3,210.4	3,210.4	0.00	0.00	0.00
14,100.0	89.90	90.00	10,875.2	-601.3	3,310.4	3,310.4	0.00	0.00	0.00
14,200.0	89.90	90.00	10,875.4	-601.3	3,410.4	3,410.4	0.00	0.00	0.00
14,300.0	89.90	90.00	10,875.5	-601.3	3,510.4	3,510.4	0.00	0.00	0.00
14,400.0	89.90	90.00	10,875.7	-601.3	3,610.4	3,610.4	0.00	0.00	0.00
14,500.0	89.90	90.00	10,875.9	-601.3	3,710.4	3,710.4	0.00	0.00	0.00
14,600.0	89.90	90.00	10,876.1	-601.3	3,810.4	3,810.4	0.00	0.00	0.00
14,700.0	89.90	90.00	10,876.2	-601.3	3,910.4	3,910.4	0.00	0.00	0.00
14,800.0	89.90	90.00	10,876.4	-601.3	4,010.4	4,010.4	0.00	0.00	0.00
14,900.0	89.90	90.00	10,876.6	-601.3	4,110.4	4,110.4	0.00	0.00	0.00
15,000.0	89.90	90.00	10,876.8	-601.3	4,210.4	4,210.4	0.00	0.00	0.00
15,100.0	89.90	90.00	10,876.9	-601.3	4,310.4	4,310.4	0.00	0.00	0.00
15,200.0	89.90	90.00	10,877.1	-601.3	4,410.4	4,410.4	0.00	0.00	0.00
15,300.0	89.90	90.00	10,877.3	-601.3	4,510.4	4,510.4	0.00	0.00	0.00
15,400.0	89.90	90.00	10,877.5	-601.3	4,610.4	4,610.4	0.00	0.00	0.00
15,500.0	89.90	90.00	10,877.6	-601.3	4,710.4	4,710.4	0.00	0.00	0.00
15,600.0	89.90	90.00	10,877.8	-601.3	4,810.4	4,810.4	0.00	0.00	0.00
15,700.0	89.90	90.00	10,878.0	-601.3	4,910.4	4,910.4	0.00	0.00	0.00
15,712.9	89.90	90.00	10,878.0	-601.3	4,923.4	4,923.4	0.00	0.00	0.00
Three Forks 2nd Bench Claystone									
15,800.0	89.90	90.00	10,878.2	-601.3	5,010.4	5,010.4	0.00	0.00	0.00
15,900.0	89.90	90.00	10,878.3	-601.3	5,110.4	5,110.4	0.00	0.00	0.00
16,000.0	89.90	90.00	10,878.5	-601.3	5,210.4	5,210.4	0.00	0.00	0.00
16,100.0	89.90	90.00	10,878.7	-601.3	5,310.4	5,310.4	0.00	0.00	0.00
16,200.0	89.90	90.00	10,878.8	-601.3	5,410.4	5,410.4	0.00	0.00	0.00
16,300.0	89.90	90.00	10,879.0	-601.3	5,510.4	5,510.4	0.00	0.00	0.00



Ryan Directional Services Planning Report



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Wellbore: Kline Federal 5300 41-18 13T2X
Design: Design #1

Local Co-ordinate Reference:
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MD Reference:
North Reference:
Survey Calculation Method:

Well Kline Federal 5300 41-18 13T2X
 WELL @ 2082.0usft (Original Well Elev)
 WELL @ 2082.0usft (Original Well Elev)
 True
 Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
16,400.0	89.90	90.00	10,879.2	-601.3	5,610.4	5,610.4	0.00	0.00	0.00
16,500.0	89.90	90.00	10,879.4	-601.3	5,710.4	5,710.4	0.00	0.00	0.00
16,600.0	89.90	90.00	10,879.5	-601.3	5,810.4	5,810.4	0.00	0.00	0.00
16,700.0	89.90	90.00	10,879.7	-601.3	5,910.4	5,910.4	0.00	0.00	0.00
16,800.0	89.90	90.00	10,879.9	-601.3	6,010.4	6,010.4	0.00	0.00	0.00
16,900.0	89.90	90.00	10,880.1	-601.3	6,110.4	6,110.4	0.00	0.00	0.00
17,000.0	89.90	90.00	10,880.2	-601.3	6,210.4	6,210.4	0.00	0.00	0.00
17,100.0	89.90	90.00	10,880.4	-601.3	6,310.4	6,310.4	0.00	0.00	0.00
17,200.0	89.90	90.00	10,880.6	-601.3	6,410.4	6,410.4	0.00	0.00	0.00
17,300.0	89.90	90.00	10,880.8	-601.3	6,510.4	6,510.4	0.00	0.00	0.00
17,400.0	89.90	90.00	10,880.9	-601.3	6,610.4	6,610.4	0.00	0.00	0.00
17,500.0	89.90	90.00	10,881.1	-601.3	6,710.4	6,710.4	0.00	0.00	0.00
17,600.0	89.90	90.00	10,881.3	-601.3	6,810.4	6,810.4	0.00	0.00	0.00
17,700.0	89.90	90.00	10,881.5	-601.3	6,910.4	6,910.4	0.00	0.00	0.00
17,800.0	89.90	90.00	10,881.6	-601.3	7,010.4	7,010.4	0.00	0.00	0.00
17,900.0	89.90	90.00	10,881.8	-601.3	7,110.4	7,110.4	0.00	0.00	0.00
18,000.0	89.90	90.00	10,882.0	-601.3	7,210.4	7,210.4	0.00	0.00	0.00
18,100.0	89.90	90.00	10,882.2	-601.3	7,310.4	7,310.4	0.00	0.00	0.00
18,200.0	89.90	90.00	10,882.3	-601.3	7,410.4	7,410.4	0.00	0.00	0.00
18,300.0	89.90	90.00	10,882.5	-601.3	7,510.4	7,510.4	0.00	0.00	0.00
18,400.0	89.90	90.00	10,882.7	-601.3	7,610.4	7,610.4	0.00	0.00	0.00
18,500.0	89.90	90.00	10,882.9	-601.3	7,710.4	7,710.4	0.00	0.00	0.00
18,600.0	89.90	90.00	10,883.0	-601.3	7,810.4	7,810.4	0.00	0.00	0.00
18,700.0	89.90	90.00	10,883.2	-601.3	7,910.4	7,910.4	0.00	0.00	0.00
18,800.0	89.90	90.00	10,883.4	-601.3	8,010.4	8,010.4	0.00	0.00	0.00
18,900.0	89.90	90.00	10,883.6	-601.3	8,110.4	8,110.4	0.00	0.00	0.00
19,000.0	89.90	90.00	10,883.7	-601.3	8,210.4	8,210.4	0.00	0.00	0.00
19,100.0	89.90	90.00	10,883.9	-601.3	8,310.4	8,310.4	0.00	0.00	0.00
19,200.0	89.90	90.00	10,884.1	-601.3	8,410.4	8,410.4	0.00	0.00	0.00
19,300.0	89.90	90.00	10,884.3	-601.3	8,510.4	8,510.4	0.00	0.00	0.00
19,400.0	89.90	90.00	10,884.4	-601.3	8,610.4	8,610.4	0.00	0.00	0.00
19,500.0	89.90	90.00	10,884.6	-601.3	8,710.4	8,710.4	0.00	0.00	0.00
19,600.0	89.90	90.00	10,884.8	-601.3	8,810.4	8,810.4	0.00	0.00	0.00
19,700.0	89.90	90.00	10,885.0	-601.3	8,910.4	8,910.4	0.00	0.00	0.00
19,800.0	89.90	90.00	10,885.1	-601.3	9,010.4	9,010.4	0.00	0.00	0.00
19,900.0	89.90	90.00	10,885.3	-601.3	9,110.4	9,110.4	0.00	0.00	0.00
20,000.0	89.90	90.00	10,885.5	-601.3	9,210.4	9,210.4	0.00	0.00	0.00
20,100.0	89.90	90.00	10,885.7	-601.3	9,310.4	9,310.4	0.00	0.00	0.00
20,200.0	89.90	90.00	10,885.8	-601.3	9,410.4	9,410.4	0.00	0.00	0.00
20,300.0	89.90	90.00	10,886.0	-601.3	9,510.4	9,510.4	0.00	0.00	0.00
20,400.0	89.90	90.00	10,886.2	-601.3	9,610.4	9,610.4	0.00	0.00	0.00
20,500.0	89.90	90.00	10,886.4	-601.3	9,710.4	9,710.4	0.00	0.00	0.00
20,600.0	89.90	90.00	10,886.5	-601.3	9,810.4	9,810.4	0.00	0.00	0.00
20,700.0	89.90	90.00	10,886.7	-601.3	9,910.4	9,910.4	0.00	0.00	0.00
20,800.0	89.90	90.00	10,886.9	-601.3	10,010.4	10,010.4	0.00	0.00	0.00
20,839.6	89.90	90.00	10,886.9	-601.3	10,050.0	10,050.0	0.00	0.00	0.00
5300 41-18 13T2X PBHL									



Ryan Directional Services
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WELL @ 2082.0usft (Original Well Elev)
WELL @ 2082.0usft (Original Well Elev)
True
Minimum Curvature

Design Targets

Target Name	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- hit/miss target	(°)	(°)	(usft)	(usft)	(usft)	(usft)	(usft)		
- Shape									
5300 41-18 13T2X PBHI	0.00	0.00	10,886.8	-801.0	10,050.0	404,097.76	1,219,984.49	48° 4' 1.262 N	103° 33' 43.975 W
- plan misses target center by 0.4usft at 20839.6usft MD (10886.9 TVD, -801.3 N, 10050.0 E)									
- Point									

Casing Points

Measured Depth	Vertical Depth		Name	Casing Diameter	Hole Diameter
(usft)	(usft)			(")	(")
11,142.0	10,870.0	7"		7	8-3/4



Ryan Directional Services

Planning Report



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 WELL @ 2082.0usft (Original Well Elev)
 WELL @ 2082.0usft (Original Well Elev)
 True
 Minimum Curvature

Formations

Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,972.0	1,972.0	Pierre			
4,646.0	4,646.0	Greenhorn			
5,057.0	5,057.0	Mowry			
5,471.0	5,471.0	Dakota			
6,486.0	6,486.0	Rierdon			
6,814.0	6,814.0	Dunham Salt			
6,929.0	6,929.0	Dunham Salt Base			
7,024.0	7,024.0	Spearfish			
7,283.0	7,283.0	Pine Salt			
7,318.0	7,318.0	Pine Salt Base			
7,374.0	7,374.0	Opeche Salt			
7,403.0	7,403.0	Opeche Salt Base			
7,605.0	7,605.0	Broom Creek (Top of Minnelusa Gp.)			
7,684.0	7,684.0	Amsden			
7,853.0	7,853.0	Tyler			
8,047.0	8,047.0	Otter (Base of Minnelusa Gp.)			
8,390.0	8,390.0	Kibbey			
8,542.0	8,542.0	Charles Salt			
9,159.0	9,159.0	UB			
9,238.0	9,238.0	Base Last Salt			
9,301.0	9,301.0	Ratcliffe			
9,454.0	9,454.0	Mission Canyon			
10,022.0	10,022.0	Lodgepole			
10,209.0	10,209.0	Lodgepole Fracture Zone			
10,761.7	10,726.0	False Bakken			
10,774.5	10,735.0	Upper Bakken			
10,798.0	10,751.0	Middle Bakken			
10,853.3	10,785.0	Lower Bakken			
10,864.0	10,791.0	Pronghorn			
10,906.9	10,813.0	Three Forks 1st Bench			
10,966.8	10,838.0	Three Forks 1st Bench Claystone			
11,007.5	10,851.0	Three Forks 2nd Bench			
15,712.9	10,878.0	Three Forks 2nd Bench Claystone			

Plan Annotations

Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,392.5	10,392.5	0.0	0.0	KOP Build 12°/100'
11,141.7	10,870.0	-233.6	415.5	EOC

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
13-3/8"	0' to 2,072'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,072'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl₂, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl₂ and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 31-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 6,101'	40	HCL-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,101'	9-5/8", 40#, HCL-80, LTC, 8rd	4230 / 2.13	5750 / 3.72	837 / 2.78

API Rating & Safety Factor

- a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c) Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **598 sks** (309 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti-foam and 0.4% fluid loss agent.

Tail Slurry: **349 sks** (72 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,664'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,664' – 10,393'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,393' – 11,142'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,664'	6,664'	7", 29#, P-110, LTC, 8rd	8530 / 2.46*	11220 / 1.19	797 / 2.07
6,664' – 10,393'	3,729'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.18*	12460 / 1.29	
6,664' – 10,393'	3,729'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,393' – 11,142'	749'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.50*	11220 / 1.15	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,870' TVD.
- c) Based on string weight in 10 ppg fluid, (283k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **157 sks** (72 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **621 sks** (172 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10,343' – 20,840'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse (psi) a	Burst (psi) b	Tension (1000 lbs) c
10,343' – 20,840'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10680 / 1.98	12410 / 1.28	443 / 2.00

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,887' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,887' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 121k lbs.) plus 100k lbs overpull.

DRILLING PLAN																																																																																																																																																				
OPERATOR		Oasis Petroleum				COUNTY/STATE		McKenzie Co., ND																																																																																																																																												
WELL NAME		Kline Federal 5300 41-18 13T2X				RIG		0																																																																																																																																												
WELL TYPE		Horizontal Three Forks																																																																																																																																																		
LOCATION		SWSW 18-153N-100W				Surface Location (survey plat):		401' fsl 237' fwl																																																																																																																																												
EST. T.D.		20,840'				GROUND ELEV:		2057 Finished Pad Elev.																																																																																																																																												
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DEVIATION:					Surf: 3 deg max, 1 deg / 100'; srvy every 500' Prod: 5 deg max, 1 deg / 100'; srvy every 100'																																																																																																																																															
DST'S:					None planned																																																																																																																																															
CORES:					None planned																																																																																																																																															
MUDLOGGING:					Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land																																																																																																																																															
BOP:					11" 5000 psi blind, pipe & annular																																																																																																																																															
Dip Rate: 0.1																																																																																																																																																				
Max. Anticipated BHP: 4714																																																																																																																																																				
Surface Formation: Glacial till																																																																																																																																																				
MUD:		Interval	Type	WT	Vis	WL	Remarks																																																																																																																																													
Surface:		0' - 2,072'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks																																																																																																																																													
Intermediate:		2,072' - 11,142'	Invert	9.5-10.4	40-50	30+HHP	Circ Mud Tanks																																																																																																																																													
Lateral:		11,142' - 20,840'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks																																																																																																																																													
CASING:		Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks																																																																																																																																												
Surface:		13-3/8"	54.5#	17-1/2"	2,072'	To Surface	12	100' into Pierre																																																																																																																																												
Dakota Contingency:		9-5/8"	40#	12-1/4"	6,101'	To Surface	12	Below Dakota																																																																																																																																												
Intermediate:		7"	29#	8-3/4"	11,142'	3,971'	24	1500' above Dakota																																																																																																																																												
Production Liner:		4.5"	13.5#	6"	20,840'	TOL @ 10,343'		50' above KOP																																																																																																																																												
PROBABLE PLUGS, IF REQ'D:																																																																																																																																																				
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI																																																																																																																																													
Surface:		2,072	2,072	401' FSL	237' FWL	SEC 18-T153N-R100W		Survey Company:																																																																																																																																												
KOP:		10,393'	10,393'	401' FSL	237' FWL	SEC 18-T153N-R100W		Build Rate: 12 deg /100'																																																																																																																																												
EOC:		11,142'	10,870'	167' FSL	653' FWL	SEC 18-T153N-R100W	119.35																																																																																																																																													
Casing Point:		11,142'	10,870'	167' FSL	653' FWL	SEC 18-T153N-R100W	119.35																																																																																																																																													
Three Forks Lateral TD:		20,840'	10,887'	200' FNL	200' FEL	SEC 20-T153N-R100W	90.00																																																																																																																																													
Comments: Request Sundry to Waive Open Hole Logs Exception well: Oasis Petroleum's Kline 5300 11-18H 35 packers, 35 sleeves, no frac string Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations. 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2) 68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-5 (Primary Name: Kerosene)																																																																																																																																																				
					Engineering: hbadar rpm 7/18/14																																																																																																																																															
Geology: M.Steed 4/23/14																																																																																																																																																				



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28654

BRANDI TERRY

Date: 6/23/2014

OASIS PETROLEUM NORTH AMERICA LLC

1001 FANNIN STE 1500

HOUSTON, TX 77002 USA

RE: CORES AND SAMPLES

Well Name: **KLINE FEDERAL 5300 41-18 13T2X** Well File No.: **28654**

Location: **LOT4 18-153-100** County: MCKENZIE

Permit Type: **Development - HORIZONTAL**

Field: **BAKER** Target Horizon: **THREE FORKS B2**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (08-2006)



Well File No.

28654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number

Kline Federal 5300 41-18 13T2X

Footages	Qtr-Qtr	Section	Township	Range
401 F S L 237 F W L	<u>Lot 4</u>	18	153 N	100 W
Field	Pool	County		
	Bakken	McKenzie		

24-HOUR PRODUCTION RATE

Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)

Address	City	State	Zip Code
---------	------	-------	----------

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum's #20275 Kline Federal 5300 11-18H (153N 100W 18 NW NW) located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company	Telephone Number	
Oasis Petroleum North America LLC	281-404-9563	
Address		
1001 Fannin, Suite 1500		
City	State	Zip Code
Houston	TX	77002
Signature	Printed Name	
<u>Heather McCowan</u>	Heather McCowan	
Title	Date	
Regulatory Assistant	May 29, 2014	
Email Address		
hmccowan@oasispetroleum.com		

FOR STATE USE ONLY

☐ Received	☒ Approved
Date	
<u>6-17-2014</u>	
By	
<u>Stephen Fried</u>	
Title	
Stephen Fried	
Geologist	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Kline Federal 5300 41-18 13T2X					
Footages	Qtr-Qtr	Section	Township	Range	
401 F S L 237 F WL	SWSW	18	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

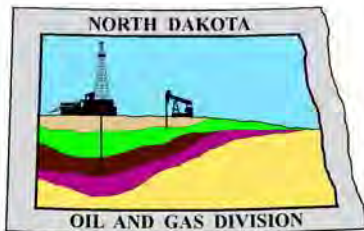
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector Richard Dunn at 701-770-3554 with spud and TD info.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date May 29, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	6/17/14
By	Nathaniel Erbele
Title	Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 17, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
KLINE FEDERAL 5300 41-18 13T2X
LOT4 Section 18-153N-100W
McKenzie County
Well File # 28654**

Dear Heather :

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement shall be located approximately down the east-west axis (**within 250' of the section line** per Commission policy) and no closer than **200 feet** to the east or west boundary within the 2560 acre spacing unit consisting of Sections 17, 18, 19, & 20 T153N R100W.

PERMIT STIPULATIONS: Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. Due to the proximity of Lake Sakakawea to the well site, a dike is required surrounding the entire location. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 151' south. Also, based on the azimuth of the proposed lateral the maximum legal coordinates from the well head are: 10052' east and 651' south.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele

Petroleum Resource Specialist



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 10 / 1 / 2013	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name KLINE FEDERAL				Well Number 5300 41-18 13T2X			
Surface Footages 401 F S L 237 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 168 F S L 652 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 233 S From WH 415 E From WH		Azimuth 119.3 °	Longstring Total Depth 11141 Feet MD 10870 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F N L 202 F E L		Qtr-Qtr LOT2	Section 20	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 601 S From WH 10050 E From WH		KOP Lateral 1 10393 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20839 Feet MD 10886 Feet TVD		
Latitude of Well Head 48 ° 04 ' 7.22 "	Longitude of Well Head -103 ° 36 ' 11.95 "	NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sections 17, 18, 19, & 20 T153N R1				
Ground Elevation 2056 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 0 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10544 Feet	South Line of Spacing/Drilling Unit 10489 Feet	East Line of Spacing/Drilling Unit 10524 Feet			West Line of Spacing/Drilling Unit 10520 Feet		
Objective Horizons Three Forks B2					Pierre Shale Top 1972		
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2072 Feet	Cement Volume 749 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11141 Feet MD 10870 Feet TVD		Cement Volume 764 Sacks	Cement Top 3971 Feet	Top Dakota Sand 5471 Feet
Base Last Charles Salt (If Applicable) 9238 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD Feet TVD		KOP Coordinates From Well Head From WH From WH	
Formation Entry Point Coordinates From Well Head From WH From WH		Bottom Hole Coordinates From Well Head From WH From WH			
KOP Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County
Bottom Hole Footages From Nearest Section Line F L F L		Qtr-Qtr	Section	Township N	Range W County

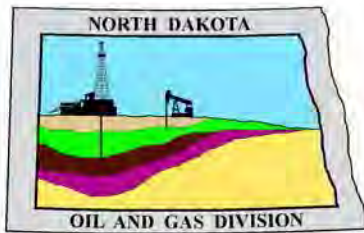
I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.			Date 5 / 29 / 2014
ePermit	Printed Name Heather McCowan	Title Regulatory Assistant	

FOR STATE USE ONLY

Permit and File Number 28654	API Number 33 - 053 - 06028
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 17 / 2014
By Nathaniel Erbele
Title Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

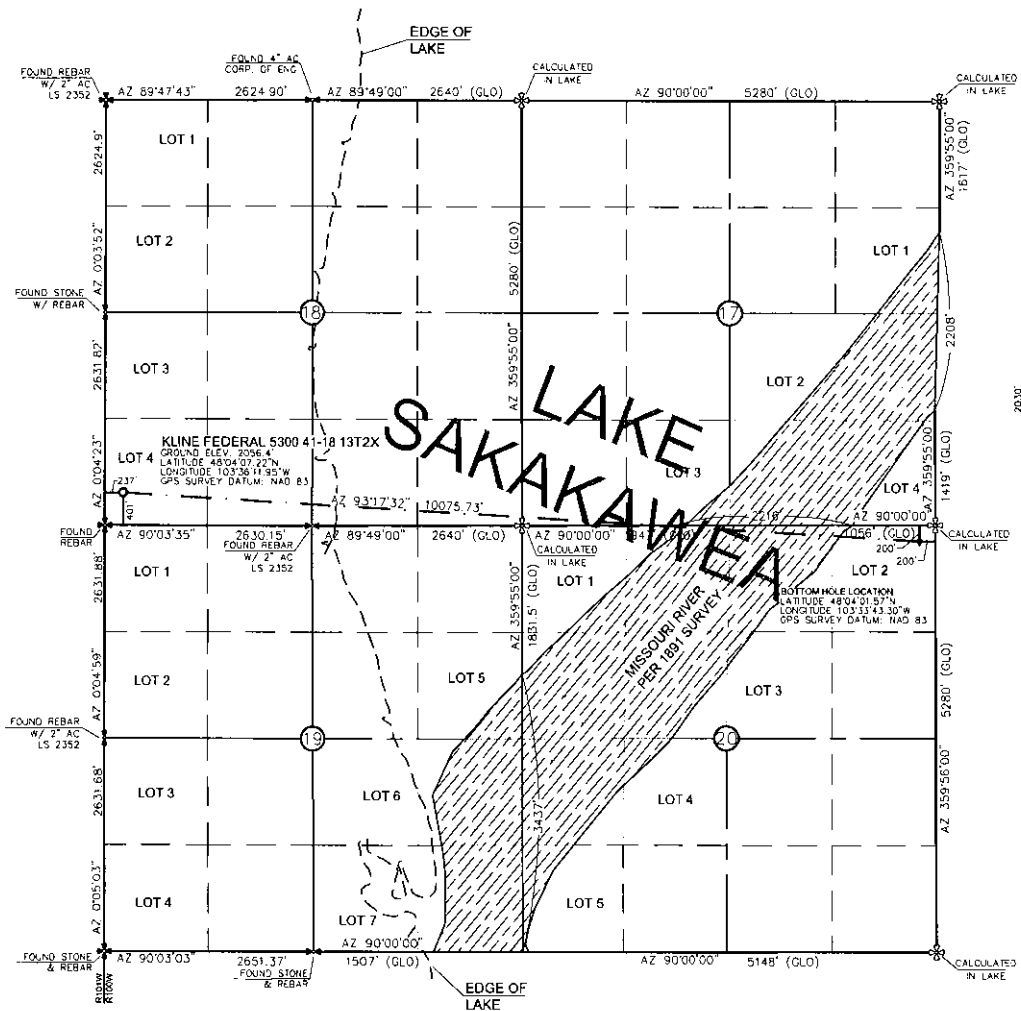
Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

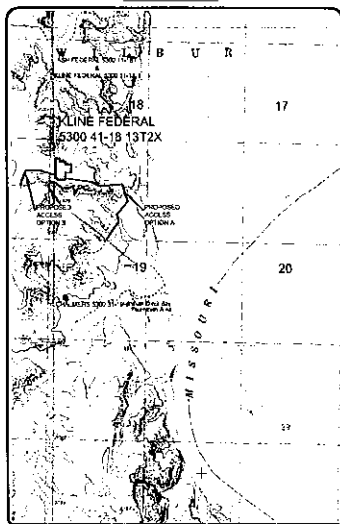
Assistant Director

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 13T2X"
 401 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

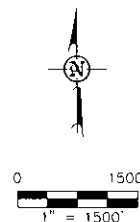


- MONUMENT - RECOVERED
- MONUMENT - NOT RECOVERED

VICINITY MAP



THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY ROBERT L. PROCIVE,
 PLS. REGISTRATION NUMBER 2884 ON
 4/15/14, AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
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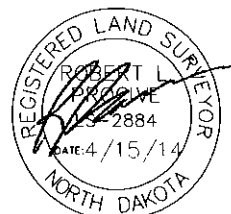


STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF

Robert L. Procive
 ROBERT L. PROCIVE 2884LS

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1/8



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59720
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com
 Other Offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: 514609-081.01
 Checked By: R.L.P. Date: APRIL 2014

Revised No.	Date	By	Description

DRILLING PLAN

OPERATOR

WELL NAME

WELL TYPE

LOCATION

EST. T.D.

Oasis Petroleum

Kline Federal 5300 41-18 13T2X

Horizontal Three Forks

SWSW 18-153N-100W

20,840'

COUNTY/STATE

RIG

McKenzie Co., ND

0

TOTAL LATERAL:

9,698' (est)

GROUND ELEV:

2057

Finished Pad Elev.

Sub Height:

25

KB ELEV:

2082

PROGNOSIS:

Based on 2,082" KB(est)

LOGS:

Type

Interval

OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC;
GR to surf; CND through the Dakota
CBL/GR: Above top of cement/GR to base of casing
MWD GR: KOP to lateral TD

DEVIATION:

Surf:

3 deg. max., 1 deg / 100'; srvy every 500'

Prod:

5 deg. max., 1 deg / 100'; srvy every 100'

DST'S:

None planned

CORES:

None planned

MUDLOGGING:

Two-Man:

8,342'

~200' above the Charles (Kibbey) to
Casing point; Casing point to TD
30' samples at direction of wellsite geologist; 10' through target @
curve land

BOP:

11" 5000 psi blind, pipe & annular

Dip Rate:

-0.1

Max. Anticipated BHP:

4714

Surface Formation:

Glacial till

MUD:

Interval

Type

WT

Vis

WL

Remarks

Surface:

0' - 2,072'

FW/Gel - Lime Sweeps

8.4-9.0

28-32

NC

Circ Mud Tanks

Intermediate:

2,072' - 11,142'

Invert

9.5-10.4

40-50

30+HtHp

Circ Mud Tanks

Lateral:

11,142' - 20,840'

Salt Water

9.8-10.2

28-32

NC

Circ Mud Tanks

CASING:

Size

Wt ppf

Hole

Depth

Cement

WOC

Remarks

Surface:

9 5/8"

36#

13.5"

2,072'

To Surface

12

100' into Pierre

Intermediate:

7"

29#

8.75"

11,142'

3,971'

24

1500' above Dakota

Production Liner:

4.5"

13.5#

6"

20,840'

TOL @ 10,343'

50' above KOP

PROBABLE PLUGS, IF REQ'D:

OTHER:

MD

TVD

FNL/FSL

FEL/FWL

S-T-R

AZI

Surface:

2,072

2,072

401' FSL

237' FWL

SEC 18-T153N-R100W

Survey Company:

KOP:

10,393'

10,393'

401' FSL

237' FWL

SEC 18-T153N-R100W

Build Rate:

12 deg /100'

EOC:

11,142'

10,870'

167' FSL

653' FWL

SEC 18-T153N-R100W

Casing Point:

11,142'

10,870'

167' FSL

653' FWL

SEC 18-T153N-R100W

Three Forks Lateral TD:

20,840'

10,887'

200' FNL

200' FEL

SEC 20-T153N-R100W

Comments:

Request Sundry to Waive Open Hole Logs

Exception well: Oasis Petroleum's Kline 5300 11-18H

35 packers, 35 sleeves, no frac string

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)

68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)

OASIS

PETROLEUM

Geology: M.Steed 4/23/14

Engineering: hbadler rpm

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 2,072'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,072'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.08	3520 / 3.63	453 / 2.76	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 64k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **449 sks** (232 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl₂ and 0.25 lb/sk lost circulation control agent

Tail Slurry: **300 sks** (62 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and 0.25 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,614'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,614' – 10,393'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,393' – 11,142'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,614'	6,614'	7", 29#, P-110, LTC, 8rd	8530 / 2.48*	11220 / 1.19	797 / 2.09
6,614' – 10,393'	3,779'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.19*	12460 / 1.29	
6,614' – 10,393'	3,779'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,393' – 11,142'	749'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.16	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside
 **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals. (Bottom of last salt 9,238' TVD)
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,870' TVD.
- c) Based on string weight in 10 ppg fluid, (283k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **170 bbls** Saltwater
 20 bbls CW8 System
 10 bbls Fresh Water

Lead Slurry: **177 sks** (82 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.7% retarder, 0.125 lb/sk lost circulation control agent, and 0.3% dispersant

Tail Slurry: **587 sks** (172 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant, 0.125 lb/sk lost circulation control agent and 0.3% retarder

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 13T2X
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10,343' – 20,840'	11.6	P-110	BTC	4.000"	3.875"			

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
10,343' – 20,840'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7560 / 1.41	10690 / 1.10	385 / 1.88	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,870' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,870' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 104k lbs.) plus 100k lbs overpull.



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 13T2X

Kline Federal 5300 41-18 13T2X

Plan: Design #1

Standard Planning Report

23 May, 2014



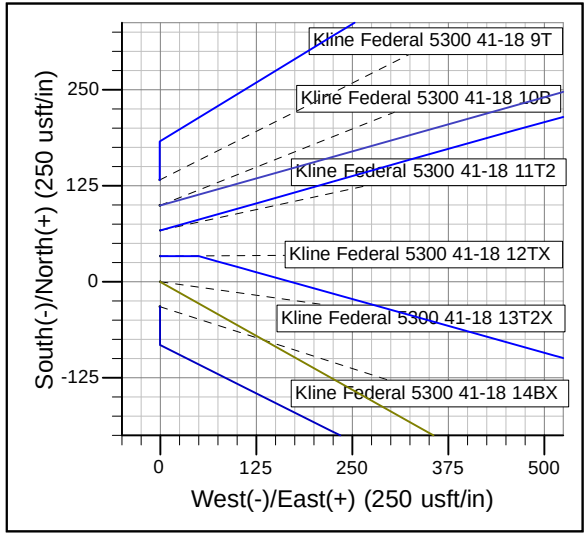
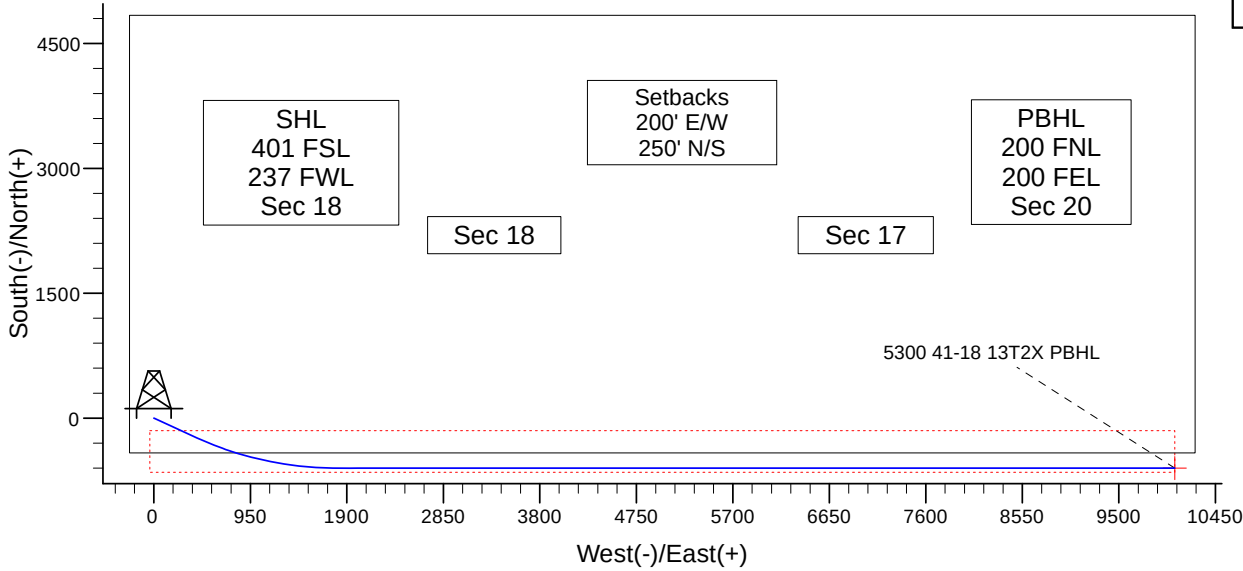
**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY

Project: Indian Hills
 Site: 153N-100W-17/18
 Well: Kline Federal 5300 41-18 13T2X
 Wellbore: Kline Federal 5300 41-18 13T2X
 Design: Design #1

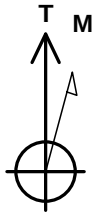


WELL DETAILS: Kline Federal 5300 41-18 13T2X

Ground Level: 2057.0
 Northing 405103.22 Easting 1209966.87 Latittude 48° 4' 7.220 N Longitude 103° 36' 11.950 W

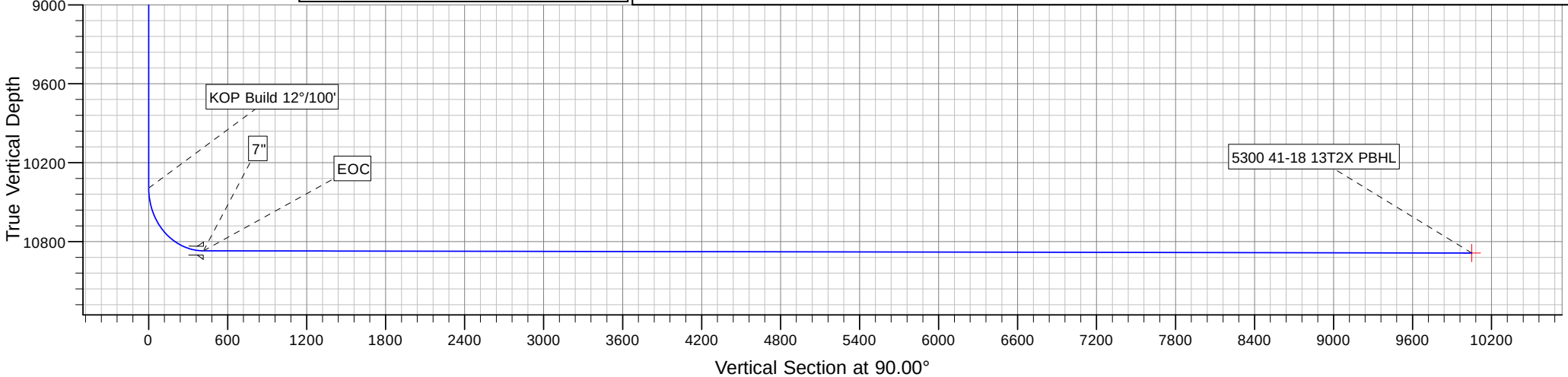


CASING DETAILS			
TVD	MD	Name	Size
10870.0	11142.0	7"	7



Azimuths to True North
 Magnetic North: 8.27°
 Magnetic Field
 Strength: 56437.0snT
 Dip Angle: 72.99°
 Date: 5/23/2014
 Model: IGRF2010

SECTION DETAILS								
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target	
0.0	0.00	0.00	0.0	0.0	0.0	0.00		
10392.5	0.00	0.00	10392.5	0.0	0.0	0.00		
11141.7	89.90	119.35	10870.0	-233.6	415.5	12.00		
12609.2	89.90	90.00	10872.6	-601.3	1819.6	2.00		
20839.6	89.90	90.00	10886.9	-601.3	10050.0	0.00	5300 41-18 13T2X PBHL	



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-17/18			
Site Position:		Northing:	408,962.44 usft	Latitude:	48° 4' 45.380 N
From:	Lat/Long	Easting:	1,210,229.18 usft	Longitude:	103° 36' 10.380 W
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 13T2X					
Well Position	+N/-S	-3,866.7 usft	Northing:	405,103.22 usft	Latitude:	48° 4' 7.220 N
	+E/-W	-106.6 usft	Easting:	1,209,966.86 usft	Longitude:	103° 36' 11.950 W
Position Uncertainty		2.0 usft	Wellhead Elevation:		Ground Level:	2,057.0 usft

Wellbore	Kline Federal 5300 41-18 13T2X				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #1			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
10,392.5	0.00	0.00	10,392.5	0.0	0.0	0.00	0.00	0.00	0.00	
11,141.7	89.90	119.35	10,870.0	-233.6	415.5	12.00	12.00	0.00	119.35	
12,609.2	89.90	90.00	10,872.6	-601.3	1,819.6	2.00	0.00	-2.00	-90.03	
20,839.6	89.90	90.00	10,886.9	-601.3	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 13T2X PE

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
10,392.5	0.00	0.00	10,392.5	0.0	0.0	0.0	0.00	0.00	0.00	
KOP Build 12°/100'										
10,400.0	0.90	119.35	10,400.0	0.0	0.1	0.1	12.00	12.00	0.00	
10,425.0	3.90	119.35	10,425.0	-0.5	1.0	1.0	12.00	12.00	0.00	
10,450.0	6.90	119.35	10,449.9	-1.7	3.0	3.0	12.00	12.00	0.00	
10,475.0	9.90	119.35	10,474.6	-3.5	6.2	6.2	12.00	12.00	0.00	
10,500.0	12.90	119.35	10,499.1	-5.9	10.5	10.5	12.00	12.00	0.00	
10,525.0	15.90	119.35	10,523.3	-9.0	15.9	15.9	12.00	12.00	0.00	
10,550.0	18.90	119.35	10,547.2	-12.6	22.4	22.4	12.00	12.00	0.00	
10,575.0	21.90	119.35	10,570.6	-16.9	30.0	30.0	12.00	12.00	0.00	
10,600.0	24.90	119.35	10,593.5	-21.8	38.7	38.7	12.00	12.00	0.00	
10,625.0	27.90	119.35	10,615.9	-27.2	48.4	48.4	12.00	12.00	0.00	
10,650.0	30.90	119.35	10,637.7	-33.2	59.1	59.1	12.00	12.00	0.00	
10,675.0	33.90	119.35	10,658.8	-39.8	70.7	70.7	12.00	12.00	0.00	
10,700.0	36.90	119.35	10,679.2	-46.9	83.4	83.4	12.00	12.00	0.00	
10,725.0	39.90	119.35	10,698.8	-54.5	96.9	96.9	12.00	12.00	0.00	
10,750.0	42.90	119.35	10,717.5	-62.6	111.3	111.3	12.00	12.00	0.00	
10,761.7	44.31	119.35	10,726.0	-66.6	118.3	118.3	12.00	12.00	0.00	
False Bakken										
10,774.5	45.83	119.35	10,735.0	-71.0	126.2	126.2	12.00	12.00	0.00	
Upper Bakken										
10,775.0	45.90	119.35	10,735.4	-71.2	126.6	126.6	12.00	12.00	0.00	
10,798.0	48.66	119.35	10,751.0	-79.5	141.3	141.3	12.00	12.00	0.00	
Middle Bakken										
10,800.0	48.90	119.35	10,752.3	-80.2	142.6	142.6	12.00	12.00	0.00	
10,825.0	51.90	119.35	10,768.2	-89.6	159.4	159.4	12.00	12.00	0.00	
10,850.0	54.90	119.35	10,783.1	-99.5	176.9	176.9	12.00	12.00	0.00	
10,853.3	55.29	119.35	10,785.0	-100.8	179.2	179.2	12.00	12.00	0.00	
Lower Bakken										
10,864.0	56.58	119.35	10,791.0	-105.1	186.9	186.9	12.00	12.00	0.00	
Pronghorn										
10,875.0	57.90	119.35	10,797.0	-109.7	195.0	195.0	12.00	12.00	0.00	
10,900.0	60.90	119.35	10,809.7	-120.2	213.8	213.8	12.00	12.00	0.00	
10,906.9	61.73	119.35	10,813.0	-123.2	219.0	219.0	12.00	12.00	0.00	
Three Forks 1st Bench										
10,925.0	63.90	119.35	10,821.3	-131.1	233.1	233.1	12.00	12.00	0.00	
10,950.0	66.90	119.35	10,831.7	-142.2	252.9	252.9	12.00	12.00	0.00	
10,966.8	68.92	119.35	10,838.0	-149.8	266.5	266.5	12.00	12.00	0.00	
Three Forks 1st Bench Claystone										
10,975.0	69.90	119.35	10,840.9	-153.6	273.2	273.2	12.00	12.00	0.00	
11,000.0	72.90	119.35	10,848.9	-165.2	293.8	293.8	12.00	12.00	0.00	
11,007.5	73.80	119.35	10,851.0	-168.7	300.0	300.0	12.00	12.00	0.00	
Three Forks 2nd Bench										
11,025.0	75.90	119.35	10,855.6	-177.0	314.8	314.8	12.00	12.00	0.00	
11,050.0	78.90	119.35	10,861.0	-189.0	336.1	336.1	12.00	12.00	0.00	
11,075.0	81.90	119.35	10,865.2	-201.1	357.5	357.5	12.00	12.00	0.00	
11,100.0	84.90	119.35	10,868.1	-213.2	379.2	379.2	12.00	12.00	0.00	
11,125.0	87.90	119.35	10,869.6	-225.5	400.9	400.9	12.00	12.00	0.00	
11,141.7	89.90	119.35	10,870.0	-233.6	415.5	415.5	11.98	11.98	0.00	
EOC										
11,142.0	89.90	119.34	10,870.0	-233.8	415.7	415.7	2.22	0.00	-2.22	
7"										
11,200.0	89.90	118.18	10,870.1	-261.7	466.6	466.6	2.00	0.00	-2.00	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
11,300.0	89.90	116.18	10,870.2	-307.4	555.5	555.5	2.00	0.00	-2.00
11,400.0	89.90	114.18	10,870.4	-349.9	646.0	646.0	2.00	0.00	-2.00
11,500.0	89.90	112.18	10,870.6	-389.3	737.9	737.9	2.00	0.00	-2.00
11,600.0	89.90	110.18	10,870.8	-425.4	831.2	831.2	2.00	0.00	-2.00
11,700.0	89.90	108.18	10,871.0	-458.3	925.6	925.6	2.00	0.00	-2.00
11,800.0	89.90	106.18	10,871.1	-487.8	1,021.2	1,021.2	2.00	0.00	-2.00
11,900.0	89.90	104.18	10,871.3	-514.0	1,117.7	1,117.7	2.00	0.00	-2.00
12,000.0	89.90	102.18	10,871.5	-536.8	1,215.0	1,215.0	2.00	0.00	-2.00
12,100.0	89.90	100.18	10,871.7	-556.2	1,313.1	1,313.1	2.00	0.00	-2.00
12,200.0	89.90	98.18	10,871.9	-572.2	1,411.8	1,411.8	2.00	0.00	-2.00
12,300.0	89.90	96.18	10,872.0	-584.7	1,511.0	1,511.0	2.00	0.00	-2.00
12,400.0	89.90	94.18	10,872.2	-593.7	1,610.6	1,610.6	2.00	0.00	-2.00
12,500.0	89.90	92.18	10,872.4	-599.3	1,710.5	1,710.5	2.00	0.00	-2.00
12,609.2	89.90	90.00	10,872.6	-601.3	1,819.6	1,819.6	2.00	0.00	-2.00
12,700.0	89.90	90.00	10,872.7	-601.3	1,910.4	1,910.4	0.00	0.00	0.00
12,800.0	89.90	90.00	10,872.9	-601.3	2,010.4	2,010.4	0.00	0.00	0.00
12,900.0	89.90	90.00	10,873.1	-601.3	2,110.4	2,110.4	0.00	0.00	0.00
13,000.0	89.90	90.00	10,873.3	-601.3	2,210.4	2,210.4	0.00	0.00	0.00
13,100.0	89.90	90.00	10,873.4	-601.3	2,310.4	2,310.4	0.00	0.00	0.00
13,200.0	89.90	90.00	10,873.6	-601.3	2,410.4	2,410.4	0.00	0.00	0.00
13,300.0	89.90	90.00	10,873.8	-601.3	2,510.4	2,510.4	0.00	0.00	0.00
13,400.0	89.90	90.00	10,874.0	-601.3	2,610.4	2,610.4	0.00	0.00	0.00
13,500.0	89.90	90.00	10,874.1	-601.3	2,710.4	2,710.4	0.00	0.00	0.00
13,600.0	89.90	90.00	10,874.3	-601.3	2,810.4	2,810.4	0.00	0.00	0.00
13,700.0	89.90	90.00	10,874.5	-601.3	2,910.4	2,910.4	0.00	0.00	0.00
13,800.0	89.90	90.00	10,874.7	-601.3	3,010.4	3,010.4	0.00	0.00	0.00
13,900.0	89.90	90.00	10,874.8	-601.3	3,110.4	3,110.4	0.00	0.00	0.00
14,000.0	89.90	90.00	10,875.0	-601.3	3,210.4	3,210.4	0.00	0.00	0.00
14,100.0	89.90	90.00	10,875.2	-601.3	3,310.4	3,310.4	0.00	0.00	0.00
14,200.0	89.90	90.00	10,875.4	-601.3	3,410.4	3,410.4	0.00	0.00	0.00
14,300.0	89.90	90.00	10,875.5	-601.3	3,510.4	3,510.4	0.00	0.00	0.00
14,400.0	89.90	90.00	10,875.7	-601.3	3,610.4	3,610.4	0.00	0.00	0.00
14,500.0	89.90	90.00	10,875.9	-601.3	3,710.4	3,710.4	0.00	0.00	0.00
14,600.0	89.90	90.00	10,876.1	-601.3	3,810.4	3,810.4	0.00	0.00	0.00
14,700.0	89.90	90.00	10,876.2	-601.3	3,910.4	3,910.4	0.00	0.00	0.00
14,800.0	89.90	90.00	10,876.4	-601.3	4,010.4	4,010.4	0.00	0.00	0.00
14,900.0	89.90	90.00	10,876.6	-601.3	4,110.4	4,110.4	0.00	0.00	0.00
15,000.0	89.90	90.00	10,876.8	-601.3	4,210.4	4,210.4	0.00	0.00	0.00
15,100.0	89.90	90.00	10,876.9	-601.3	4,310.4	4,310.4	0.00	0.00	0.00
15,200.0	89.90	90.00	10,877.1	-601.3	4,410.4	4,410.4	0.00	0.00	0.00
15,300.0	89.90	90.00	10,877.3	-601.3	4,510.4	4,510.4	0.00	0.00	0.00
15,400.0	89.90	90.00	10,877.5	-601.3	4,610.4	4,610.4	0.00	0.00	0.00
15,500.0	89.90	90.00	10,877.6	-601.3	4,710.4	4,710.4	0.00	0.00	0.00
15,600.0	89.90	90.00	10,877.8	-601.3	4,810.4	4,810.4	0.00	0.00	0.00
15,700.0	89.90	90.00	10,878.0	-601.3	4,910.4	4,910.4	0.00	0.00	0.00
15,712.9	89.90	90.00	10,878.0	-601.3	4,923.4	4,923.4	0.00	0.00	0.00
Three Forks 2nd Bench Claystone									
15,800.0	89.90	90.00	10,878.2	-601.3	5,010.4	5,010.4	0.00	0.00	0.00
15,900.0	89.90	90.00	10,878.3	-601.3	5,110.4	5,110.4	0.00	0.00	0.00
16,000.0	89.90	90.00	10,878.5	-601.3	5,210.4	5,210.4	0.00	0.00	0.00
16,100.0	89.90	90.00	10,878.7	-601.3	5,310.4	5,310.4	0.00	0.00	0.00
16,200.0	89.90	90.00	10,878.8	-601.3	5,410.4	5,410.4	0.00	0.00	0.00
16,300.0	89.90	90.00	10,879.0	-601.3	5,510.4	5,510.4	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
16,400.0	89.90	90.00	10,879.2	-601.3	5,610.4	5,610.4	0.00	0.00	0.00
16,500.0	89.90	90.00	10,879.4	-601.3	5,710.4	5,710.4	0.00	0.00	0.00
16,600.0	89.90	90.00	10,879.5	-601.3	5,810.4	5,810.4	0.00	0.00	0.00
16,700.0	89.90	90.00	10,879.7	-601.3	5,910.4	5,910.4	0.00	0.00	0.00
16,800.0	89.90	90.00	10,879.9	-601.3	6,010.4	6,010.4	0.00	0.00	0.00
16,900.0	89.90	90.00	10,880.1	-601.3	6,110.4	6,110.4	0.00	0.00	0.00
17,000.0	89.90	90.00	10,880.2	-601.3	6,210.4	6,210.4	0.00	0.00	0.00
17,100.0	89.90	90.00	10,880.4	-601.3	6,310.4	6,310.4	0.00	0.00	0.00
17,200.0	89.90	90.00	10,880.6	-601.3	6,410.4	6,410.4	0.00	0.00	0.00
17,300.0	89.90	90.00	10,880.8	-601.3	6,510.4	6,510.4	0.00	0.00	0.00
17,400.0	89.90	90.00	10,880.9	-601.3	6,610.4	6,610.4	0.00	0.00	0.00
17,500.0	89.90	90.00	10,881.1	-601.3	6,710.4	6,710.4	0.00	0.00	0.00
17,600.0	89.90	90.00	10,881.3	-601.3	6,810.4	6,810.4	0.00	0.00	0.00
17,700.0	89.90	90.00	10,881.5	-601.3	6,910.4	6,910.4	0.00	0.00	0.00
17,800.0	89.90	90.00	10,881.6	-601.3	7,010.4	7,010.4	0.00	0.00	0.00
17,900.0	89.90	90.00	10,881.8	-601.3	7,110.4	7,110.4	0.00	0.00	0.00
18,000.0	89.90	90.00	10,882.0	-601.3	7,210.4	7,210.4	0.00	0.00	0.00
18,100.0	89.90	90.00	10,882.2	-601.3	7,310.4	7,310.4	0.00	0.00	0.00
18,200.0	89.90	90.00	10,882.3	-601.3	7,410.4	7,410.4	0.00	0.00	0.00
18,300.0	89.90	90.00	10,882.5	-601.3	7,510.4	7,510.4	0.00	0.00	0.00
18,400.0	89.90	90.00	10,882.7	-601.3	7,610.4	7,610.4	0.00	0.00	0.00
18,500.0	89.90	90.00	10,882.9	-601.3	7,710.4	7,710.4	0.00	0.00	0.00
18,600.0	89.90	90.00	10,883.0	-601.3	7,810.4	7,810.4	0.00	0.00	0.00
18,700.0	89.90	90.00	10,883.2	-601.3	7,910.4	7,910.4	0.00	0.00	0.00
18,800.0	89.90	90.00	10,883.4	-601.3	8,010.4	8,010.4	0.00	0.00	0.00
18,900.0	89.90	90.00	10,883.6	-601.3	8,110.4	8,110.4	0.00	0.00	0.00
19,000.0	89.90	90.00	10,883.7	-601.3	8,210.4	8,210.4	0.00	0.00	0.00
19,100.0	89.90	90.00	10,883.9	-601.3	8,310.4	8,310.4	0.00	0.00	0.00
19,200.0	89.90	90.00	10,884.1	-601.3	8,410.4	8,410.4	0.00	0.00	0.00
19,300.0	89.90	90.00	10,884.3	-601.3	8,510.4	8,510.4	0.00	0.00	0.00
19,400.0	89.90	90.00	10,884.4	-601.3	8,610.4	8,610.4	0.00	0.00	0.00
19,500.0	89.90	90.00	10,884.6	-601.3	8,710.4	8,710.4	0.00	0.00	0.00
19,600.0	89.90	90.00	10,884.8	-601.3	8,810.4	8,810.4	0.00	0.00	0.00
19,700.0	89.90	90.00	10,885.0	-601.3	8,910.4	8,910.4	0.00	0.00	0.00
19,800.0	89.90	90.00	10,885.1	-601.3	9,010.4	9,010.4	0.00	0.00	0.00
19,900.0	89.90	90.00	10,885.3	-601.3	9,110.4	9,110.4	0.00	0.00	0.00
20,000.0	89.90	90.00	10,885.5	-601.3	9,210.4	9,210.4	0.00	0.00	0.00
20,100.0	89.90	90.00	10,885.7	-601.3	9,310.4	9,310.4	0.00	0.00	0.00
20,200.0	89.90	90.00	10,885.8	-601.3	9,410.4	9,410.4	0.00	0.00	0.00
20,300.0	89.90	90.00	10,886.0	-601.3	9,510.4	9,510.4	0.00	0.00	0.00
20,400.0	89.90	90.00	10,886.2	-601.3	9,610.4	9,610.4	0.00	0.00	0.00
20,500.0	89.90	90.00	10,886.4	-601.3	9,710.4	9,710.4	0.00	0.00	0.00
20,600.0	89.90	90.00	10,886.5	-601.3	9,810.4	9,810.4	0.00	0.00	0.00
20,700.0	89.90	90.00	10,886.7	-601.3	9,910.4	9,910.4	0.00	0.00	0.00
20,800.0	89.90	90.00	10,886.9	-601.3	10,010.4	10,010.4	0.00	0.00	0.00
20,839.6	89.90	90.00	10,886.9	-601.3	10,050.0	10,050.0	0.00	0.00	0.00
5300 41-18 13T2X PBHL									



Ryan Directional Services

Planning Report



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

Design Targets									
Target Name									
- hit/miss target	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- Shape	(°)	(°)	(usft)	(usft)	(usft)	(usft)	(usft)		
5300 41-18 13T2X PBHI	0.00	0.00	10,886.8	-601.0	10,050.0	404,097.76	1,219,984.49	48° 4' 1.262 N	103° 33' 43.975 W
- plan misses target center by 0.4usft at 20839.6usft MD (10886.9 TVD, -601.3 N, 10050.0 E)									
- Point									

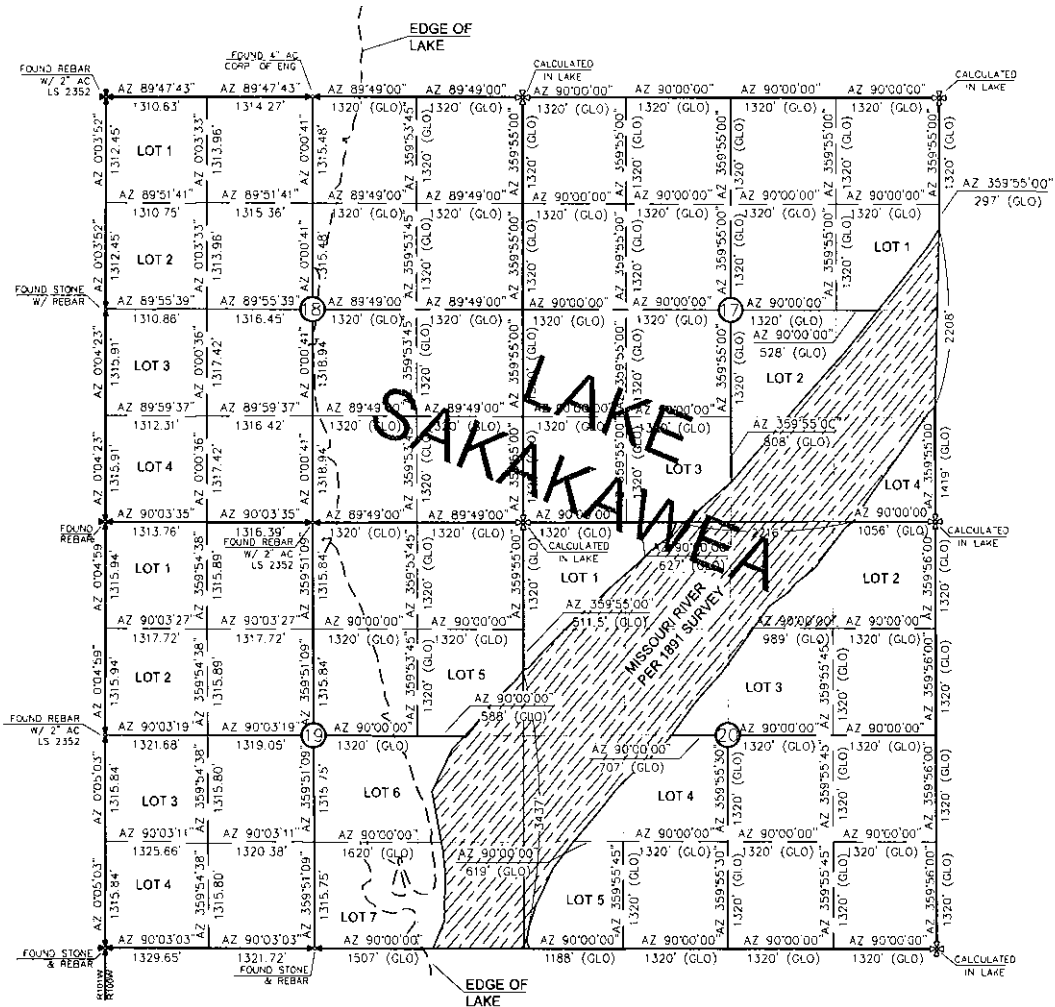
Casing Points					
	Measured Depth	Vertical Depth			
	(usft)	(usft)	Name	Casing Diameter	Hole Diameter
				(")	(")
	11,142.0	10,870.0	7"	7	8-3/4

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 13T2X
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 13T2X	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 13T2X		
Design:	Design #1		

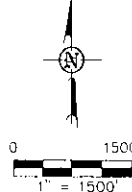
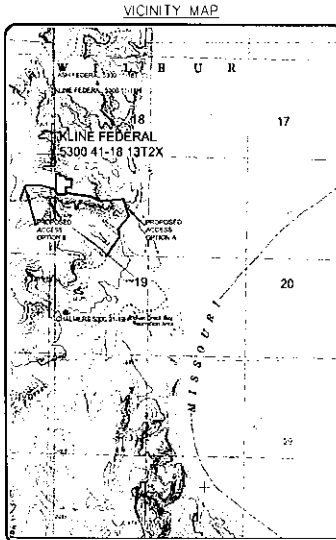
Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,972.0	1,972.0	Pierre				
4,646.0	4,646.0	Greenhorn				
5,057.0	5,057.0	Mowry				
5,471.0	5,471.0	Dakota				
6,486.0	6,486.0	Rierdon				
6,814.0	6,814.0	Dunham Salt				
6,929.0	6,929.0	Dunham Salt Base				
7,024.0	7,024.0	Spearfish				
7,283.0	7,283.0	Pine Salt				
7,318.0	7,318.0	Pine Salt Base				
7,374.0	7,374.0	Opeche Salt				
7,403.0	7,403.0	Opeche Salt Base				
7,605.0	7,605.0	Broom Creek (Top of Minnelusa Gp.)				
7,684.0	7,684.0	Amsden				
7,853.0	7,853.0	Tyler				
8,047.0	8,047.0	Otter (Base of Minnelusa Gp.)				
8,390.0	8,390.0	Kibbey				
8,542.0	8,542.0	Charles Salt				
9,159.0	9,159.0	UB				
9,238.0	9,238.0	Base Last Salt				
9,301.0	9,301.0	Ratcliffe				
9,454.0	9,454.0	Mission Canyon				
10,022.0	10,022.0	Lodgepole				
10,209.0	10,209.0	Lodgepole Fracture Zone				
10,761.7	10,726.0	False Bakken				
10,774.5	10,735.0	Upper Bakken				
10,798.0	10,751.0	Middle Bakken				
10,853.3	10,785.0	Lower Bakken				
10,864.0	10,791.0	Pronghorn				
10,906.9	10,813.0	Three Forks 1st Bench				
10,966.8	10,838.0	Three Forks 1st Bench Claystone				
11,007.5	10,851.0	Three Forks 2nd Bench				
15,712.9	10,878.0	Three Forks 2nd Bench Claystone				

Plan Annotations					
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates			
		+N/-S (usft)	+E/-W (usft)	Comment	
10,392.5	10,392.5	0.0	0.0	KOP Build 12°/100'	
11,141.7	10,870.0	-233.6	415.5	EOC	

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 13T2X"
 401 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

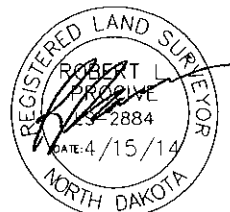


- ⊕ - MONUMENT - RECOVERED
- ⊗ - MONUMENT - NOT RECOVERED



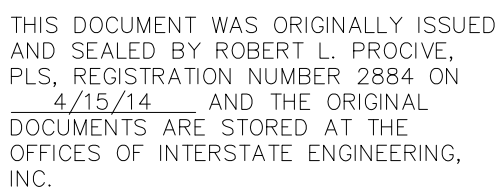
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ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY 0°03'.

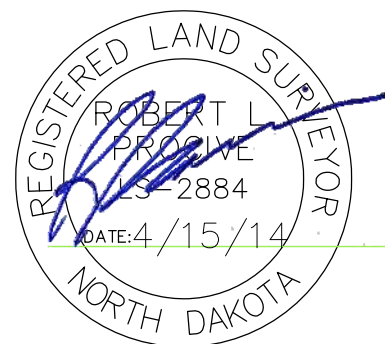


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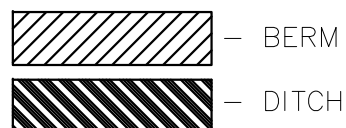
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 9T"
533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

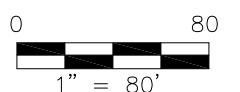
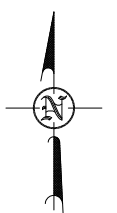


———— Proposed Contours
- - - - - Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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SHEET NO.



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Ph (406) 433-5617
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www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

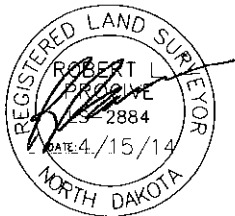
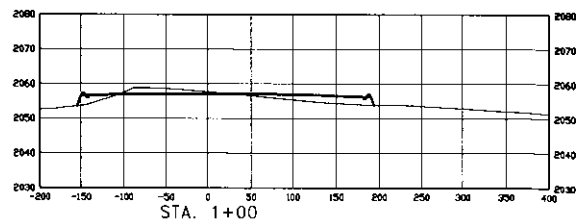
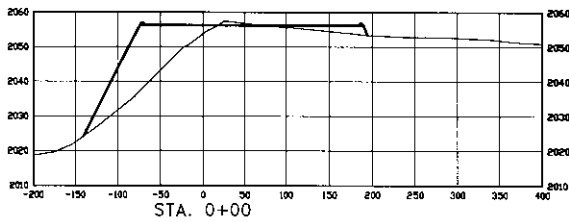
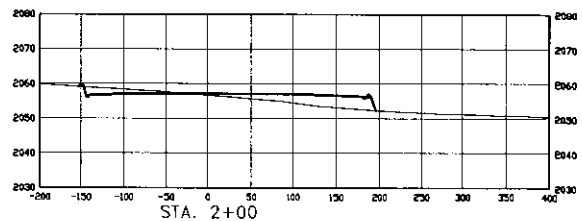
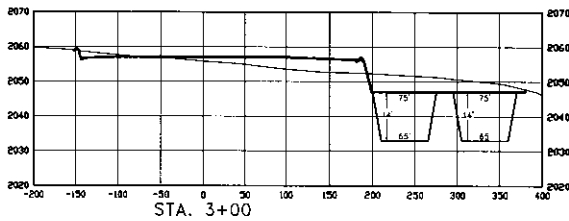
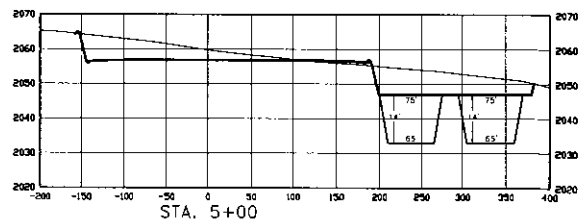
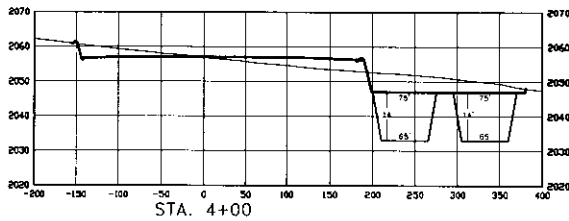
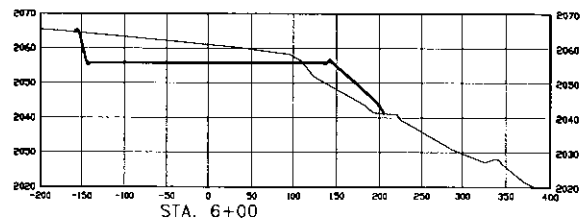
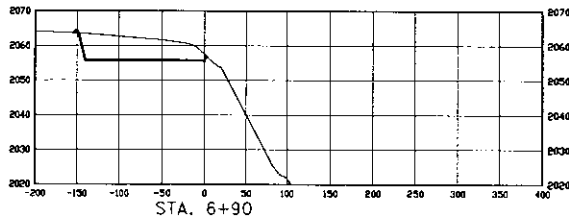
Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081.05</u>
Checked By: <u>D.D.K.</u>	Date: <u>APRIL 2014</u>

Revision No.	Date	By	Description

© 2014/04-09-08/05 Doss Petroleum -- Kline Federal 5300 41-18 91

Q:\2014\S14-09-081.05 Oasis Petroleum - Kline Federal 5300 41-18 9T
Location\CAD\Kline Federal 5300 41-18 9T.dwg - 4/15/2014 3:59 PM josh schmierer

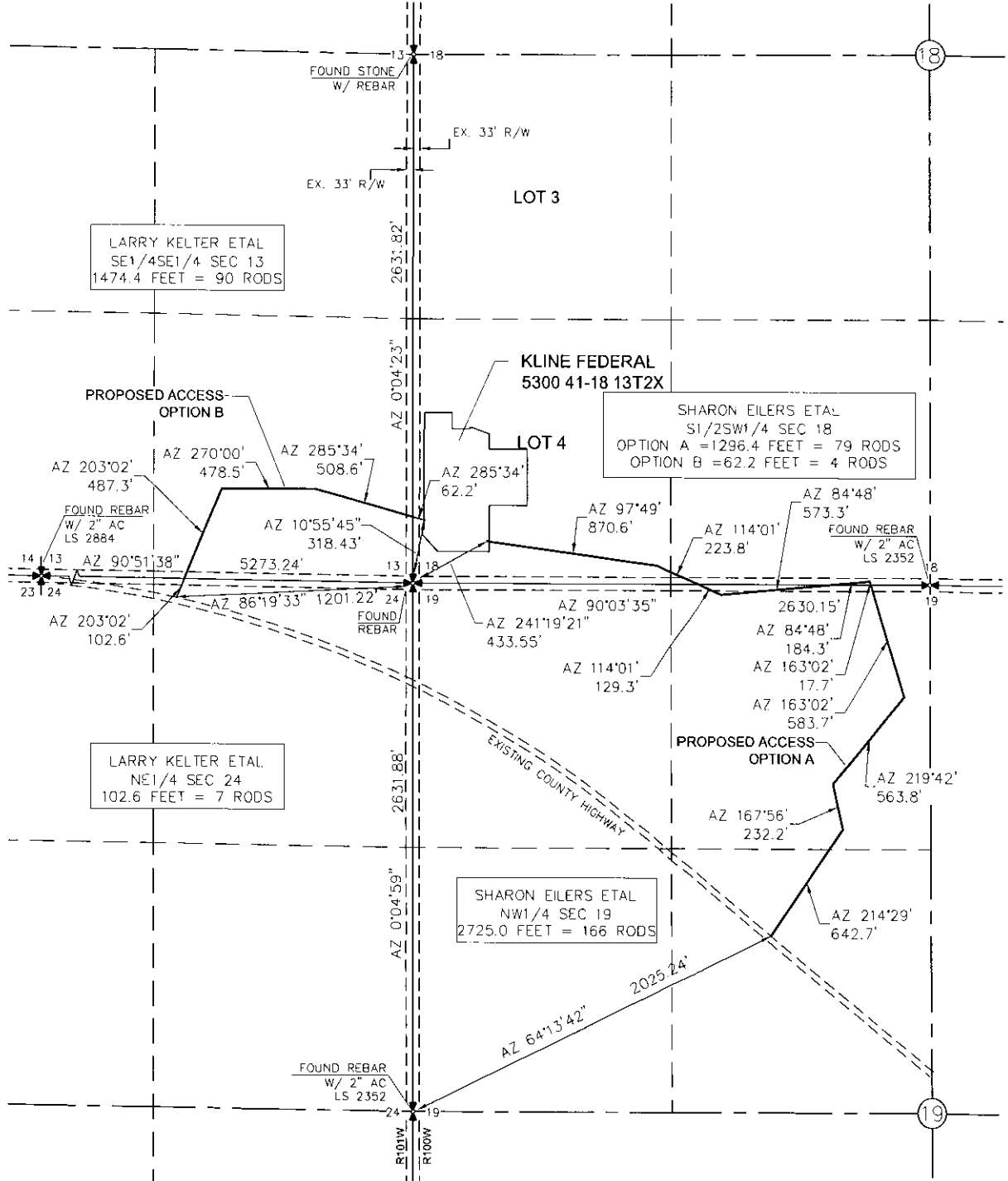
CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 3300 41-18 1372X"
 401 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



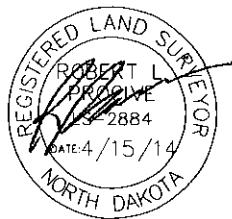
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SCALE
 HORIZ 1"=140'
 VERT 1"=35'

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Interstate Engineering, Inc.
P.O. Box 648
425 East Main Street
Sidney, Montana 59270
Ph (406) 433-5617
Fax (406) 433-5618

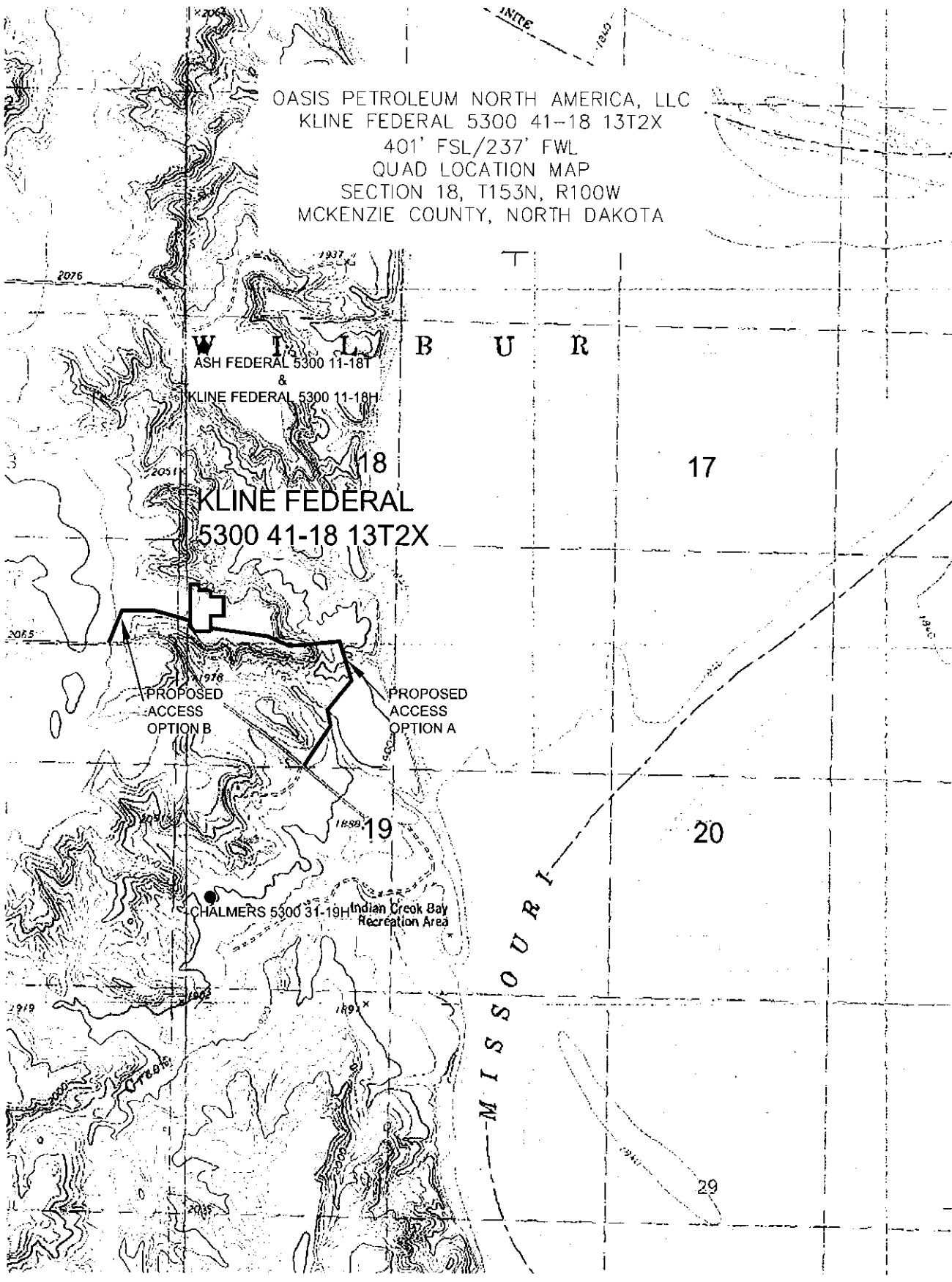
Other Offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
ACCESS APPROACH
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081.01</u>
Checked By: <u>R.L.P.</u>	Date: <u>APRIL 2014</u>

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
 KLINE FEDERAL 5300 41-18 13T2X
 401' FSL/237' FWL
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA



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SHEET NO.



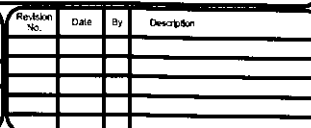
Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph (406) 433-5617
 Fax (406) 433-5618
 www.interstateeng.com
 Drawn in Montana, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 QUAD LOCATION MAP
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.01
 Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

401 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



401 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



5/29/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: **Kline Federal 5300 41-18 9T**
 Kline Federal 5300 41-18 10B
 Kline Federal 5300 41-18 11T2
 Kline Federal 5300 41-18 12TX
 Kline Federal 5300 41-18 13T2X
 Kline Federal 5300 41-18 14BX
 Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 41-18 9T, Kline Federal 5300 41-18 10B, Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 12TX, Kline Federal 5300 41-18 13T2X, and Kline Federal 5300 41-18 14BX. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Heather McCowan". The signature is fluid and cursive, with the first name "Heather" and last name "McCowan" clearly distinguishable.

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

Surface Damage Affidavit

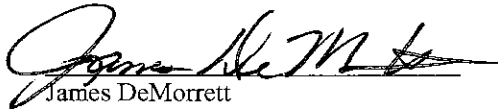
STATE OF NORTH DAKOTA)
) ss.
COUNTY OF WILLIAMS)

James DeMorrett, being duly sworn, states as follows:

The below described lands are not located in a wellhead protection/source water protection area per the website provided by the NDIC

and the adjacent landowner is fully aware that Oasis Petroleum will be drilling and operating multiple wells on spacing unit from a surface location in Section 18: Lot 4 a/k/a SW¼SW¼, Township 153 North, Range 100 West, 5th P.M., McKenzie County, North Dakota. (totaling 6.52 acres). The adjacent surface landowner is fully aware that the above referenced location will consist of the well bores and pumping units only and will not have cuttings pits, storage tanks or flare pits.

Dated this 16th day of June, 2014

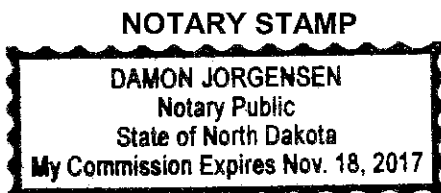

James DeMorrett

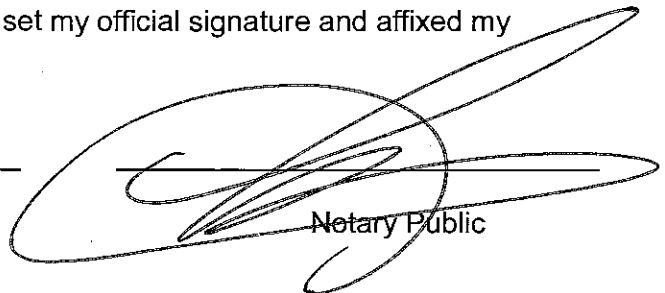
STATE OF NORTH DAKOTA }
 } ss.
COUNTY OF WILLIAMS }

BE IT REMEMBERED, That on this 16 day of June, 2014 before me, a Notary Public, in and for said County and State, personally appeared James DeMorrett, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires: Nov 18, 2017




Notary Public